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# Eloralintide Boc-Fragment 3, LSN4182928, SPPS Risk Assessment and PAR Report

Byoung Min, Riccardo Candeago, Chen-Chun Chen, Shu Zheng, Jenn Stockdill, Nil Tandogan Gray

Synthetic Molecule Design and Development

Eloralintide Boc-Fragment 3, LSN4182928

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## Abbreviations

| Abbreviation/Term | Definition |
|---|---|
| 2-CTC | 2-chlorotrityl chloride resin |
| AA | Amino acid (see table below for AA abbreviations) |
| adduct | Impurities with addition of certain functional group |
| API EM | Active pharmaceutical ingredient external manufacturing |
| Boc | tert-Butyloxycarbonyl |
| CIPC | Critical in-process control |
| CPP | Critical process parameter |
| CoA | Certificate of Analysis |
| CQA | Critical quality attribute |
| DCM | Dichloromethane |
| des | Impurities with loss of amino acids or functional groups |
| DIC | Diisopropylcarbodiimide |
| DIPEA | *N*,*N*-Diisopropylethylamine |
| DMF | Dimethylformamide |
| eq, equiv | Molar Equivalents |



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| | |
|---|---|
| EtOAc | Ethyl acetate |
| FIO | For information only |
| Fmoc | Fluorenylmethyloxycarbonyl |
| HPLC | High performance liquid chromatography |
| IPA | Isopropyl alcohol |
| IPAC | Isopropyl acetate |
| IPC | In-process control |
| UPLC | ultra-performance liquid chromatography |
| LOD | Loss on drying |
| MeOH | Methanol |
| MS | Mass spectrometry/spectrometer |
| MTBE | methyl *tert*-butyl ether |
| NLT | Not less than |
| NMT | Not more than |
| Oxyma | Ethyl cyano(hydroxyimino)acetate |
| PAR | Proven acceptable range |
| PrAR | Predicted acceptable range |
| PyBOP | Benzotriazol-1-yloxytripyrrolidinophosphonium hexafluorophosphate |
| RRT | Relative retention time |
| SM | Starting material |
| SMDD | Synthetic Molecule Design & Development |



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| | |
|---|---|
| SPPS | Solid-phase peptide synthesis |
| TFA | Trifluoroacetic acid |
| TFA-lutidine salts | Lutidinium trifluoroacetate |
| *t*Bu | tert-Butyl |
| Trt | Trityl |
| UHPLC | Ultra high performance liquid chromatography |
| UV | Ultraviolet |
| Vol | Volumes (L / kg basis) |

| **AA** | **Single Letter Abbreviation** | **Three or more Letters Abbreviation** |
|---|---|---|
| Glycine | G | Gly |
| Alanine | A | Ala |
| Threonine | T | Thr |
| Glutamic acid | E | Glu |
| Cysteine | C | Cys |
| Asparagine | N | Asn |



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| **Density** | **g/mL** |
|---|---|
| DMF | 0.948 |
| Piperidine | 0.862 |

## 1. Abstract

The following report describes the risk assessment for the solid phase peptide synthesis (SPPS) step used in the manufacture of eloralintide Boc-Fragment 3 (Boc-L3-OH, LSN4182928). The initial risk assessment was based on SPPS process knowledge with the output being potential risk parameter determination for experimental investigations. Parameters were combined into a series of bracketing experiments and directionally perturbed with an additive impact for a given CQA as a worst-case scenario, to determine PARs and characterize the level of risk (i.e., criticality) associated with those individual parameters. Where possible, parameters were tested to the full range of perturbation that can reasonably be expected in manufacturing, defined as six times normal parametric operating variability (6σ).

Stability studies were conducted during this work to assess exposure times during hold time between unit operation step of the SPPS process. As a result of analyses performed in this effort, 23 parameters are identified as CPPs, including temperature related parameters along with parameters with less than 6 sigma ranges and a couple of unspecified impurity related parameters.

## 2. Definitions

The following definitions have been standardized across all Synthetic Molecule Design and Development (SMDD) portfolio projects and regulatory submissions. Figure 1 illustrates how the risk level (*i.e.*, no risk, low, medium and high risk) is determined for each process parameter based on the product purity specification and results from baseline studies.

- Critical Quality Attribute (CQA): A physical, chemical, biological, or microbiological property or characteristic that should be within an appropriate limit, range or distribution



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to ensure the desired product quality. CQAs are the most important attributes to achieving desired product performance in the intended patient population, i.e., "the desired quality, safety, and efficacy."

- Process Parameters: Variables (e.g., time, temperature, molar equivalents of a reactant, operational equipment setpoint, etc.) associated with a unit operation within a manufacturing process. Typically include a target set-point and a proven acceptable range (PAR) of operation.

- Proven Acceptable Range (PAR): A characterized range of a process parameter for which operation within this range will result in producing a material meeting forward processing criteria or CQAs.

- Predicted Acceptable Range (PrAR): A predicted range of a process parameter based on a model or calculation for which operation within this range will result in producing a material meeting forward processing criteria or CQAs.

- No Risk Parameter: A parameter whose perturbation over a range of ±6σ of the setpoint showed no meaningful impact to the quality criterion (less than 10% the difference between the specification limit for the quality criterion and the baseline process response, blue dots in Figure 1).[1] In some instances, scientific rationale may be used to assign a parameter as no risk.

- Low Risk Parameter: A parameter whose perturbation over a range of ±6σ of the setpoint showed minimal impact to the quality criterion (greater than 10% and less than half the difference between the specification limit for the quality criterion and the baseline process response, green triangles in Figure 1).

- Medium Risk Parameter: A parameter whose perturbation over a range of ±6σ of the setpoint has an impact to the quality criterion of less than the specification limit and

[1] Seibert K. D. et al., "The Use of Routine Process Capability for the Determination of Process Parameter Criticality in Small-molecule API Synthesis," J Pharm Innov (2008) 3:105–112.



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greater than half the difference between the specification limit and the baseline process response (yellow squares in Figure 1.

- Critical Process Parameters (CPPs): A parameter whose variability has a high likelihood of impacting a CQA when outside the PAR. Therefore, it should be monitored or controlled to ensure the process produces drug substance that meets the CQAs. Medium and high-risk parameters are defined as critical due to their relatively narrow PAR, and/or the high likelihood of producing product that fails to meet the CQAs when operating beyond the PAR.

- Critical In-Process Control: A test or measurement performed during production to monitor and, if appropriate, to adjust the process parameters or inputs to ensure drug substance CQAs are met.

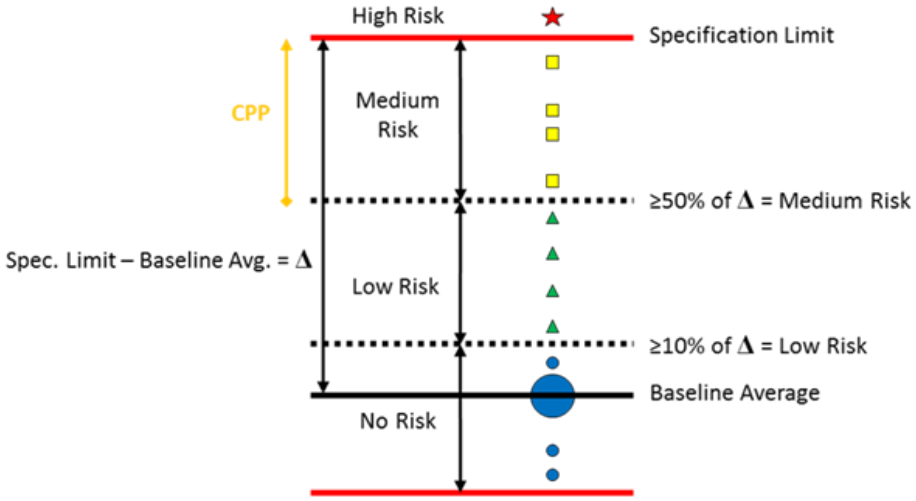



**Figure 1. Representation of no risk, low, medium and high risk parameters.**

## 3. Risk Assessment Methodology

The isolated Boc-Fragment 3 intermediate CQAs and critical in-process controls were used to assess the impact of a process parameter or material attribute on Boc-Fragment 3 intermediate quality.



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CQAs are established using clinically relevant outcomes based on an understanding of the drug substance's chemical and physical properties. Intermediate specifications and IPCs are developed iteratively based upon input material quality, spiking studies, process optimization, and by assessing the impact of parametric perturbations on the CQAs of the drug substance. The specifications for Boc-Fragment 3 were established to assure high purity material post downstream coupling and purification. Therefore, the CQAs utilized for this study are the release specifications for Boc-Fragment 3[2].

### 3.1. Engineering Controls

The process is defined such that engineering controls are in place in certain parts of the operations to eliminate the risk from parametric variability through manufacturing systems including procedural controls and/or mechanical set-up.

### 3.2. Parametric Controls

Once CQAs have been established and the synthetic routes have been finalized, the drug substance production processes are optimized around process parameter set points and acceptable ranges. The activities include:

- Use of iterative risk assessments to categorize process parameters
- Determine proven acceptable ranges (PARs)
- Process parameter risk categorization
- Study of the appropriate process parameters and input material properties in a multivariate manner to ensure any potential interactions will be accounted for and an acceptable design space established so further process refinement may occur without risk to product CQAs.

The generalized approach to parametric categorization is shown in Figure 2.

[2] RPT-1193758, Specification Rationale for Boc-Fragment L3 LSN4182928



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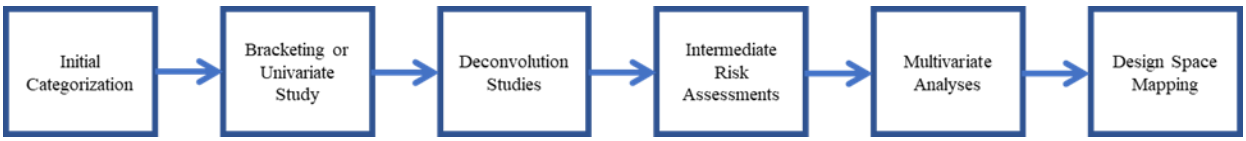



**Figure 2. Generalized approach to parametric categorization.**

### 3.3. Initial Parameter Categorization

For the initial parameter categorization, prior knowledge, experience with the current process, and experience with related processes allow for an initial characterization of the risk level for process parameters. Process parameters are characterized in this initial risk assessment as no-risk or potential risk. The initial determination of risk is conservative to ensure inclusion of all parameters that may pose risk to meeting the critical quality attributes of the drug substance.

### 3.4. Bracketing and/or Univariate Parametric Studies

After performing the initial risk assessment, PARs are determined for any parameter designated as a 'potential risk' parameter. Due to the complexity and the duration of synthesis for synthetic peptide processes, the study of individual parameters in a fully univariate sense may not be practical. A conservative approach is the selective pooling of univariate studies into grouped bracketing studies. Bracketing studies consist of the collective perturbation of several parameters in the same study, with a deliberate directional perturbation. This approach results in an additive impact to the requisite critical quality attribute. While experimentally more efficient, this methodology has the potential to be overly conservative in assigning parametric risk. Results from a bracketing study of elevated risk implicate all studied parameters, and result in all parameters carrying the same elevated risk when individual parametric contributions to a CQA impact cannot be determined from the collective assessment. As a result, deconvolution studies may be performed to elucidate specific contributions to a CQA impact.

The majority of the parametric ranges were studied across ranges considered statistically improbable under common cause operational variability, or beyond, but not necessarily studied to the limits of failure for the process. The breadth of the ranges covered by parametric studies, performed either univariately or in bracketing studies, is tied to the typical operational variability determined for the set of unit operations and equipment type utilized. Typical operational



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variability (σ) is defined as the level of variation from a set point that is expected from an equipment set due to the natural dynamics of the system. Studies were performed to analyze the equipment used for various processes, at various scales and for a variety of unit operations, to determine the expected σ. Initial estimates of σ were based upon both internal and external benchmarking efforts aimed at determining worst-case control scenarios for typical operating parameters. The values for σ are established as baseline performance criteria expected of manufacturing process equipment. The strategy by which the manufacturing data were analyzed to determine σ is the subject of several external publications.[1,3]

Assuming all other parameters are held at their target set points, the probability of a parameter falling outside a weighted operational variability range (denoted by $\pm k\sigma$), when subject to common cause variability, is very small for large values of *k*. After an examination of a variety of statistical distributions, *k* = 6 was chosen as a conservative weighting factor. The value of *k* = 6 ensures that at least 99.5% of the values of a given process parameter fall within the range used to evaluate the criticality of that parameter, even given severely skewed, non-normal distributions of parameter values. Therefore, the risk-based approach for determining parametric risk for the eloralintide Boc-Fragment 3 manufacturing process is extremely conservative.

If, through mining of historical Boc-Fragment 3 data, it was determined that the range for a parameter was already studied to an appropriate extent, and resulted in intermediate or drug substance meeting specification, the results are documented and used to establish the PAR. For situations in which historical data was unavailable, mathematical modeling and/or experimental studies were performed to establish PARs. For PARs determined through modeling or bracketing analysis, the responses are multivariate in nature, and therefore conservative in their estimation of the univariate impact to an individual CQA.

[3] Mitchell, et al., "Unit Operations Characterization Using Historical Manufacturing Performance," Ind. Eng. Chem. Res., 47 (17) 6612–6621, 2008.



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### 3.5. Deconvolution Studies

As discussed, bracketing studies resulting in elevated risk may result in further targeted univariate studies to determine the univariate response a particular processing parameter may have on a specific CQA. These deconvolution studies assist in the proper assignment of risk rather than assigning an elevated risk level to all the parameters in a bracketing study when that risk is likely being driven by a particular factor or subset of factors in that bracketing study.

### 3.6. Intermediate Risk Assessments

After performing the bracketing and/or univariate studies for determining PARs for any parameter designated as a 'potential risk' parameter, a more comprehensive determination of the parametric risk associated with an individual parameter is done through scientific judgment, analysis of the PARs relative to the pre-determined operating capability, or analysis of responses from the failure mode and effects analysis (FMEA). The magnitude of the impact to relevant quality attributes of isolated and non-isolated intermediates or to the Boc-Fragment 3 CQAs are used to categorize the parameter as either high, medium, low, or no risk (Table 1).

**Table 1.** Parameter risk levels

| Risk Level | Description | Criticality Level |
|---|---|---|
| **High** | A parameter with a narrow proven acceptable range (PAR), where operation outside the PAR may occur under common cause variability. Operation outside the PAR has a high probability of producing Boc-Fragment 3 that fails to meet CQAs. | **Critical** |
| **Medium** | A parameter with a sufficiently large PAR where operation outside the PAR is statistically improbable based on common cause variability. Operation outside the PAR has a moderate probability of producing Boc-Fragment 3 that fails to meet CQAs. | |
| **Low** | A parameter with a sufficiently large PAR where operation outside the PAR is statistically improbable based on common cause variability. Operation outside the PAR has a low probability of producing Boc-Fragment 3 that fails to meet CQAs. | **Non-critical** |



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| | | |
|---|---|---|
| **No** | A parameter with a sufficiently large PAR where operation outside the PAR is statistically improbable based on common cause variability. Operating at the limits of the PAR has negligible impact to relevant quality attributes compared to baseline conditions. | |

Nearly all process parameters have the potential to impact CQAs to some extent when evaluated at extreme ranges. The process by which a parameter is deemed "critical" is based on a fixed set of criteria, considering both the breadth of a PAR and the relative impact to a critical quality attribute. Internally, parameters may be designated as no, low, medium, or high risk based on this analysis, but all parameters that are designated as medium or high risk are considered critical process parameters (CPPs).

To differentiate between no, low, medium, and high-risk parameters, an analysis of the univariate or bracketed responses across the PAR are compared to a fixed quality criterion to determine the parametric risk category. A comparison of the baseline response (process at target operating conditions) to the impact on a quality criterion when operating at a minimum distance of 6σ from process target serves as the basis for categorizing parameters Figure 3.



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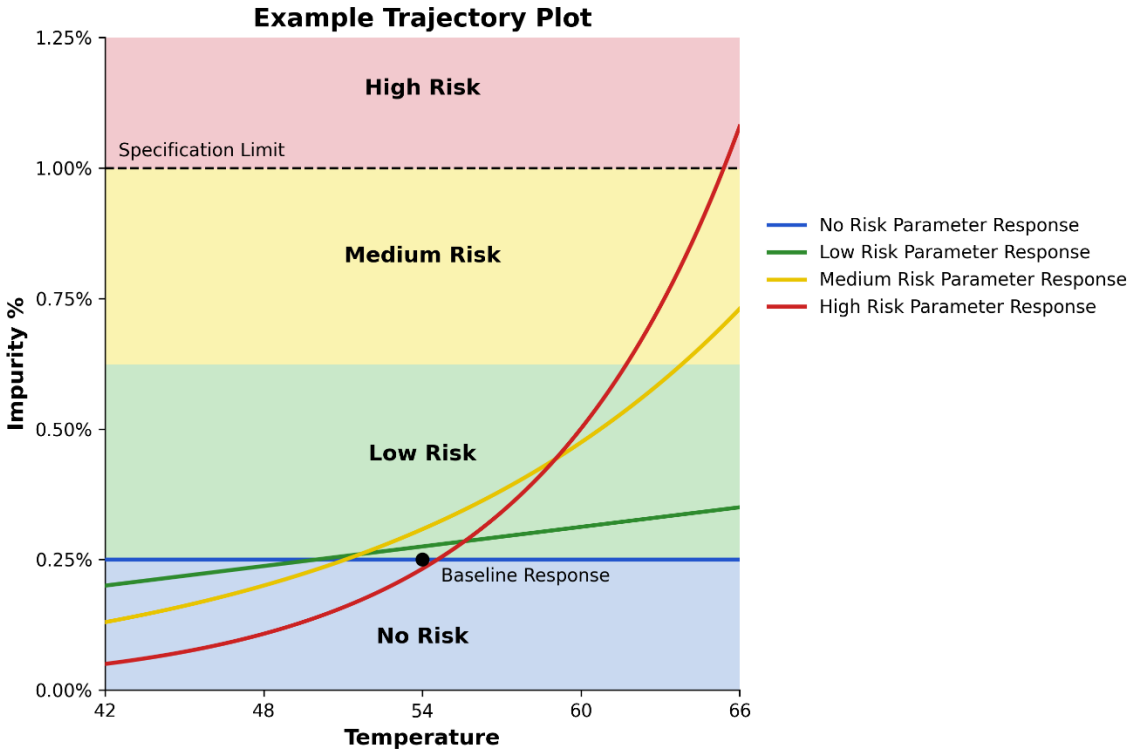



**Figure 3.** Example trajectory plot - graphical representation of the regions comprising impurity response leading to a no, low, medium, or high-risk designation for an example process parameter.

A **high-risk parameter** (red line in Figure 3) is a parameter whose perturbation to six times the operating variability would fail to meet the quality criterion.

A **medium-risk parameter** (yellow line in Figure 3) is a parameter for which the response across the PAR showed an impact to the quality criterion of less than the specification limit and greater than half the difference between the specification limit and the baseline process response.

A **low-risk parameter** (green line in Figure 3) is a parameter for which the response across the PAR studied showed minimal impact to the quality criterion of less than half the difference but more than 10% of the difference between the specification limit for the quality criterion and the baseline process response.

A **no-risk parameter** (blue line in Figure 3) is a parameter for which the response across the PAR studied showed no impact to the quality criterion defined as 10% or less of the difference between the specification limit for the quality criterion and the baseline process response.



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The analysis of the trajectory responses of the process parameters allowed identification of parameters with negligible impact to quality attributes across the entire studied range relative to expected baseline performance. The conservative inclusion of parameters for further study from the initial risk assessment resulted in the inclusion of parameters in this category during the intermediate risk assessment. All parameters categorized as medium and high were designated as critical process parameters, low risk parameters were considered non-critical. Parameters identified as no-risk are also considered non-critical and were not necessarily further studied as part of the multivariate analysis.

When including multiple non-interacting parameters into a single study (i.e., bracketing study), the results from this one study serve as a surrogate for several univariate studies as described above. While the analysis is the same, the visualization of the data includes a summary of the bracketing studies and the net impact to the CQA from the collective set of perturbed parameters in the form of a Limit Plot (Figure 4). The designated point in the plot represents the net impact to the CQA from all parameters collectively studied at their PAR limits. Collective responses resulting in a series of parameters being categorized as having elevated risk can then be deconvoluted with targeted univariate experimental studies.

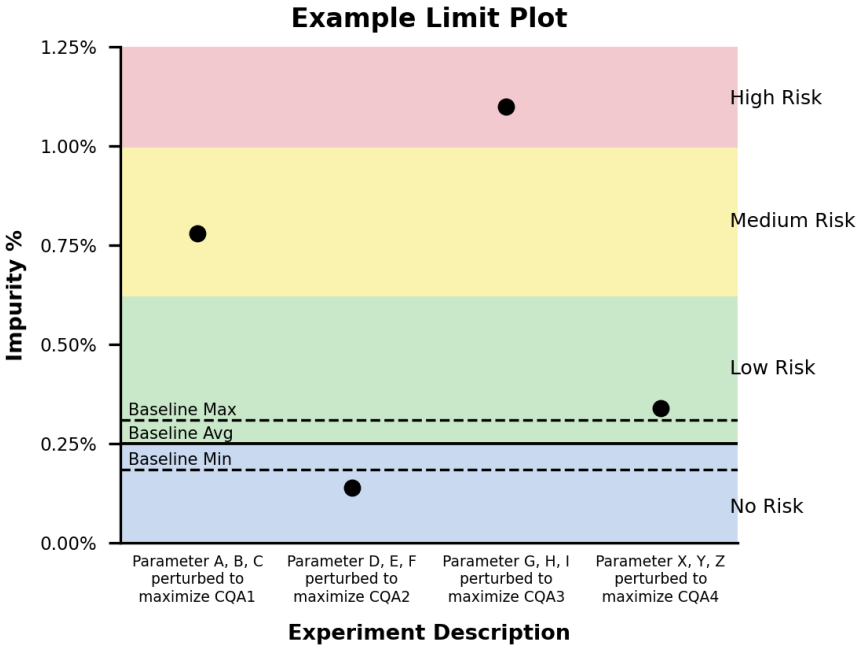



**Figure 4.** Example limit plot used to assess parametric risk for a grouped collection of parametric studies



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## 4. Eloralintide Boc-Fragment 3 Process Description and General Methodology for Establishment of PARs

This section describes the work to determine the proven acceptable ranges (PARs) for SPPS steps of Boc-Fragment 3 intended for validation (Scheme 1). The amount of 2-CTC resin charged is the basis for the SPPS process. The theoretical loading of ~1.16 mmol/g of resin using 2-CTC resin with 1.54 mmol/g is used as molar basis for equivalents charge for all steps. The entire SPPS process can be broken down into 8 sub-unit operations as depicted in Figure 7 and corresponding equipment flow diagram is shown in Figure 5. For a more detailed summary of the process, please see the lab model section (7.1).

**Resin loading (Step 1);**

The process starts with loading of Fmoc-Gly-OH onto 2-chlorotrityl chloride (2-CTC) resin.

The resin is charged to a jacketed reactor, swelled with DMF, and drained. Fmoc-Gly-OH is dissolved in DMF with DIPEA and transferred to the resin. Following loading, the resin is washed with DMF and remaining active sites are capped with MeOH and DIPEA in DMF.

**Fmoc deprotection (all cycles)**

Fmoc deprotection is accomplished using piperidine in DMF. For all amino acid positions a single piperidine/DMF treatment is applied. The resin is washed with iPrOAc and DMF following each deprotection step.

**Amino acid coupling (Steps 2–10)**

Amino acid coupling conditions vary by residue. Fmoc-L-Thr(tBu)-OH (Step 2) is activated using PyBOP, Oxyma, and DIPEA prior to coupling. All remaining Fmoc-amino acids (Steps 3–9: Fmoc-L-Ala-OH·$H_2O$, Fmoc-L-Cys(Trt)-OH, Fmoc-L-Thr(tBu)-OH, Fmoc-L-Ala-OH·$H_2O$, Fmoc-L-Thr(tBu)-OH, Fmoc-L-Asn(Trt)-OH, and Fmoc-L-Cys(Trt)-OH) are coupled using DIC and Oxyma in DMF. After each coupling, the resin is washed with DMF. In the final backbone step (Step 10), Boc-L-Glu-OtBu is coupled using PyBOP, Oxyma, and DIPEA, followed by a DMF wash.

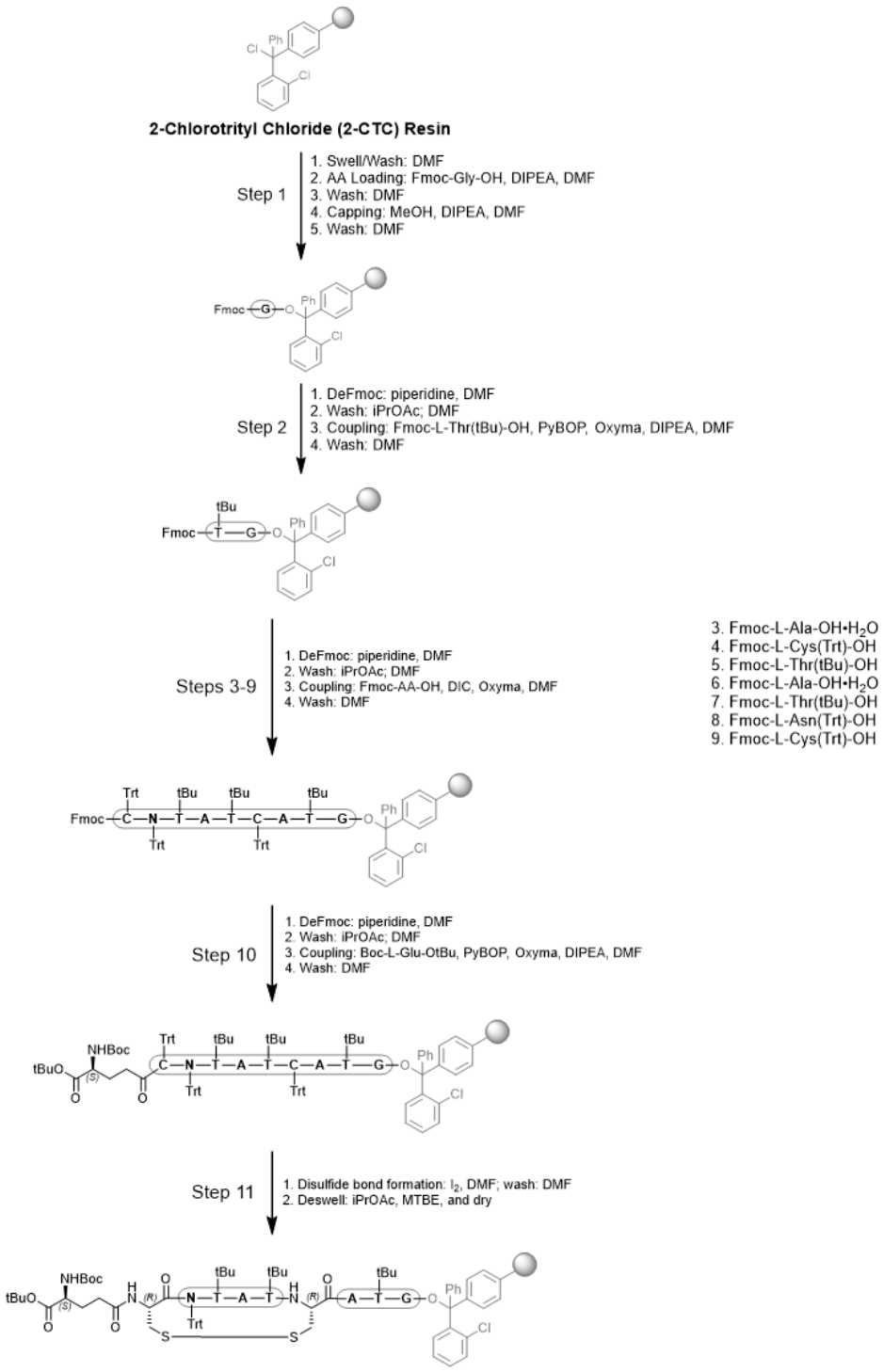
**Disulfide formation, de-swell, and dry (Step 11)**

The intramolecular disulfide bond is formed by treatment with $I_2$ in DMF, followed by a DMF wash. The resin is then de-swelled and dried via sequential washes with iPrOAc and MTBE, and dried under vacuum with $N_2$ flow to the target LOD.



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**Scheme 1. Eloralintide Boc-Fragment 3 (LSN4182928) synthetic route.[4]**

[4] LY3841136.sodium-pSRJUL2026; Boc-Fragment 3 (LSN4182928) is the product generated after L3 Step 3 while L2 Step 1 & 2 yields Boc-[L3SS]-OH (LSN4214882).



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The equipment flow diagram outlining the production of Boc-Fragment 3 on resin is provided below (Figure 5). Activation of the incoming amino acid with OxymaPure and PyBOP is performed in the Activation Tank, which also receives DIPEA, DIC, and DMF charges prior to transfer of the solution to the synthesis reactor. Dry 2-CTC resin is charged directly to the synthesis reactor through the top charge port; it is not dissolved or transferred from a dissolution tank. The synthesis reactor then contains the 2-CTC resin with the growing peptide sequence. A dissolution tank is used to prepare the DIC/DMF solution charged in the coupling steps. A second dissolution tank is used to prepare bulk quantities of the 20% piperidine/DMF solution for use in the Fmoc removal steps. A third dissolution tank supplies the DMF and $I_2$ charges. The product collected at the end of the sequence is Boc-Fragment 3 disulfide on resin.



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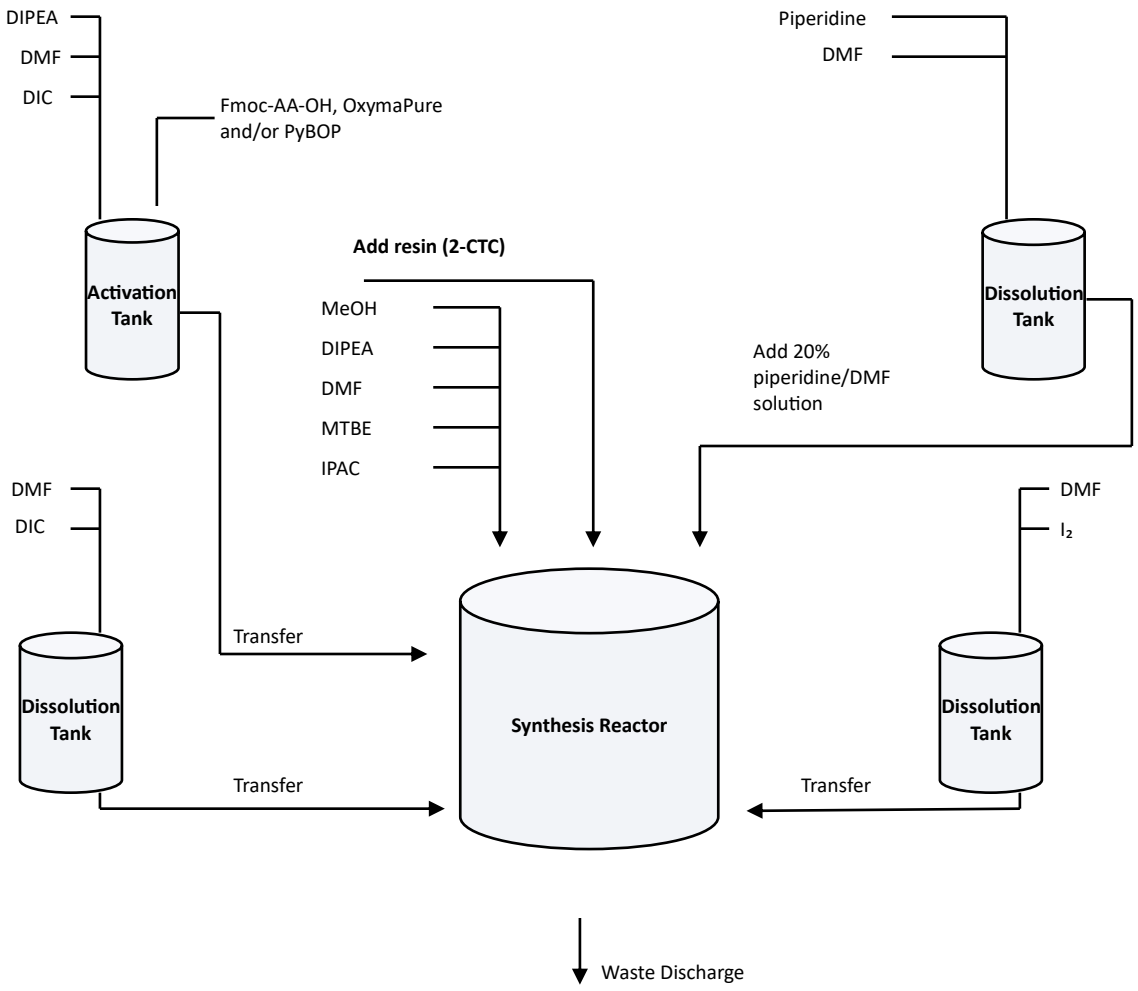



**Figure 5.** Process diagram for eloralintide Boc-Fragment 3 disulfide solid phase synthesis

## 5. Quality Metrics and Evaluation of CQAs

The critical quality attributes assessed for parametric impact in the intermediate fragment synthesis process include the various impurities formed during the SPPS, cleavage and dithioacetal formation ("dithioacetalization") steps. The predominant (most conservative) measure of the impact of a perturbation under parametric control is to forward process all materials to eloralintide crude API, purify under baseline conditions, and assess product quality via impurity analysis post chromatography purification using the preliminary drug substance release tests and specifications. Boc-Fragment 3 is a registered intermediate with release specifications and is multiple steps away from the chromatographic purification assessment.



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Therefore, material generated from parametric perturbations during the synthesis of Boc-Fragment 3 was assessed against the intermediate fragment release specification (see **Table 4**).

Each of the process unit operations and associated parameters (*e.g.*, charge quantity, temperature, stir time, etc.) were considered routine operational variables for the manufacturing equipment (see Section 6 for a more in-depth discussion of the ranges used), and their potential to negatively impact on the quality of LSN4182928 was assessed. Experiments were generally designed to cover the range of routine operational variables.

Several classes of impurities are commonly present in traditional solid phase peptide synthesis. Considering that SPPS related impurities form through well-known pathways that can occur at each amino acid, the impurities can be grouped into categories to enable the process risk assessment. Table 2 lists the common impurities and how they are formed during the SPPS, resin cleavage, and/or elaboration to dithioacetal. These impurities in the eloralintide fragment synthesis process are controlled via synthesis parameters (e.g., temperature, stoichiometry, reaction time, concentration), by in-process controls, through specifications on the amino acid starting materials, and at the GMP fragment intermediate specifications. [2]

**Table 2.** Impurities formed during SPPS process for eloralintide L3.

| SPPS-Related Impurity | Cause(s) |
|---|---|
| **Addition** | • Premature Fmoc removal by exposure to base (incomplete base removal during resin washing, excess base addition, long exposure time to base)<br>• AA starting material contains H-AA-OH or Fmoc-AA-AA-OH<br>• H-$(AA)_n$-OH present during coupling (insufficient piperidine removal during post-deFmoc washing leads to deFmoc of incoming Fmoc-AA-OH, impurity in the starting material, or premature peptide cleavage from resin during coupling). |
| **Deletion** | • Incomplete coupling<br>• Incomplete de-Fmoc |
| **Truncation** | • **After incomplete coupling:** halted chain elongation most likely due to sequence-specific aggregation or inaccessible N terminus due to conformation within the resin matrix<br>• **After incomplete deFmoc:** If deFmoc is not followed by any additional deFmoc steps, truncation is observed rather than typical deletion products |
| **D-Isomer** | • D-Isomer impurity **in RSM**<br>• **Racemization** during Fmoc-AA-OH activation<br>• **Epimerization** during chain elongation (due to repeated base exposure); commonly seen with Thr, Cys |



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| SPPS-Related Impurity | Cause(s) |
|---|---|
| **Degradation Products (often unknown structure)** | • Exposure to high temperature<br>• pH Excursions<br>• Exposure to oxygen/oxidants<br>• Exacerbated by long exposure time to above conditions |
| **Acetyl capping** | • **Planned:** Capping performed after a challenging coupling to avoid deletion impurities<br>• **Unplanned:** AcOH present during coupling reaction (example impurity source: impurity in IPAC used for washing steps) |
| **Protecting group loss** | • Removal of acid-labile protecting group due to HI formation during oxidative cleavage of Trt |

| Soft Cleavage-Related Impurity | Cause(s) |
|---|---|
| **Degradation Products** | • Exposure to acidic conditions or heat over a long period of time |
| **Protecting group loss (or migration)** | • Protecting group on the AA side-chain functional group is removed in presence of acid (loss of *t*Bu, Trt, Boc) with possible re-incorporation on other nucleophilic residues<br>• May be followed by other side reactions (e.g., loss of Trt from Asn) |

| Dithioacetalization-Related Impurity | Cause(s) |
|---|---|
| **Disulfide** | • Incomplete reduction of disulfide during conversion of disulfide to dithiol |
| **Linear dithiol** | • Incomplete alkylation during conversion of dithiol to dithioacetal |
| **L3 Dimers** | • Reaction of Cys from one L3 molecule with iodomethyl-Cys intermediate from another L3 molecule followed by the closure of the large ring by forming a second transannular dithioacetal linkage<br>• Occurs when localized concentration of L3 linear dithiol is high relative to diiodomethane (addition rate and mixing control)<br>• Can be head-to-head (Cys2-$CH_2$-Cys2' and Cys7-$CH_2$-Cys7') or head-to-tail dimerization (Cys2-$CH_2$-Cys7' and Cys7-$CH_2$-Cys2') |



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### 5.1. Discussion of impurities observed during SPPS, cleavage, and dithioacetalization to form Boc-Fragment 3

Both the upstream SPPS process and downstream cleavage processes contribute to formation of eloralintide fragment impurities. The SPPS process can form impurities such as single amino-acid addition, single amino-acid deletion, D-isomers, acetyl capping on the N terminus, and degradation to form unknown impurities. The oxidative cleavage of the trityl groups in the fully protected, linear, resin-bound peptide can also lead to an exothermic temperature excursion, which can cause loss of acidic protecting groups and premature resin-cleavage. The acidic cleavage (dilute TFA) of the protected L3 disulfide (Boc-L3SS-OH) sequence from the resin would not impact the levels of these impurities once formed. During subsequent conversion of the disulfide to the final Boc-L3-OH structure, a reduction of the disulfide is followed by immediate alkylation with diiodomethane. Incomplete reaction of these steps results in disulfide and dithiol impurities, respectively. High local concentrations of the L3 dithiol during dithioacetalization can lead to double dithioacetal-linked L3 dimers. Eloralintide Boc-Fragment 3 is made up of 10 amino acid residues. For each residue, there is the possibility of the formation of several of these impurities. Deletion impurities are among the most common impurities in SPPS. They can arise if the preceding Fmoc deprotection reaction or the amino acid coupling reaction does not go to completion. Acetyl capped impurities arise from the presence of residual acetic acid in coupling reactions. Acetic acid can be activated with DIC and Oxyma and react with the free amine of the growing peptide chain. All of the impurities described above contain a C-terminal carboxylic acid and are therefore capable of forward processing through the condensation with H-L21-$NH_2$ and global deprotection, presenting a possible risk to final API purity.

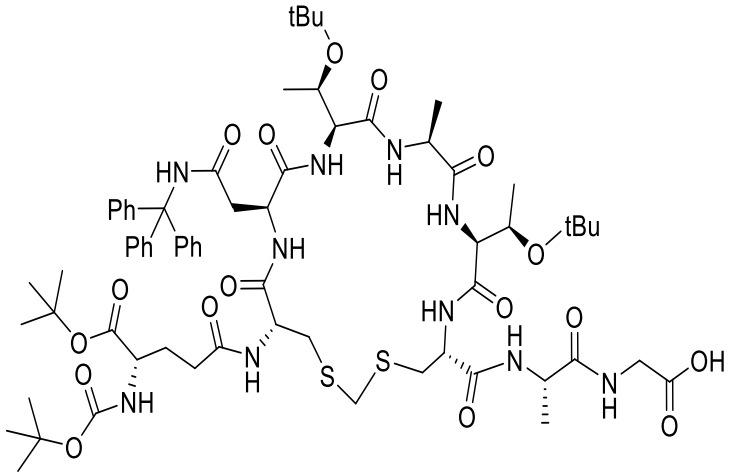
Individual impurities were analyzed by a UHPLC-UV method and SPPS process parameter ranges were tightened to meet the proposed specifications. However, even with a well-controlled process, a number of low-level impurities could be present at the end of synthesis. To ensure specifications were appropriate, detectability at the fragment stage was evaluated. Accordingly, PAR experiments for SPPS and cleavage were also forward processed to the final Boc-L3-OH product. There are numerous potential impurities associated with SPPS, and the most frequently observed impurities for Boc-Fragment 3 are summarized in Table 3.



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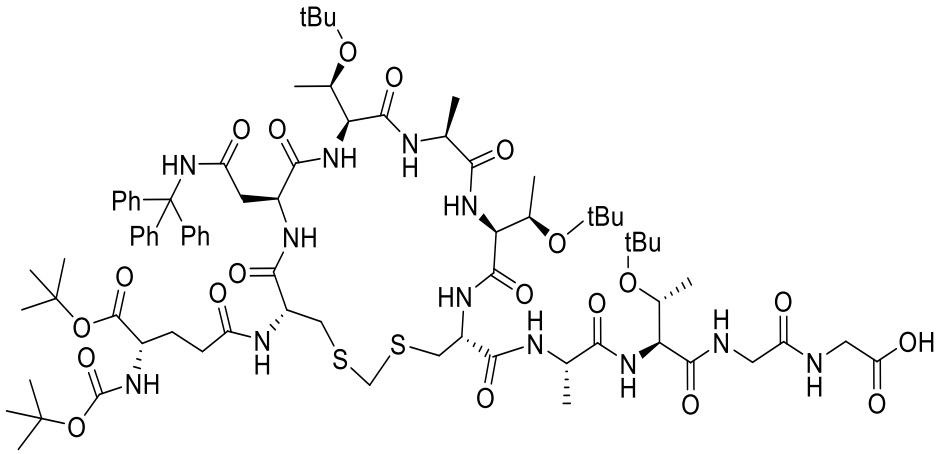
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**Table 3.** Impurity types formed during SPPS of Boc-Fragment 3, their points of control, and a representative example of each.

| Impurity Type | Observed Species | Point(s) of Control |
|---|---|---|
| **Deletion** | L3 des-Asn3 @ RRT 0.64<br>L3 Fmoc-2-10 des-Asn @ RRT 0.77<br>L3 des-Thr4 @ RRT 0.82<br>L3 des-Thr9 @ RRT 0.90 | • **Fmoc deprotection:** piperidine concentration, reaction time and temperature, post-deFmoc hold time<br>• **Coupling reaction:** amino acid eq., PyBOP eq, DIPEA eq, Oxyma eq, DIC eq, DMF charge quantity (dilution), reaction time and temperature<br>• **Capping reaction:** $Ac_2O$ eq, pyridine eq, DMF charge (dilution), reaction time and temperature |
| **Representative Example:**<br>  <br>L3 des-Thr9 | | |

| Impurity Type | Observed Species | Point(s) of Control |
|---|---|---|
| **Addition** | L3 endo-Ala @ RRT 0.90<br>L3 endo-Ala @ RRT 0.95<br>L3 endo-Gly10 @ RRT 0.96<br>L3 endo-γGlu1 @ RRT 1.04<br>L3 endo-Thr @ RRT 1.11 | • **Fmoc deprotection:** Wash temperature, wash volume, number of washes<br>• **Coupling reaction:** amino acid eq., Oxyma eq, DIC eq, coupling time, wash temperature, wash volume, number of washes<br>• **Raw material:** amino acid specification |

**Representative example:**




L3 endo-Gly10

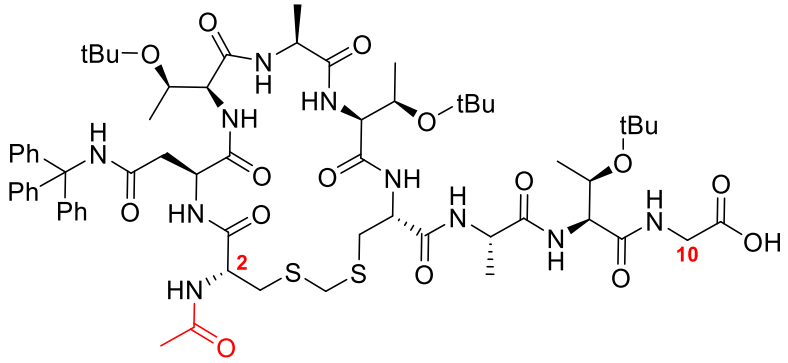


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| Impurity Type | Observed Species | Point(s) of Control |
|---|---|---|
| **Acetyl capping** | L3 Ac-2-10 @ RRT 0.75 | • **Raw Material:** Acetic acid specification in IPAC |

**Representative Structure:**



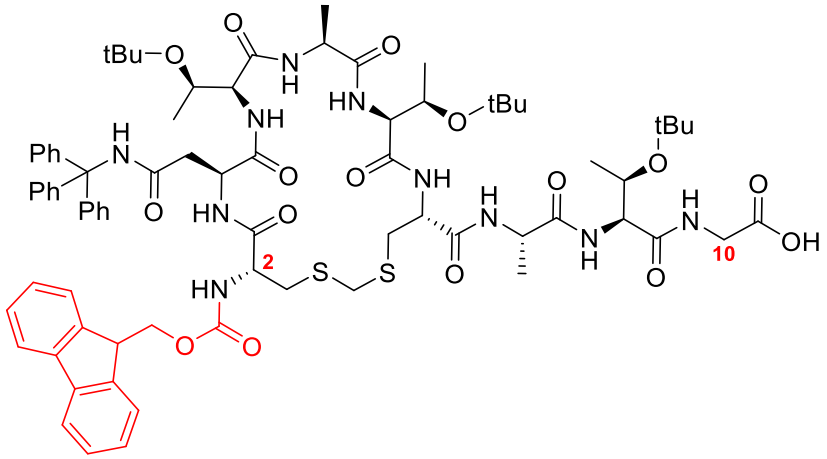

L3 Ac-2-10@ RRT 0.742



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| Impurity Type | Observed Species | Point(s) of Control |
|---|---|---|
| **Truncation** | L3 Fmoc-2-10 @ RRT 1.06 | • **Fmoc deprotection:** deFmoc solution temperature, and volume, reaction time, agitation speed |

**Representative Structure:**



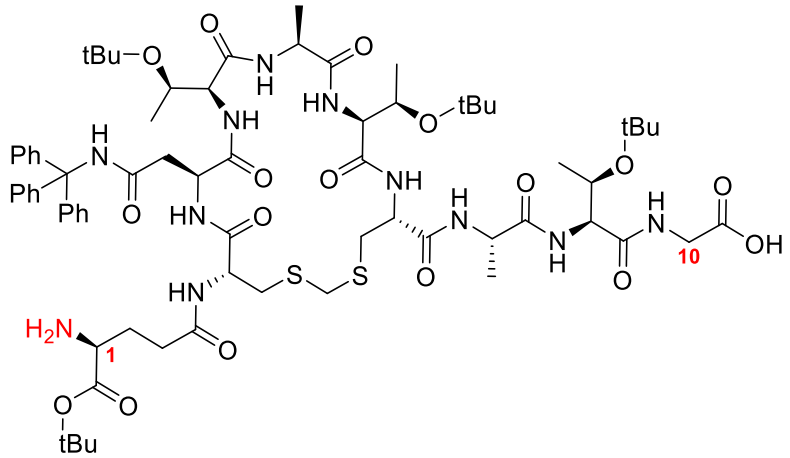

L3 Fmoc-2-10



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| Impurity Type | Observed Species | Point(s) of Control |
|---|---|---|
| **Protecting group modification/loss** | L3 ɣGlu1(des-Boc) @ RRT 0.56<br>L3 des-tBu @ RRT 0.82<br>L3 des-tBu @ RRT 0.88 | • **Des-Boc or Des-tBu:** cleavage temperature, TFA concentration in cleavage, cleavage time (exposure time to TFA), cleavage wash temperature |

**Representative Example:**




L3 γGlu1(des-Boc)

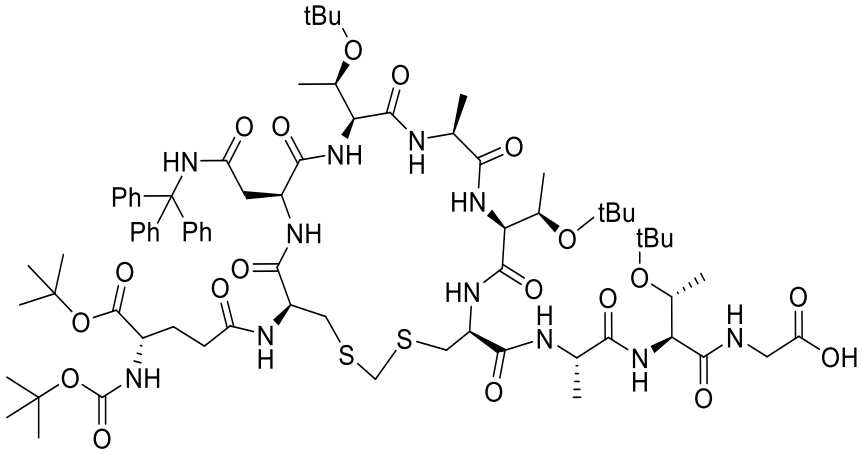


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| Impurity Type | Observed Species | Point(s) of Control |
|---|---|---|
| **D-Isomers** | L3 D-Cys7 @ RRT 0.98<br>L3 D-Cys2&7 @ RRT 1.03 | • **Coupling reaction:** Activation time and temperature, base concentration (if relevant)<br>• **DeFmoc reaction:** piperidine concentration, deprotection time and temperature, concentration<br>• **Raw material specification:** D-isomer ≤ 0.2% |

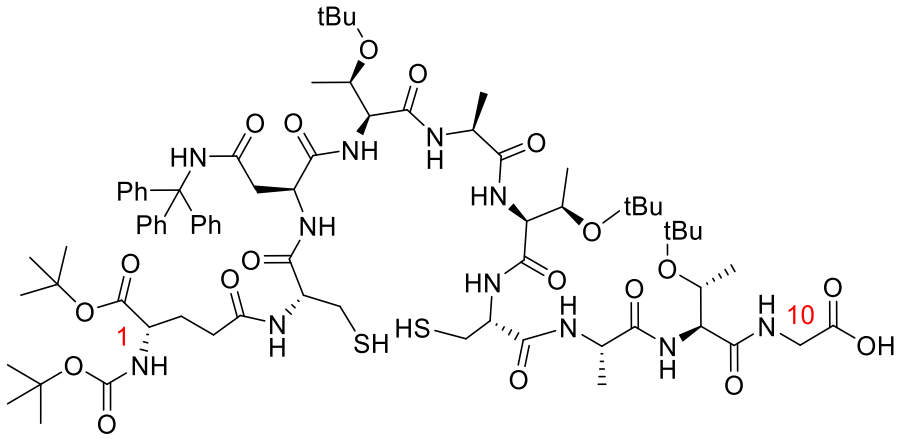
**Representative Structure:**




L3 D-Cys2&7 @ RRT 1.03



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| Impurity Type | Observed Species | Point(s) of Control |
|---|---|---|
| **Step 3 impurity** | L3 Boc-L3-SS @ RRT 1.01<br>L3 linear dithiol @ RRT 0.92 | • **TCEP reduction reaction;** TCEP stoichiometry, temperature and reaction time, and IPC<br>• **Thioacetal reaction;** temperature, reaction time, and IPC |
| **Representative Structure:**<br>  <br>L3 linear dithiol @ RRT 1.03 | | |

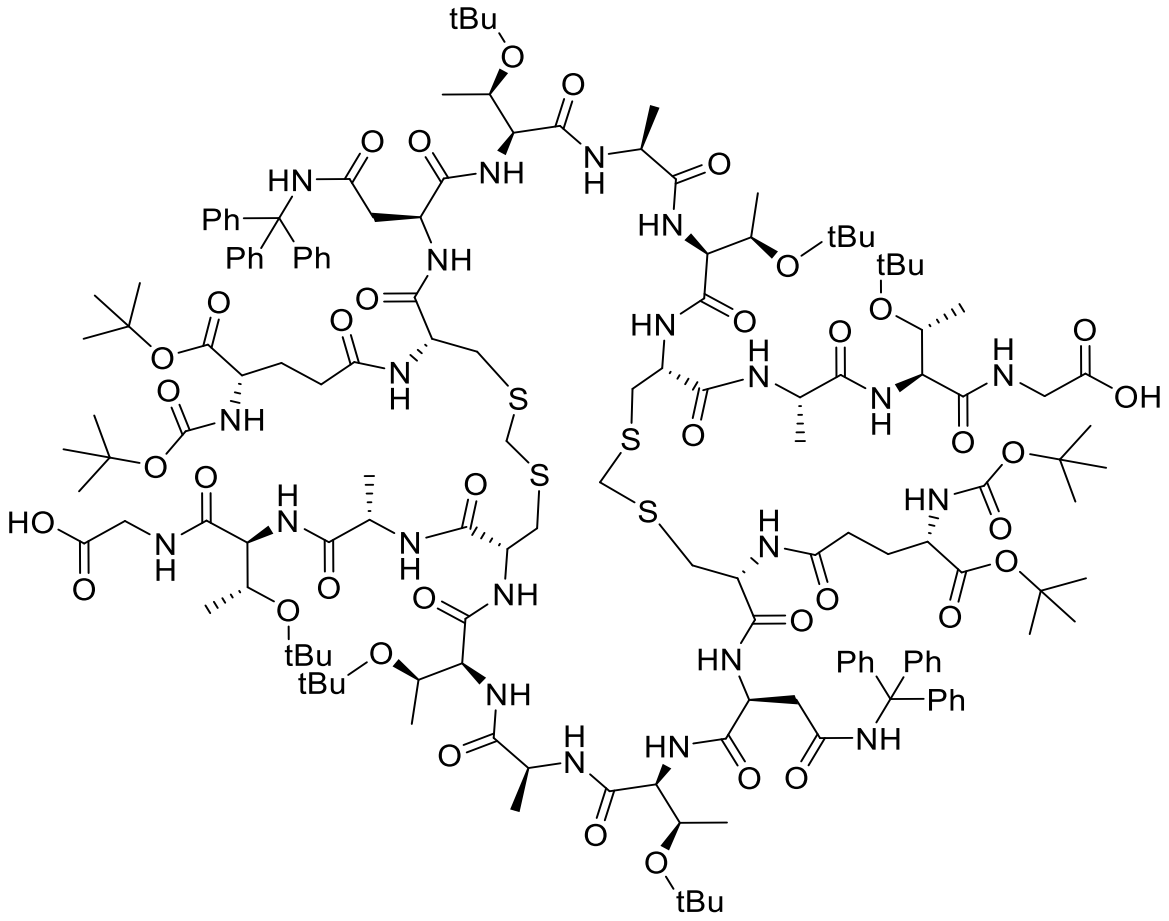


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| Impurity Type | Observed Species | Point(s) of Control |
|---|---|---|
| **Dimer** | L3 Dimer @ RRT 1.34<br>L3 Dimer (single D-Cys) @ RRT 1.37<br>L3 Dimer (single D-Cys) @ RRT 1.39 | • **Disulfide Formation:** iodine concentration, temperature, and rate of addition |

**Representative Structure:**



L3 Dimer @ RRT 1.34



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| Impurity Type | Observed Species | Point(s) of Control |
|---|---|---|
| **Unknown** | L3 Imp @ RRT 0.49<br>L3 M+294 Da @ RRT 0.69<br>L3 M+262 Da @ RRT 0.69<br>L3 M-257 Da @ RRT 0.74<br>L3 M-522 Da @ RRT 0.81<br>L3 M-213 Da @ RRT 0.83<br>L3 M+46 Da @ RRT 1.01<br>L3 M+48 Da @ RRT 1.31 | **Parametric** controls at relevant SPPS sub-steps. |



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### 5.2. L3 SPPS Control Strategy

The control strategy shown in **Figure 6** was followed for all relevant impurities. Impurities that have been observed to be an out-of-specification (OOS) risk for an unspecified impurity limit of 0.5% would require either tightening of parameters or reclassification as a specified impurity. To raise a specification limit (from the 0.5% any unspecified impurity target), material containing an impurity at that level would need to be

a) forward processed through the rest of the API synthesis process (LPPS steps and chromatography related operations) or

b) spiked into the crude API dissolution and subsequent chromatography to ensure that sufficient impurity purge is observed to provide API that meets purity specifications after the second chromatography purification step.

All impurities from the PAR studies and any spiking studies will be compared to baseline levels and the risk level categorized as defined in Section 2.

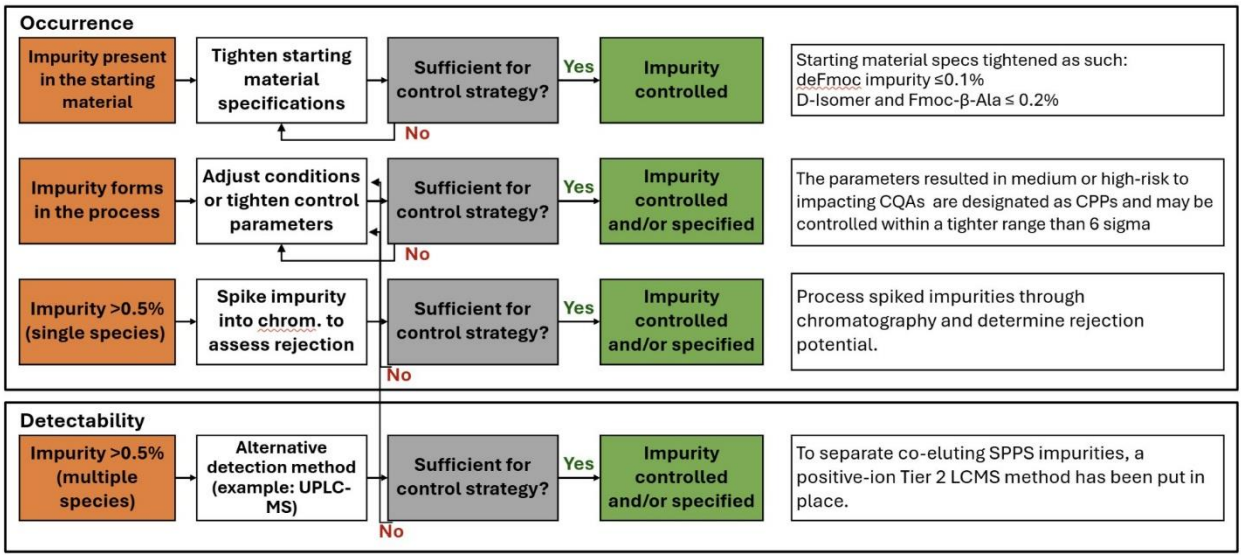



**Figure 6. Decision tree for handling known process impurities based on occurrence and detectability.**



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### 5.3. UPLC-UV and UPLC-MS Release Methods

Post-cleavage, the isolated material is first analyzed by UHPLC-UV. If any impurity peak determined by the UHPLC-UV method is above the specification criteria, additional release testing via a UHPLC-MS method will be performed to determine the composition of the impurity peak. Once the composition of the peak of interest is determined, the UV peak area percent is corrected and reported. Table 4 lists proposed specifications (via UHPLC-UV) for purity, specified impurities, and any unspecified impurity limits of Boc-Fragment 3 (LSN4182928) isolated solid. These specifications are appropriate for this stage of development based on process knowledge. Specifications can be revised prior to submission based on additional development experiments or manufacturing experience. LC-MS analysis was performed for each PAR experiment to better understand the impurity profile and impurity identification, which is critical for setting specifications. Moving forward, MS analysis will be triggered only when the UHPLC peak area% exceeds the proposed limit. The MS data will be used to correct the UHPLC UV data and ensure the components remain within the specified limits.

**Table 4. Specification for the LSN4182928, Boc-Fragment 3[2]**

| Test | Analytical Procedure | Acceptance Criteria |
|---|---|---|
| **Identification Test** | | |
| Identity | UHPLC-UV | Retention time compares favorably with that of the reference standard |
| **Potency Test** | | |
| Assay | UHPLC-UV | Not less than 80.0% w/w |
| **Purity Tests** | | |
| LSN4182928 Main Peak Purity | UHPLC-UV | Not less than 90.0% |
| **Impurities** | | |
| Total | UHPLC-UV | Not more than 10.0% area |
| L3 Dimer (LSN5973656) | UHPLC-UV | Not more than 1.5% area |
| L3 endo-Gly10 (LSN5673654) | UHLC-UV | Not more than 0.5% area |
| Any Unspecified Impurity | UHPLC-UV | Not more than 0.5% area |
| Description | Visual | White to yellow powder |

### 5.4. Enantiomeric purity

While HPLC and UPLC-MS are highly valuable for separating and identifying impurities, diastereomers generated via amino acid racemization during activation/couplings and



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epimerization of the residues embedded in the synthesized fragment are challenging to identify since there is no change in mass from the target peptide. Additionally, a specific method for analyzing D-isomers contained in the final fragment is used to assess the impact of stressed conditions. In some selected PAR studies, samples are hydrolyzed in the presence of deuterium to enable detection of the level of D-isomer present for each amino acid using derivatized LCMS analysis. Data from previous SPPS batches has demonstrated good process control for all D-isomer impurities. Therefore, the enantiomeric purity method is a characterization test that will be used in assessing the impact of process stresses and amino acid supply changes.

## 6. Initial Risk Assessment Process

### 6.1. Initial Risk Assessment for the SPPS Process

For the development of a synthetic sequence for a drug substance, a robust control strategy is needed. Development of the control strategy includes a thorough understanding of the numerous parameters and the risk that variations of these parameters will result in product that will not meet the CQAs of the final drug substance. For the Boc-Fragment 3 process, the fragment release specifications were set such that the final drug substance will meet all CQAs. The risk assessment was performed systematically to determine the level of detail required for studying process parameters and raw materials.

The initial risk assessment for SPPS of Boc-Fragment 3 (LSN4182928) consisted of assembling a SMDD-standard initial risk assessment spreadsheet that described all unit operations to be performed in the proposed registered process (Scheme 1). Standard parameters were assigned to each unit operation and a group of scientists with expertise on the fragment SPPS process and requisite unit operations were assembled for analysis. The initial risk assessment looked at all steps involved in the SPPS of eloralintide Boc-Fragment 3 (LSN4182928) (Figure 7). Both development as well as manufacturing representatives from SMDD and API EM were present or consulted to ensure all perspectives were considered when assessing risk. The team evaluated the risk of each parameter and its effect on CQAs for the drug substance given a univariate perturbation in that parameter. As a gauge of the magnitude of the proposed



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perturbation to be considered, generally a factor of six times the operational variability for any individual parameter was used.[1] The output from the initial risk assessment was the identification of process parameters with potential risk which required further study.[5]

In the following sections, 6.2 through 6.13 summarize the results of the initial risk assessment and only show parameters perceived to have potential risk through the initial risk assessment or parameters needing acceptable ranges proven experimentally. The team's findings were related to risks associated primarily with the purity of the isolated Boc-Fragment 3. The output of the initial risk assessment identified 609 parameters as having a potential risk. After further analysis and combining redundant parameters, the relevant controls for each potential impacting parameter were categorized as requiring in-process control, parametric control, or an alternative engineering control. Engineering controls eliminate the risk from parametric variability through manufacturing systems including procedural controls and/or mechanical set-up. The parameters for further parametric studies were identified. These parameters are discussed in detail in the subsections below.

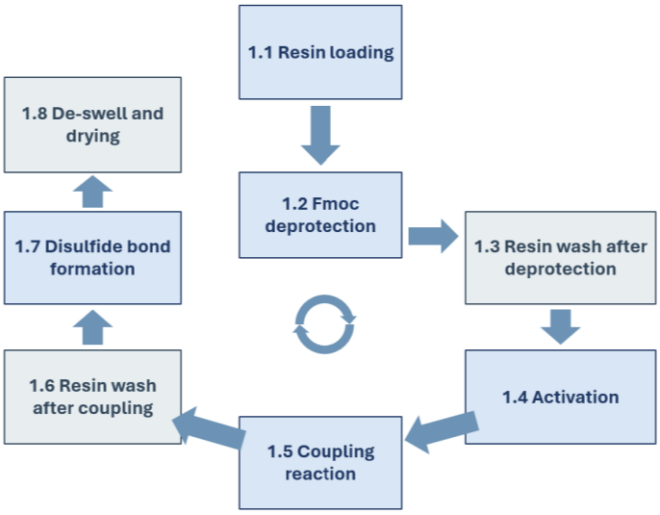



**Figure 7**. Sub-unit breakdown for fragment SPPS process.

## 6.2. Calculation of Target Ranges for Risk Assessment and Identification of PAR Data Gaps

[5] eLN Signals: Nil Tandogan Gray BLR-D01392-026



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Sigma (σ) has been defined as the routine operational variability for Lilly manufacturing equipment. Values for σ for key process parameters such as temperature, pressure, charge amounts, flow rates, and time have been determined previously (**Table 5**).

**Table 5**. Testing range for assessing parameter criticality.[6]

| Parameter Type | Range Assessed for Criticality |
|---|---|
| Temperature | ± 12 °C |
| Charge Quantity (by scale) | ± 6% |
| Charge Quantity (by mass flow meter) | ± 12% |
| Batch Quantity (by Radar Level Detector) | ± 18% |
| Pressure | ± 18% |
| Time (Manual Operation) | ± 1 hour |
| Time (Computer Controlled Operation) | ≈ seconds |

±6σ is conservative and inclusive from previous literature and the Lilly risk assessment process. The probability of a parameter falling outside of this range when operating under common cause variability for parameters that are described by normal, as well as significantly skewed distributions is very small (< 0.5%). Therefore, the minimum range tested for each parameter was at least ±6σ from the target set point for that process parameter, unless determined beforehand that the full 6σ range posed severe risk to product quality and parameters automatically designated as critical. For reaction temperature, the team chose to establish a conservative ±4.0 σ and assign temperature as critical process parameters due to known sensitivity of impurity formation due to reaction temperature. Those value ranges have been confirmed to be appropriate for API EM and Asymchem manufacturing operations. One parameter's 6σ value, 20% piperidine, was derived from the above table. Piperidine concentration depends upon two parameters, piperidine charge quantity and DMF charge quantity to the bulk tank. A variation of ± 6% was applied to each charge and a concentration range was calculated. For a 20% (v/v) piperidine/DMF target, the 6σ range is 17.6 – 22.4%.

[6] Sethuraman and Seibert, PILARS Object Name: TR060227EP93217; Jeffery, S. "Kinsale Equipment Capability Report", KIN-ENG-TECH-RPT-COMM-Kinsale Equipment Capability Report.



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### 6.3. Step 1.1: Resin charge, wash, and pre-swell

Of the five identified parameters of potential risk (**Table 6**), no parameters ultimately require parametric control within the resin charge, resin wash and pre-swell sub-unit as all parameters were deemed to be handled procedurally. 2-CTC resin charge is deemed no risk as it is basis for all charges and resin loading is controlled by CoA.

**Table 6**. Summary of the initial risk assessment for resin charge, wash, and pre-swell[5]

| Unit operation | Process Parameter | Parameter or Engineering Control | Control |
|---|---|---|---|
| Charge 2-CTC resin | Resin loading and charge Quantity | Engineering | Basis for all charges. Resin substitution value (mmol/g) controlled by CoA. |
| DMF Resin wash and pre-swell | Mixing time | Engineering | Procedural control confirming the resin is stirred sufficiently during wash and pre-swell |
| DMF Resin wash and pre-swell | Jacket Temperature | Engineering | Procedural control; ambient temperature for swelling shown to have no impact |
| DMF Resin wash and pre-swell | Water content | Engineering | CoA of Solvent |
| DMF Resin wash and pre-swell | Solvent Spec | Engineering | |

For each SPPS step, water content and DMF solvent specifications are initially treated as potential risks but are controlled through engineering controls via CoA. Accordingly, the remaining sections omit these two parameters from the risk assessment table.

### 6.4. Step 1.2: Resin Loading with Fmoc-Gly-OH and Capping

Of the 25 identified parameters of potential risk (**Table 7**), fifteen parameters ultimately require parametric control within the resin loading and capping sub-unit. Parameters were tested



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within bracketing studies across the proposed range for determining criticality, and the responses were analyzed for impact to the Boc-Fragment 3 specifications (**Table 4**).

**Table 7.** Summary of the initial risk assessment for resin loading and capping

| Unit operation | Process Parameter | Parameter or Engineering Control | Control |
|---|---|---|---|
| Activation Fmoc-Gly-OH | DMF quantity | Parameter | **Parametric,** total DMF activation volume |
| | Gly10 quantity | Parameter | **Parametric** |
| | Dissolution temperature | Parameter | **Parametric** |
| | DIPEA quantity | Parameter | **Parametric** |
| | Agitation speed | Engineering | Procedural control ensuring the solution is mixed well |
| | Activation time | Parameter | **Parametric** |
| Loading Fmoc-Gly-OH | Reaction temperature | Parameter | **Parametric** |
| | Reaction time | Parameter | **Parametric*** |
| | DMF quantity | Parameter | **Parametric,** total DMF loading volume |
| | Agitation speed | Engineering | Procedural control ensuring the reaction solution is mixed well |
| Wash post-loading | DMF quantity | Parameter, Engineering | **Parametric,** min wash volume tested Procedural control ensuring sufficient solvent is charged |
| | Stir time | Engineering | Procedural control ensuring sufficient time is allocated |
| Capping | DMF quantity | Parameter | **Parametric,** total DMF capping quantity |
| | Methanol quantity | Parameter | **Parametric** |
| | Methanol – water content | Engineering | Solvent Specification |



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| | | | |
|---|---|---|---|
| | Methanol – Solvent spec | Engineering | Solvent Specification |
| | DIPEA quantity | Parameter | **Parametric** |
| | DIPEA – water content | Engineering | Solvent Specification |
| | DIPEA – Solvent spec | Engineering | Solvent Specification |
| | Reaction time | Parameter | **Parametric** |
| | Agitation speed | Engineering | Procedural control ensuring the reaction solution is mixed well |
| | Reaction temperature | Parameter | **Parametric** |
| Wash post-capping | DMF quantity | Parameter, Engineering | **Parametric,** min wash volume tested Procedural control ensuring sufficient solvent is charged |
| | Stir time | Engineering | Procedural control ensuring sufficient time is allocated |
| | Number of wash cycles | Engineering | Procedural control ensuring sufficient number of washes complete |

*Reaction time is parametric but is not studied as incomplete reaction is controlled by capping the empty sites on resin post coupling reaction, resulting in lower yield but no impact to CQAs

The parameters in the above table were studied within bracketing or stability studies across the proposed range for determining criticality, and the responses were analyzed for impact to the Boc-Fragment 3 specifications. A summary for each parameter listing the target value and range studied is given in Table 8.

**Table 8.** Parametric studies for resin loading and capping

| Step | Unit Operation | Process Parameter | Units | Target | Range Studied |
|---|---|---|---|---|---|
| **1.2.** | Resin loading | Dissolution - DMF quantity | kg/kg | 6.63 | 5.81 – 7.43 |
| | | Gly10 quantity | eq. | 2.25 | 1.95 – 2.53 |
| | | DIPEA quantity | eq. | 6.00 | 5.18 – 6.76 |
| | | Dissolution time | min | 5 | 0 – 65 |
| | | Dissolution temperature | °C | 15.0 | 2.9 – 37.0 |
| | | Loading – reaction temperature | °C | 25.0 | 17.0 – 33.0 |



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| Step | Unit Operation | Process Parameter | Units | Target | Range Studied |
|---|---|---|---|---|---|
| | | Loading – reaction time | min | 180 | 120 - 480 |
| | | Loading – total DMF quantity | kg/kg | 7.58 | 6.64 – 8.49 |
| | Post – resin loading wash | DMF wash quantity | kg/kg | 7.58 | 6.63 – 7.62 |
| | Capping | Total DMF quantity | kg/kg | 8.53 | 7.48 – 9.63 |
| | | Methanol quantity | eq. | 10.00 | 8.75 – 11.41 |
| | | DIPEA quantity | eq. | 5.00 | 4.37 – 5.61 |
| | | Capping time | min | 60 | 0 - 120 |
| | | Capping temperature | °C | 25.0 | 13.0 – 37.0 |
| | Post – capping wash | DMF wash quantity | kg/kg | 7.58 | 6.62 - 8.50 |

### 6.5. Step 1.3a: Fmoc deprotection: Piperidine/DMF solution preparation

Of the five identified parameters of potential risk (Table 9), two parameters ultimately require parametric control within the piperidine/DMF solution preparation sub-unit. The two parametric control parameters were combined into a single parameter, piperidine concentration. Parameters were tested within bracketing studies across the proposed range for determining criticality, and the responses were analyzed for impact to the Boc-Fragment 3 specifications (Table 4).

**Table 9.** Summary of the initial risk assessment for Piperidine/DMF solution preparation

| Unit operation | Process Parameter | Parameter or Engineering Control | Control |
|---|---|---|---|
| Charge piperidine | Piperidine quantity | Parameter | **Parametric**, combined into single parameter – piperidine concentration |
| Charge DMF | DMF quantity | Parameter | |
| Deprotection solution | Agitation speed | Engineering | Procedural control confirming piperidine is sufficiently mixing |



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| Unit operation | Process Parameter | Parameter or Engineering Control | Control |
|---|---|---|---|
| Deprotection solution | Stir time | Engineering | Procedural control ensuring the solution is stirred until it is homogeneous. |
| Deprotection solution | Temperature | Engineering | Procedural control ensuring the setting is attained before solution transfer |

Water content and solvent specification of piperidine are controlled via engineering control through CoA. The combined parameters were studied experimentally in a multivariate study as part of a larger bracketing study (a more conservative estimation of the univariate response) to determine PARs. The initial study ranges for the piperidine/DMF solution make-up are summarized in **Table 10**.

**Table 10.** Parametric studies for Piperidine/DMF solution preparation

| Step | Unit Operation | Parameter | Unit | Target Value | Range studied |
|---|---|---|---|---|---|
| 1.3a | Fmoc deprotection | Piperidine concentration | % (v/v) | 20.00 | 14.99 – 24.99 |

### 6.6. Step 1.3b: Fmoc deprotection: Resin treatment with piperidine solution

Of the five identified parameters of potential risk (**Table 11**), three parameters require parametric control within the deprotection reaction sub-unit. Two of those were combined into one parameter, Fmoc deprotection reaction temperature. Piperidine exposure time refers to the duration during which both the peptide and resin are subjected to piperidine, beginning with the addition of the piperidine solution and concluding with the removal of the solution after the reaction is complete.



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**Table 11.** Summary of the initial risk assessment for the Fmoc deprotection reaction.

| **Unit Operation** | **Parameter** | **Parameter or Engineering Control** | **Control** |
|---|---|---|---|
| Fmoc deprotection reaction | Piperidine/DMF solution quantity | Parameter | **Parametric** |
| | Reaction Temperature | Parameter | **Parametric** |
| | Exposure time to piperidine | Parameter | **Parametric** |
| | Agitation | Engineering | Procedural control confirming agitation is sufficient |
| | Reaction cycle | Engineering | Procedural control confirming the number of Fmoc deprotection cycles are performed |

The temperature adjustment prior to the Fmoc deprotection reaction is not studied separately, since it is covered by the overall Fmoc deprotection reaction temperature. The DMF rinses performed during the reaction to wash the resin off the vessel walls are not included in the table, as they are accounted for by the total piperidine solution concentration and solution charge amounts.

The parameters in the above table were studied within bracketing or stability studies across the proposed range for determining criticality, and the responses were analyzed for impact to the Boc-Fragment 3 specifications. A summary for each parameter listing the target value and range studied is given in **Table 12**.



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**Table 12.** Parametric studies for Fmoc deprotection reaction.

| Step | Unit Operation | Process Parameter | AA cycle | Units | Target | Range Studied |
|---|---|---|---|---|---|---|
| 1.3b | Fmoc deprotection | Piperdine/DMF Chaging amount | All AAs | Kg/kg | 7.44 | 6.58 – 8.05 |
| | | Reaction Time | All AAs | min | 120 | 60 -180 |
| | | Reaction temperature | All AAs | °C | 25.0 | 13.0 - 37.0 |

### 6.7. Step 1.4: Resin wash and Re-swell after Fmoc deprotection

Of the eight identified parameters of potential risk (**Table 13**), two parameters require parametric control within the resin wash and re-swell after Fmoc deprotection sub-unit. These parameters were studied within bracketing and stability studies across the proposed range for determining criticality, and the responses were analyzed for impact to the Boc-fragment 3 specifications. A summary for each parameter listing the target value and range studied is given in **Table 14**.

**Table 13.** Summary of the initial risk assessment for Resin wash after Fmoc deprotection.

| Unit Operation | Parameter | Parameter or Engineering Control | Control |
|---|---|---|---|
| Post-Fmoc deprotection IPAC wash | IPAC quantity | Parameter | **Parametric** |
| | Wash cycles | Engineering | Procedural control confirming sufficient wash cycles apply to remove piperidine |



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| | | | |
|---|---|---|---|
| | IPAC wash – temperature | Parameter | Parametric, bundled study with reaction batch temperature [7] |
| Re-swell – DMF wash | DMF quantity | Parameter | **Parametric** |
| | Wash cycles | Engineering | Procedural control confirming sufficient wash cycles apply to remove piperidine |
| | DMF wash – temperature | Parameter | Parametric, bundled study with reaction batch temperature |
| Resin wash | Time | Engineering | Procedural control confirming wash is performed and stirred sufficiently[8] |
| Resin wash | Agitation | Engineering | Procedural control confirming agitation is sufficient |

**Table 14.** Parametric studies for Resin wash and re-swell after Fmoc deprotection.

| Step | Unit Operation | Process Parameter | AA Cycle | Units | Target | Range Studied |
|---|---|---|---|---|---|---|
| **1.4** | Resin wash and re-swell after Fmoc deprotection | IPAC wash quantity | All AA | kg/kg | 6.98 | 6.14 – 7.82 |
| | | DMF wash quantity | All AA | Kg/kg | 7.58 | 6.67 – 8.49 |

[7] Although the parametric ranges were not studied for IPAC washes, the jacket temperature was controlled at 25 °C for all other AA.

[8] Minimum resin wash time confirmed at scale and PAR studies to ensure cake stirrability– Non-saleable Eng run MBR (docu ID# MOR-TJ1414002101.03).



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### 6.8. Step 1.5: Activation

Of the twelve identified parameters of potential risk (Table 15), nine parameters require parametric control within the activation sub-unit. These parameters were studied within bracketing and stability studies across the proposed range for determining criticality, and the responses were analyzed for impact to the Boc-fragment 3 specifications. A summary for each parameter listing the target value and range studied is given in Table 16.

**Table 15** Summary of the initial risk assessment for Activation step.

| Unit Operation | Parameter | Parameter or Engineering Control | Control |
|---|---|---|---|
| Charge amino acid | Quantity | Parameter | **Parametric** |
| Charge Oxyma | Quantity | Parameter | **Parametric** |
| Charge PyBOP | Quantity | Parameter | **Parametric** |
| Charge DMF | DMF Quantity | Parameter | **Parametric** |
| Dissolve PyBOP and/or Oxyma, amino acid | Dissolution - Stir time | Engineering | Procedural control to confirm dissolution |
| Dissolve PyBOP and/or Oxyma, amino acid | Agitation | Engineering | Procedural control to confirm dissolution |
| Charge DIC | Quantity | Parameter | **Parametric** |
| Charge DIPEA | Quantity | Parameter | **Parametric** |
| Dissolution temperature | Temperature | parameter | **Parametric** |
| Amino acid activation | Temperature | Parameter | **Parametric** |
| Amino acid activation | Reaction time | Parameter | **Parametric** |
| Amino acid activation | Agitation speed | Engineering | Procedural control confirming agitation is sufficient |



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**Table 16.** Parametric studies for activation step.

| Step | Unit Operation | Process Parameter | AA cycles | Units | Target | Range Studied |
|---|---|---|---|---|---|---|
| **1.5.** | Activation | AA quantity | Standard AA (Ala8, Thr6, Ala5, Thr4, Asn3) | eq. | 2.00 | 1.85 – 2.13 |
| | | | Thr9 | eq. | 2.00 | 1.88 – 2.12 |
| | | | Cys7, Cys2 | eq. | 2.00 | 1.73 – 2.24 |
| | | | γGlu1 | eq. | 2.00 | 1.85 – 2.12 |
| | | Oxyma quantity | Standard AA | eq. | 2.00 | 1.73 – 2.25 |
| | | | Thr9 | eq. | 2.00 | 1.73 – 2.25 |
| | | | Cys7, Cys2 | eq. | 3.00 | 2.61 – 3.22 |
| | | | γGlu1 | eq. | 2.00 | 0.07 – 2.25 |
| | | Dissolution jacket temperature[9] | Standard AA | °C | 20.0 | 8.0 – 32.0 |
| | | | Thr9 | °C | 20.0 | 8.0 – 32.0 |
| | | | Cys7, Cys2 | °C | 20.0 | 12.0 – 32.0 |
| | | | γGlu1 | °C | 5.0 | -7.0 – 17.0 |
| | | PyBOP quantity | Thr9 | eq. | 2.00 | 1.73 – 2.25 |
| | | DIPEA quantity | Thr9 | eq. | 3.00 | 2.59 – 3.42 |
| | | | γGlu1 | | 3.00 | 2.62 -3.41 |
| | | DIC quantity | Standard AA | eq. | 2.20 | 1.90 – 2.47 |
| | | | Cys7, Cys2 | eq. | 3.30 | 2.85 – 3.72 |
| | | Activation - Total DMF quantity | Standard AA | kg/kg | 6.63 | 5.83 – 7.39 |
| | | | Thr9 | kg/kg | 4.74 | 4.14 – 5.27 |
| | | | Cys7, Cys2 | kg/kg | 6.63 | 5.81 – 7.40 |
| | | | gGlu1 | kg/kg | 6.63 | 5.80 – 7.48 |
| | | Activation time | Standard AA | min | 20 | 0 – 80 |
| | | | Thr9 | min | 20 | 0 – 80 |
| | | | Cys7, Cys2 | min | 5 | 0 – 65 |
| | | | γGlu1 | min | 20 | 0 – 80 |

[9] Dissolution jacket temperature is the temperature set before DMF, and amino acid charged into the activation temperature. It was studied together unidirectionally with the activation temp.



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| | | | | | | |
|---|---|---|---|---|---|---|
| | | Activation temperature | Standard AA | °C | 20.0 | 8.0 – 32.0 |
| | | | Thr9 | °C | 5.0 | -7.0 – 17.0 |
| | | | Cys7, Cys2 | °C | 5.0 | -7.0 – 17.0 |
| | | | γGlu1 | °C | 20.0 | 8.0 – 32.0 |

### 6.9. Step 1.6 Coupling reaction

Out of the six identified parameters of potential risk (**Table 17**), four parameters require parametric control within the coupling sub-unit. These parameters were studied within bracketing and stability studies across the proposed range for determining criticality, and the responses were analyzed for impact to the Boc-fragment 3 specifications. A summary for each parameter listing the target value and range studied is given in **Table 18**.

**Table 17.** Summary of the initial risk assessment for the Coupling reaction step.

| Unit Operation | Parameter | Parameter, In-Process or Engineering Control | Control |
|---|---|---|---|
| Transfer activated solution | Quantity | Engineering | Procedural control to confirm solution transferred |
| DMF – wall rinse | Quantity | Parameter | **Parametric**, combined with DMF charge in activation to report as total DMF charge - coupling |
| DMF charge to SPPS reactor | Quantity | Parameter, Engineering | **Parametric**, combined with DMF charge in activation to report as total DMF charge - coupling<br><br>**Procedural control** to ensure the DMF charged |



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| Unit Operation | Parameter | Parameter, In-Process or Engineering Control | Control |
|---|---|---|---|
| | | | prior to the transfer activated solution |
| Coupling reaction | Temperature | Parameter | **Parametric** |
| Coupling reaction | Reaction time | Parameter, IPC | **Parametric control:** Time **In Process Control:** Colorimetric test for end of reaction. |
| Coupling reaction | Agitation | Engineering | Procedural control confirming agitation is sufficient |

**Table 18.** Parametric studies for the Coupling reaction step.

| Step | Unit Operation | Process Parameter | | Units | Target | Target Range Studied |
|---|---|---|---|---|---|---|
| 1.6 | Amino acid coupling | Coupling Temperature | Standard AA | °C | 25.0 | 17.0 – 33.0 |
| | | | Thr9 | °C | 25.0 | 17.0 – 33.0 |
| | | | Cys7, Cys2 | °C | 20.0 | 12.0 – 28.0 |
| | | | γGlu1 | °C | 25.0 | 17.0 – 33.0 |
| | | Coupling – total DMF quantity | Standard AA | kg/kg | 7.58 | 6.68 – 8.45 |
| | | | Thr9 | kg/kg | 8.53 | 7.45 – 9.49 |
| | | | Cys7, Cys2 | kg/kg | 8.53 | 7.71 – 9.28 |
| | | | γGlu1 | kg/kg | 7.58 | 6.63 – 8.59 |
| | | Oxyma Charge | γGlu1 | eq | 2.00 | 1.75 – 2.25 |
| | | PyBOP Charge | | eq | 2.00 | 1.74 – 2.25 |
| | | DIPEA Charge | | eq | 3.00 | 2.62 – 3.41 |
| | | Reaction time[10] | Standard AA | min | 240 | 180 – 600 |

[10] The coupling reactions do not all have the same target time. At the end of the specified target time, an IPC sample for reaction completion is taken and analyzed, while the reaction continues. Upon passing result of two consecutive samples, the reaction is stopped. If IPC fails again reaction mixture is stirred for additional 1 h and then retested. See



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| Step | Unit Operation | Process Parameter | | Units | Target | Target Range Studied |
|---|---|---|---|---|---|---|
| | | | Thr9 | min | 240 | 180 – 600 |
| | | | Cys7, Cys2 | min | 240 | 180 – 600 |
| | | | γGlu1 – coupling total time | min | 240 | 180 – 600 |

### 6.10. Step 1.7 Resin wash after coupling

Out of the six identified parameters of potential risk (**Table 19**), one parameter requires parametric control within the resin wash after coupling sub-unit operation. This parameter was studied within bracketing studies across the proposed range for determining criticality, and the responses were analyzed for impacts to the Boc-fragment 3 specifications. A summary for each parameter listing the target value and range studied is given in **Table 20**.

**Table 19.** Summary of initial risk assessment for the Resin wash after coupling

| Unit Operation | Parameter | Parameter or Engineering Control | Control |
|---|---|---|---|
| Charge DMF wash to activation vessel | Quantity | Engineering | Procedural control ensuring sufficient volume and surface covered for washing |
| Transfer DMF wash from activation vessel to synthesizer | Quantity | Parameter, Engineering | Parametric: Minimum resin wash volume tested<br>Procedural control to confirm solution transferred |

Appendix III: Coupling and Deprotection Reaction Time Targets and PARs for a full listing of ranges studied for AA coupling reaction time.



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| | | | |
|---|---|---|---|
| Number of wash cycle[11] | Wash cycle | Engineering | Procedural control confirming sufficient wash cycles apply to remove piperidine |
| Resin wash | Time | Engineering | Procedural control confirming wash is stirred sufficiently |
| Resin wash | Agitation | Engineering | Procedural control confirming agitation sufficient |
| Resin wash | Temperature | Parameter | Parametric, bundled study with reaction batch temperature |

**Table 20.** Parametric studies for the resin wash after coupling.

| Step | Unit Operation | Process Parameter | Units | Target | Range Studied |
|---|---|---|---|---|---|
| 1.7 | Resin wash after coupling | Post loading/coupling DMF wash quantity (Gly10) | kg/kg | NLT 7.58 | NLT 6.63 |
| | | DMF post-coupling wash (Thr9, Ala8, Cys7, Thr6) | kg/kg | NLT 7.58 | NLT 6.68 |
| | | DMF post-coupling wash (Ala5, Thr4, Asn3, Cys2, gGlu1) | kg/kg | NLT 9.48 | NLT 8.32 |

### 6.11. Step 1.8 – 1.9 Oxidative Trt removal and disulfide formation

The intramolecular Cys2–Cys7 disulfide bond is formed on resin by iodine-mediated oxidative removal of the Cys(Trt) protecting groups, followed by pre-cooled (−5 °C) and ambient-temperature (20 °C) of DMF washes to remove residual iodine and oxidation by-products. Step 1.8 repeats with iodine solution preparation, oxidation and post-oxidation wash sequence with

[11] 3 DMF post-coupling wash cycles were performed for all PAR studies and at manufacturing scale, except Gly10 which includes one post-coupling DMF wash prior to capping. However, no upper limit is required because additional DMF only improves removal of residual activated amino acid, coupling reagent and urea by-products, and the resin is filtered between charges, so an excess charge carries no quality risk.



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identical charges and set points; the parameters and controls identified in the initial risk assessment are therefore common to both steps, and completion of oxidation is confirmed at the end of each cycle by the in-process control for residual L3(Trt)$_2$. Of the fourteen identified parameters of potential risk (Table 21), nine parameters require parametric control within the iodine solution preparation, oxidation and post-oxidation wash sub-unit operations; the remaining parameters are addressed through procedural or in-process controls. These parameters were studied within bracketing and deconvolution studies (PAR 11 – PAR 14) across the proposed range for determining criticality, and the responses were analyzed for impact to the Boc-Fragment 3 specifications. The iodine transfer time, oxidation exposure time and oxidation reaction time, which were not identified as parameters of potential risk in the initial assessment, were also varied in these studies to characterize the iodine contact profile. A summary for each parameter listing the target value and range studied is given in Table 22.

**Table 21.** Summary of the initial risk assessment for the oxidative Trt removal and disulfide formation

| **Unit Operation** | **Parameter** | **Parameter, In-Process or Engineering Control** | **Control** |
|---|---|---|---|
| Iodine solution preparation: DMF charge | Quantity | Parameter | **Parametric** |
| Iodine solution preparation: iodine charge | Quantity | Parameter | **Parametric** |
| Iodine solution preparation (cool and stir) | Temperature | Parameter | **Parametric** |
| Oxidation: DMF charge to SPPS reactor | Quantity | Parameter | **Parametric** |
| Oxidation (cool, $N_2$ sparge and iodine solution transfer) | Temperature | Parameter | **Parametric** (oxidation temperature) |



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| Unit Operation | Parameter | Parameter, In-Process or Engineering Control | Control |
|---|---|---|---|
| Iodine solution transfer | Quantity | Engineering | Procedural control to confirm complete transfer of the iodine solution |
| Iodine solution transfer | Transfer rate | Engineering | Procedural control confirming addition over 50 ± 10 min to avoid local iodine excess; studied parametrically as iodine transfer time |
| Post-oxidation cold DMF wash | Quantity | Parameter | **Parametric**, combined with warm DMF wash to report as DMF post-oxidation wash |
| Post-oxidation cold DMF wash | Temperature | Parameter | **Parametric** (post-oxidation wash temperature) |
| Post-oxidation cold DMF wash | Wash cycles | Engineering | Procedural control confirming 2 wash cycles at −5 ± 5 °C |
| Post-oxidation warm DMF wash | Quantity | Parameter | **Parametric**, combined with cold DMF wash to report as DMF post-oxidation wash |
| Post-oxidation warm DMF wash | Temperature | Parameter | **Parametric**, bundled study with reaction batch temperature |
| Post-oxidation warm DMF wash | Wash cycles | Engineering | Procedural control confirming 4 wash cycles at 20 ± 5 °C |
| Resin sampling after post-oxidation wash | Residual $L3(Trt)_2$ | In-Process Control | IPC: $L3(Trt)_2 \leq 0.5\%$; if > 0.5%, oxidation module is repeated |

**Table 22.** Parametric studies for the oxidative Trt removal and disulfide formation



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| Step | Unit Operation | Process Parameter | Units | Target | Range Studied |
|---|---|---|---|---|---|
| 1.11 / 1.12 | Oxidative Trt removal | DMF – iodine solution preparation quantity | kg/kg | 3.79 | 3.30 – 4.27 |
| | | Iodine quantity | eq | 5.00 | 4.36 – 5.62 |
| | | DMF – oxidation reaction quantity | kg/kg | 11.37 | 9.93 – 12.79 |
| | | Oxidation temperature | °C | −5 | −15 – 5 |
| | | Iodine transfer time | min | 50 | 20 – 100 |
| | | Oxidation exposure time | min | 50 | 20 – 130 |
| | | Oxidation reaction time | min | 0 | 0 – 30 |
| | Post-oxidation wash | Post-oxidation wash temperature | °C | −5 | −7 – 5 |
| | | DMF post-oxidation wash quantity (6×) | kg/kg | 9.48 | 8.19 – 10.65 |

### 6.12. Step 1.9 De-swell and drying

Of the seventeen identified parameters of potential risk with MTBE and toluene washes combined into five parameters (**Table 23**), three parameters require parametric control within the de-swell and drying unit operation. These parameters were studied within bracketing and stability studies across the proposed range for determining criticality, and the responses were analyzed for impact to the Boc-Fragments 3 specifications. A summary for each parameter listing the target value and range studied is given in **Table 24**.

**Table 23.** Summary of the initial risk assessment for the De-swell and drying step.

| Unit Operation | Parameter | Parameter, In-Process or Engineering Control | Control |
|---|---|---|---|
| De-Swelling with IPAC wash | Quantity | Parameter | **Parametric** |
| | Temperature | Parameter | Parametric, bundled study with reaction batch temperature |



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| Unit Operation | Parameter | Parameter, In-Process or Engineering Control | Control |
|---|---|---|---|
| | Wash cycles | Engineering | Procedural control confirming NLT 5 wash cycles |
| | Wash time | Engineering | Procedural control confirming wash is stirred sufficiently |
| | Agitation | Engineering | Procedural control confirming agitation sufficient |
| De-Swelling with alternating IPAC and MTBE wash | Quantity | Parameter | **Parametric** |
| | Temperature | Parameter | Parametric, bundled study with reaction batch temperature |
| | Wash cycles | Engineering, IPC | Procedural control confirming alternating IPAC/ MTBE washes.<br>IPC: to ensure residual piperidine meets criteria |
| | Wash time | Engineering | Procedural control confirming wash is stirred sufficiently |
| | Agitation | Engineering | Procedural control confirming agitation sufficient |
| Intermediate drying | Time | Engineering, IPC | Procedural control to ensure end of drying reached<br>IPC: end of drying when LOD ≤ 10.0% |
| Intermediate drying | Jacket temperature | Parameter | **Parametric** |

**Table 24.** Parametric studies for the resin de-swell and drying step

| Step | Unit operation | Parameters | Units | Target | Range studied |
|---|---|---|---|---|---|
| 1.9 | De-swell and drying step | IPAc wash quantity | kg/kg | 8.73 | 7.67 – 8.80 |
| | | MTBE wash quantity | kg/kg | 7.41 | 6.50 – 7.57 |



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| Step | Unit operation | Parameters | Units | Target | Range studied |
|---|---|---|---|---|---|
| | | Drying jacket temperature | °C | NMT 25 | NMT 42 |

### 6.13. Methods for PAR Determination and Parameter Criticality Evaluation Depending on Risk

Generally, four methods are used for determining the risk level associated with each parameter and mapping the trajectory responses for CQAs:

1) *Univariate perturbation of a single parameter.* This method perturbs individual parameters to establish an acceptable processing range, and uses that range to assess the associated risk to impacting a CQA. This is the most time-intensive method but may be required when the parameter has significant impact to a CQA.
2) *Bracketing studies.* A group of parameters will be combined in one experiment to test their impact to a CQA. This method allows for minimal experimental investment and results in PARs for multiple variables at the same time, but can overestimate risk for convoluted parameters. Deconvolution studies may be required when too many parameters are pooled in a single bracketed study.
3) *Prior knowledge* from existing data generated as part of a development and optimization effort may be leveraged to justify parameter ranges when the process has not materially changed from the experiments used to generate the aforementioned data.
4) *First principles model build.* First principles or mechanistic models can be used to predict PARs and multivariate interactions for all parameters where appropriate. This is the most desirable approach given the flexibility and broad applicability of the information generated from the model output, but may required significant studies to accurately predict the outcome of a chemical transformation including the totality of the impact on all critical quality attributes. In this report, first principles model build was not used.



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Notably, care must be taken when performing bracketing studies because individual factor effects in these studies can potentially be confounded with each other. Thus, all parameters evaluated in the same bracketing study will be classified with the same risk level, which may result in parameters that are actually low risk being categorized as having “medium risk” or “high risk.” Alternatively, if any bracketing study were to result in high risk or medium risk bracketed parameters, additional deconvolution studies can be performed to determine the criticality of the individual parameters contributing to the medium- or high-risk parameter(s), or modeling can be used to determine a tighter parametric control range. Conversely, if the bracketing study were to demonstrate that the grouped parameters collectively had a low risk of impacting the material quality, then the tested range of each parameter could be used to establish the univariate PAR for each of the grouped parameters.

Experiments were designed to assess the criticality of all process parameters with respect to each identified CQA in the initial risk assessment and are described in subsequent sections. These experiments were also used for the PAR determination. If, through a data-mining exercise, it was already determined that an experiment or manufacturing lot existed which was perturbed to at least ±6σ and resulted in product which met the proposed registration acceptance criteria, then the result was documented and used to support an acceptable PAR limit. Alternatively, if the ±6σ values for a specific process parameter had not been demonstrated, then a PAR experiment was performed to close the data gap.

Due to the rapid evolution of the manufacturing process, few previous experiments were available that represented the current process. Therefore, the team decided to investigate all potential risk parameters at this stage for completeness. In addition, several process targets had not undergone rigorous optimization due to the speed of development. For certain parameters, narrower ranges were studied — in some cases deliberately, where prior process understanding indicated a greater impact on the CQAs, and in others as a serendipitous consequence of experimental execution of the PAR study.

## 7. Experimental Setup and Procedure



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### 7.1. Lab Model[31]

The lab model is run at laboratory scale using a 2 L jacketed synthetic flask for the resin-bound operations and a separate 1 L jacketed activation flask for amino acid activation, with all operations conducted under nitrogen; solids are charged through a solid feeding funnel and liquids through a liquid addition funnel. A bulk 20% v/v piperidine/DMF deprotection solution is prepared in advance in a 10 L flask by charging piperidine to DMF and stirring for 30 min at 20 °C, then transferred under nitrogen displacement into a PTFE bottle and stored at 20 °C. Before the resin is charged, the equipment train is rinsed twice by charging DMF (1 V) to the activation flask, transferring it to the synthetic flask, bubbling with nitrogen for NLT 8 min and draining. Volumes below are quoted relative to the resin charge, where 1 V corresponds to 0.948 g/g DMF.

**Resin loading**

2-Chlorotrityl chloride resin (1.0 g, basis) is charged to the 2 L synthetic flask and the jacket set to 25 °C. The resin is swelled with DMF (7 V) with nitrogen bubbling (for mixing) for NLT 5 min, the flask wall rinsed with DMF (1 V), the mixture adjusted to 25 °C and stirred for NLT 30 min at 25 °C, then drained; the resin is then washed with DMF (5 V) under nitrogen for NLT 10 min and drained. The first residue, Gly10, is loaded by charging DMF (5 V) to the activation flask at a 25 °C jacket setting, adding Fmoc-L-Gly-OH (2.25 eq.) and stirring until dissolution is visually complete, rinsing the feeding funnel and flask wall with DMF (1 V), then adding DIPEA (6.00 eq.) while holding the internal temperature at NMT 15 °C (reference addition time at production scale 15–30 min) and rinsing again with DMF (1 V), giving a total activation volume of 7 V. The solution is adjusted to 15 °C and stirred for 5 min. With the synthetic flask jacket at 25 °C the activated solution is transferred, the activation flask wall rinsed with DMF (1 V) and the rinse transferred, giving a total coupling volume of 8 V; the mixture is bubbled with nitrogen for NLT 5 min, adjusted to 25 °C and stirred for 180 min before draining. The resin is washed by rinsing the activation flask with DMF (8 V) and using the wash liquor to wash the resin and synthetic flask walls, stirring for 10 min under nitrogen. Unreacted resin sites are then capped: DMF (8 V) is charged, MeOH (10.00 eq.) added and the mixture cooled to 20 °C, DIPEA (5.00 eq.) added at NMT 20 °C (reference addition time at production scale 30–45 min), the mixture adjusted to 25 °C and the flask wall



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rinsed with DMF (1 V), giving a total capping volume of 9 V; after nitrogen bubbling the mixture is warmed to 25 °C and stirred for 60 min. The capped resin is washed with DMF (8 V × 3, NLT 7 min each) delivered through the activation flask.

**Fmoc Deprotection**

Fmoc deprotection is carried out by charging 20% v/v piperidine/DMF solution (8 V) to the synthetic flask. The mixture is bubbled with nitrogen for 5 min and the agitator started, with the piperidine addition time recorded as the deprotection start time. The mixture is adjusted to 25 °C, the time at which the internal temperature reaches 25 °C is recorded, and the mixture is stirred for 120 min at 25 °C before draining. A single piperidine treatment is applied at every position, from Thr9 through gGlu1 prior to coupling reaction.

**Resin wash after deprotection**

After each deprotection the resin is washed with isopropyl acetate, each charge being used to wash the flask walls: 8 V × 4 from pre-Thr9 through pre-Ala5, and 10 V × 4 from pre-Thr4 through pre-gGlu1. Each wash is bubbled with nitrogen for 5 min and stirred for NLT 10 min before draining. The resin is then re-swelled with DMF, 8 V × 2 from pre-Thr9 through pre-Ala5 and 10 V × 2 from pre-Thr4 through pre-gGlu1, each charge bubbled with nitrogen for 5 min and stirred for NLT 15 min. The wash jacket temperature is set to 25 °C for the pre-Thr9 wash and 20 °C for all subsequent cycles.

**Activation**

Standard amino acid activation is performed in the activation flask with the jacket set to 20 °C: DMF (5 V) is charged, the Fmoc-amino acid (2.00 eq.) and OxymaPure (2.00 eq.) added through the solid feeding funnel, the funnel and flask wall rinsed with DMF (1 V), and the mixture stirred until dissolution is visually complete. DIC (2.20 eq.) is then added while holding the internal temperature at NMT 20 °C, the funnel and flask wall rinsed with DMF (1 V) — giving a total activation volume of 7 V — and the mixture adjusted to 20 °C and stirred for 20 min. These conditions are applied to Ala8 and Ala5 [Fmoc-L-Ala-OH·H2O], Thr6 and Thr4 [Fmoc-L-Thr(tBu)-OH] and Asn3 [Fmoc-L-Asn(Trt)-OH]. Three positions use a modified activation:



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Thr9 [Fmoc-L-Thr(tBu)-OH, 2.00 eq.] is dissolved in DMF (3 V) at a 20 °C jacket setting, the mixture cooled to an internal temperature of 5 °C, and PyBOP (2.00 eq.) and OxymaPure (2.00 eq.) charged at 5 ± 5 °C followed by a DMF rinse (1 V). DIPEA (3.00 eq.) is added at NMT 5 °C with a further DMF rinse (1 V), giving a total activation volume of 5 V, and the mixture is adjusted to 5 °C and stirred for 20 min.

Cys7 and Cys2 [Fmoc-L-Cys(Trt)-OH, 2.00 eq.] are not pre-activated. The amino acid and OxymaPure (3.00 eq.) are dissolved in DMF (5 V) under nitrogen, the feeding funnel and flask wall rinsed with DMF (1 V × 2), and the solution adjusted to 5 °C and stirred for 5 min, giving a total dissolution volume of 7 V. In parallel, DIC (3.30 eq.) is combined with DMF (1 V) in the activation flask and stirred for NLT 2 min; this solution is charged to the synthetic flask after the amino acid solution, so that activation takes place in the presence of the resin.

gGlu1 [Boc-L-Glu-OtBu, 2.00 eq.] is dissolved in DMF (5 V) and the activation jacket set to 5 °C, then PyBOP (2.00 eq.) and OxymaPure (2.00 eq.) charged at 5 °C followed by a DMF rinse (1 V). DIPEA (3.00 eq.) is added at NMT 5 °C with a further DMF rinse (1 V), giving a total activation volume of 7 V, and the mixture is adjusted to 20 °C and stirred for 20 min.

**Coupling reaction**

For coupling, the activated solution is transferred from the activation flask to the synthetic flask and the activation flask wall rinsed with DMF (1 V), the rinse being transferred with it. For Thr9, DMF (3 V) is charged to the synthetic flask and adjusted to 25 °C before the transfer, giving a total coupling volume of 9 V; for all other residues the total coupling volume is 8 V. Cys7 and Cys2 differ in that the DIC/DMF solution is charged to the synthetic flask after the amino acid solution while holding the internal temperature at NMT 20 °C, with the mixture then adjusted to 20 °C.

After the transfer the mixture is bubbled with nitrogen for NLT 5 min, the stirrer started and the temperature adjusted to the coupling set point — 25 °C for all residues except Cys7 and Cys2, which couple at 20 °C. Timing starts when the internal temperature reaches 20 °C. The mixture is sampled every 60 min for Kaiser (ninhydrin) analysis until two consecutive samples are colorless or nearly colorless; if two samples pass before 4 h is reached, stirring is continued until



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4 h, after which the reaction solution is drained. The Gly10 loading reaction is the exception, being run for a fixed 180 min without Kaiser monitoring.

**Resin wash after coupling**

After each coupling the activation flask is rinsed with DMF and the wash liquor used to wash the resin and synthetic flask walls: 8 V × 3 from Thr9 through Thr6 and 10 V × 3 from Ala5 through γGlu1. Each wash is bubbled with nitrogen for 5 min and stirred for 10 min before draining.

**Oxidative Trt removal and disulfide formation**

Following the γGlu1 coupling and post-coupling wash, the Cys(Trt) protecting groups are removed oxidatively and the Cys2–Cys7 disulfide formed in a single operation. An iodine solution is prepared by charging DMF (4 V) to the activation flask under nitrogen and adding iodine in 10 g portions at NMT one portion per minute, then cooling to −5 °C. DMF (12 V) is charged to the synthetic flask, bubbled with nitrogen for NLT 5 min and cooled to an internal temperature of −5 °C. The iodine solution is transferred over 50 min with the internal temperature maintained at NMT −5 °C, and the reaction solution is drained immediately on completion of the addition. The resin is then washed with pre-cooled DMF (10 V × 2, stirred 8–12 min, target 10 min) followed by ambient temperature DMF (10 V × 4, stirred 10–15 min, target 12.5 min).

**De-swell and drying**

On completion of the sequence the jacket is set to 20–25 °C and the resin de-swelled with isopropyl acetate (10 V × 3), each charge used to wash the flask walls, bubbled with nitrogen for 5 min and stirred for NLT 10 min before draining. The same cycle is then repeated with MTBE (10 V × 5). The resin is dried with a nitrogen sweep above and vacuum below at a jacket temperature of NMT 25 °C; after 12 h the solid is sampled for loss on drying every 6 ± 4 h until the LOD is ≤ 10.0% or two consecutive results differ by ≤ 1.0%, with the solid stirred for 2–5 min at 5 ± 5 rpm every 4 ± 1 h from 6 h into the drying. The dried resin is collected into tared brown glass jars and stored at 2–8 °C.

**Cleavage & Thioacetal formation procedure for analysis of SPPS PAR experiments**



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Peptidyl resin from each PAR experiment was carried through small lab-scale use-tests of the two downstream steps — Step 2 soft cleavage and Step 3 dithioacetal cyclization — so that the quality of each SPPS run could be assessed as isolated LSN4182928 rather than as peptidyl resin alone. Both use-tests were run to the baseline procedure for every PAR experiment, so that any difference in the final product profile is attributable to the SPPS parameters under study.

In Step 2, the peptide is cleaved from the 2-chlorotrityl resin under mild acidic conditions (dilute TFA in DCM) at low temperature, and the cleavage filtrate is immediately quenched into a pyridine-containing aqueous solution to limit acid exposure. After an aqueous work-up, including an antioxidant (ascorbic acid) wash, the product is solvent-exchanged into ethyl acetate, crystallized by addition of n-heptane at low temperature, filtered and dried to give the Step 2 intermediate Boc-Fragment 3-disulfide.

In Step 3, the Step 2 intermediate is first reduced with TCEP in aqueous acetonitrile to open the Cys2–Cys7 disulfide, then treated with diiodomethane in the presence of base to close the dithioacetal bridge. The reaction is followed to defined in-process limits for residual disulfide and linear peptide before work-up. The product is solvent-exchanged into 2-MeTHF, acidified, washed with brine and water, then seeded and crystallized by addition of n-heptane. The isolated cake is washed to remove residual diiodomethane and dried to give LSN4182928, which was analyzed to generate the Step 3 purity data reported for each PAR experiment.

### 7.2. Lab model experimental setup

The lab model used for the on-resin fragment isolation step is sufficiently representative of production scale processing. SPPS is performed at two resin scales: 45.0 g of 2-chlorotrityl chloride resin (substitution 1.54 mmol/g) for the baseline and PAR experiments, and a 200 g baseline batch (Run 2) run to generate material for the L3 cleavage PAR studies.

For Fmoc deprotection, the 20% piperidine/DMF solution is prepared in bulk in a three-neck flask and stored in a PTFE bottle; it is measured with a glass graduated cylinder, charged to



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the synthesizer through a funnel, and drained under nitrogen by vacuum through the bottom pipeline into a 5 L collection flask, with the post-deprotection washes following the same measure, charge, and drain sequence. In the activation step, the amino acid and Oxyma are weighed on a balance and charged to the 1 L four-neck jacketed activation vessel through a funnel, while DIC/DIPEA is delivered by graduated dropper; the contents are mixed with the overhead agitator, and the vessel walls are rinsed down with DMF from a squirt bottle and stirred for 1 to 2 minutes, with the rinse repeated three times. The activated solution is transferred to the synthesizer through a funnel for the coupling reaction, mixed with the overhead agitator in the jacketed vessel, sampled by drawing an aliquot into a test tube with a graduated dropper, and drained under vacuum through the bottom pipeline; the subsequent coupling washes again follow the same measure, charge, and drain sequence.

For the 45 g scale PAR experiments the activation reaction was performed in a 1 L jacketed activation flask and the coupling reaction in a 2 L jacketed synthetic flask, giving a working reaction volume of approximately 1440 mL. Agitation was provided by a PTFE anchor stirrer.

The SPPS equipment set-up is shown in Figure 8. The stirring speed for activation reactor is 120 rpm to homogenize the solution. Other unit operations were stirred at 90 rpm to ensure resin is fully suspended for coupling reaction.



**Figure 8.** Set up for PAR Experiments.



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Drying: the well-drained resin is dried in the synthetic flask with a nitrogen sweep above and vacuum below at a jacket temperature of NMT 25 °C, and is stirred for 2–5 min at 5 ± 5 rpm every 4 ± 1 h from 6 h into the drying.

### 7.3. Raw Materials

The raw material lots used for the PAR experiments are listed in Table 25. Each raw material was typical of the quality to be used in manufacturing.

**Table 25.** Intermediate and Raw Materials.

| No. | Raw material | Lot no. | Supplier | |
|---|---|---|---|---|
| | | | Supplier | Additional lot and supplier |
| 1. | 2-CTC resin | L125050494 | Sunresin New Materials Co., Ltd | L225111931<br>Tianjin Nankai Hecheng Science & Technology Co., Ltd |
| 2. | Fmoc-Gly-OH | L225120048 | Chengdu ZY Biochemical Technology Co., Ltd | L225111702 – Liaoning Asymchem Laboratories Co., Ltd. |
| 3. | Fmoc-L-Thr(tBu)-OH | L225120052 | Chengdu ZY Biochemical Technology Co., Ltd | L225111436 – Liaoning Asymchem Laboratories Co., Ltd. |
| 4. | Fmoc-L-Ala-OH·H2O | L225111437 | Liaoning Asymchem Laboratories Co., Ltd. | L225120047 – Chengdu ZY Biochemical Technology Co., Ltd |
| 5. | Fmoc-L-Cys(Trt)-OH | L225120051 | Chengdu ZY Biochemical Technology Co., Ltd | L125050476 – Liaoning Asymchem Laboratories Co., Ltd |



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| No. | Raw material | Lot no. | Supplier | |
|---|---|---|---|---|
| | | | Supplier | Additional lot and supplier |
| 6. | Fmoc-L-Asn(Trt)-OH | L225120049 | Chengdu ZY Biochemical Technology Co., Ltd | – |
| 7. | Boc-L-Glu-OtBu | L225120050 | Chengdu ZY Biochemical Technology Co., Ltd | – |
| 8. | PyBOP | L225111986 | Suzhou Highfine Biotech Co., Ltd | – |
| 9. | DIPEA | L125110445 / FJL123080119-002 | Changzhou Tianhua Pharmaceutical Co., Ltd. | – |
| 10. | Oxyma | L225120161 | Changzhou Tianhua Pharmaceutical Co., Ltd. | – |
| 11. | DIC | L225120001 | Zhejiang Bulk Chemical Co., Ltd | – |
| 12. | I2 | L125060545 | Loops (Tianjin) Chemical Technology Co., Ltd | – |
| 13. | DMF | L125120281-m-TF01 | Tianjin Tairong Chemical Trading Co., Ltd | L225122247 – Tianjin Yubo Fine Chemical Co., Ltd |
| 14. | IPAC | L125090783-m-TF03 | Tianjin Yubo Fine Chemical Co., Ltd | L225121662 – Tianjin Yubo Fine Chemical Co., Ltd |
| 15. | MTBE | L125120033 | Tianjin Tairong Chemical Trading Co., Ltd | – |
| 16. | MeOH | L125110195 | Tianjin Yubo Fine Chemical Co., Ltd. | – |



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## 8. Baseline and PAR Experimental Conditions

The Boc-Fragment 3 SPPS conditions were investigated using a series of parametric range studies to understand the impact of parametric perturbations on product quality. The following sections list the SPPS parametric studies, consisting of sixteen experiments to test one hundred and two sets of bracketed parameters, colloquially referred to as PAR experiments, designed to determine parameter criticality against relevant quality attributes and establish proven acceptable ranges for the process, listed in **Table 26** below.



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**Table 26**. Overview of Executed PAR experimental conditions for SPPS process.[12,13]

| Parmeter | Target | Units | Sigma | -6 sigma | +6 sigma | B1 | B2 | B3 | P3 | P4 | P5 | P6 | P8 | P9 | P10 | P11 | P12 | P13 | P14 | P15 | P16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CTC Resin quantity[14] | 45.00 | g | N/A (basis) | N/A (basis) | N/A (basis) | 45.02 | 200.0 | 44.92 | 45.03 | 45.55 | 45.00 | 45.01 | 45.00 | 44.87 | 45.00 | 45.21 | 45.09 | 45.06 | 45.08 | 45.03 | 45.05 |
| % Piperidine in DMF (v/v) | 20.00 | % | 2.83% | 16.61 | 23.39 | 19.99 | 19.99 | 20.00 | 24.99 | 20.01 | 14.99 | 20.00 | 19.99 | 19.99 | 20.00 | 20.02 | 19.99 | 20.01 | 20.01 | 20.00 | 20.00 |
| Dissolution jacket temperature (Gly10) | 25.0 | °C | 2.0 | 13.0 | 37.0 | 25.0 | 25.1 | 24.9 | 25.0 | 25.0 | 25.0 | 37.0 | 25.0 | 12.9 | 25.0 | 25.0 | 25.0 | 25.0 | 25.0 | 37.0 | 24.9 |
| DMF dissolution total quantity (Gly10) | 6.63 | kg/kg | 2.00% | 5.84 | 7.43 | 6.65 | 6.64 | 6.66 | 5.81 | 7.43 | 6.64 | 6.64 | 6.68 | 6.67 | 6.63 | 6.64 | 6.65 | 6.67 | 6.64 | 6.63 | 6.63 |
| AA quantity (Gly10) | 2.25 | eq | 2.00% | 1.98 | 2.52 | 2.25 | 2.25 | 2.27 | 2.53 | 1.95 | 2.25 | 1.97 | 2.25 | 2.27 | 2.28 | 2.24 | 2.25 | 2.26 | 2.25 | 2.25 | 1.97 |
| DIPEA quantity (Gly10) | 6.00 | eq | 2.00% | 5.28 | 6.72 | 6.00 | 5.99 | 6.00 | 6.76 | 5.18 | 6.00 | 5.24 | 6.00 | 6.03 | 6.01 | 5.96 | 5.98 | 6.08 | 6.07 | 6.00 | 5.33 |
| Activation temperature (Gly10) | 15.0 | °C | 2.0 | 3.0 | 27.0 | 15.0 | 15.1 | 13.6 | 15.0 | 15.0 | 15.0 | 27.0 | 15.0 | 2.9 | 15.0 | 15.0 | 15.0 | 15.0 | 15.0 | 27.0 | 15.0 |
| Activation time (Gly10) | 5 | min | 10 | 0 | 65 | 5 | 6 | 5 | 65 | 0 | 5 | 65 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| DMF coupling/loading total quantity (Gly10) | 7.58 | kg/kg | 2.00% | 6.67 | 8.49 | 7.60 | 7.58 | 7.61 | 6.64 | 8.49 | 7.59 | 7.59 | 7.63 | 7.62 | 7.58 | 7.61 | 7.59 | 7.62 | 7.60 | 7.58 | 7.58 |
| Coupling/loading temperature (Gly10) | 25.0 | °C | 2.0 | 13.0 | 37.0 | 25.0 | 25.1 | 25.0 | 25.0 | 25.0 | 25.0 | 33.0 | 25.0 | 17.0 | 25.1 | 25.0 | 25.0 | 25.0 | 25.0 | 33.0 | 25.0 |
| Coupling/loading time (Gly10) | 180 | min | 10 | 120 | 240 | 180 | 180 | 180 | 243 | 120 | 180 | 240 | 180 | 180 | 480 | 180 | 180 | 180 | 180 | 180 | 180 |
| DMF post-loading wash (Gly10) | 7.58 | kg/kg | 2.00% | 6.67 | 8.49 | 7.58 | 7.58 | 7.62 | 7.58 | 7.43 | 7.58 | 7.58 | 7.62 | 7.60 | 6.63 | 7.55 | 7.57 | 7.59 | 7.58 | 7.47 | 7.55 |
| DMF quantity (resin capping) | 8.53 | kg/kg | 2.00% | 7.50 | 9.55 | 8.53 | 8.54 | 8.57 | 9.63 | 8.46 | 7.48 | 8.54 | 8.55 | 8.55 | 8.54 | 8.49 | 8.51 | 8.53 | 8.61 | 8.46 | 8.50 |
| MeOH quantity (resin capping) | 10.00 | eq | 2.00% | 8.80 | 11.20 | 9.97 | 10.00 | 10.03 | 11.41 | 10.01 | 8.75 | 10.14 | 10.00 | 10.16 | 10.03 | 9.96 | 10.05 | 9.95 | 10.10 | 10.20 | 9.91 |
| DIPEA quantity (resin capping) | 5.00 | eq | 2.00% | 4.40 | 5.60 | 5.01 | 5.03 | 5.01 | 5.61 | 4.96 | 4.37 | 4.95 | 5.00 | 5.00 | 4.99 | 4.98 | 4.99 | 5.09 | 5.08 | 5.01 | 5.00 |
| DIPEA addition temperature (resin capping) | 20.0 | °C | 2.0 | 8.0 | 32.0 | 20.0 | 20.0 | 20.0 | 25.0 | 20.0 | 15.0 | 20.0 | 19.8 | 19.9 | 20.0 | 20.0 | 20.0 | 20.0 | 20.0 | 20.0 | 20.0 |
| Resin capping temperature | 25.0 | °C | 2.0 | 13.0 | 37.0 | 25.0 | 25.0 | 25.0 | 37.1 | 25.0 | 13.0 | 25.0 | 25.0 | 25.0 | 25.0 | 25.0 | 25.0 | 25.0 | 25.0 | 25.0 | 25.0 |
| Resin capping time | 60 | min | 10 | 0 | 120 | 60 | 60 | 60 | 120 | 60 | 0 | 60 | 60 | 60 | 60 | 60 | 60 | 60 | 60 | 60 | 60 |
| DMF post-resin capping wash | 7.58 | kg/kg | 2.00% | 6.67 | 8.49 | 7.63 | 7.58 | 7.60 | 7.58 | 7.54 | 7.61 | 7.59 | 7.58 | 7.61 | 7.60 | 8.50 | 6.62 | 7.60 | 7.60 | 7.53 | 7.60 |
| Piperidine in DMF quantity pre-(Thr9, Ala8, Cys7, Thr6, | 7.44 | kg/kg | 2.00% | 6.55 | 8.34 | 7.43 | 7.45 | 7.46 | 8.39 | 7.40 | 6.58 | 7.44 | 7.48 | 7.47 | 7.45 | 7.41 | 7.43 | 7.47 | 7.46 | 7.41 | 7.42 |

[12] R. Candeago Signals notebook: D4U-D03050-106 - Signals, "LAARA L3 SPPS Stats Table (equiv and kg-kg) - MS-V9-23SEP2026", last modified Sep 23, 2026, 05:20 PM ET

[13] Blue and red box correspond to low and high parameter values, respectively, in comparison with baseline.

[14] R. Candeago Signals notebook: D4U-D03050-116 > Process > open each PAR run (eg "LAARA L3 SPPS PAR Template_1(Baseline 1).xlsx"), sheet "L3 SPPS PAR Template", cell I14



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| Parmeter | Target | Units | Sigma | -6 sigma | +6 sigma | B1 | B2 | B3 | P3 | P4 | P5 | P6 | P8 | P9 | P10 | P11 | P12 | P13 | P14 | P15 | P16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Ala5, Thr4, Asn3, Cys2, gGlu1) | | | | | | | | | | | | | | | | | | | | | |
| Fmoc deprotection temperature pre-(Thr9, Ala8, Cys7, Thr6, Ala5, Thr4, Asn3, Cys2, gGlu1) | 25.0 | °C | 2.0 | 13.0 | 37.0 | 25.0 | 25.0 | 25.0 | 25.1 | 25.0 | 13.0 | 37.0 | 25.0 | 25.0 | 25.0 | 25.0 | 25.0 | 25.0 | 25.0 | 25.0 | 37.0 |
| Total Fmoc deprotection time pre-(Thr9, Ala8, Cys7, Thr6, Ala5, Thr4, Asn3, Cys2, gGlu1) | 120 | min | 10 | 60 | 180 | 120 | 120 | 120 | 120 | 120 | 60 | 180 | 120 | 120 | 120 | 121 | 120 | 120 | 120 | 120 | 180 |
| IPAC quantity - post-Fmoc deprotection wash pre-(Thr9, Ala8, Cys7, Thr6, Ala5) | 6.98 | kg/kg | 2.00% | 6.14 | 7.82 | 6.97 | 6.98 | 7.00 | 6.98 | 6.04 | 6.99 | 7.04 | 6.97 | 7.03 | 6.98 | 6.95 | 6.97 | 6.99 | 6.98 | 6.96 | 6.97 |
| DMF quantity - reswell pre-(Thr9, Ala8, Cys7, Thr6, Ala5) | 7.58 | kg/kg | 2.00% | 6.67 | 8.49 | 7.59 | 7.59 | 7.61 | 7.59 | 7.50 | 7.67 | 7.58 | 7.59 | 6.67 | 7.61 | 7.55 | 7.58 | 7.60 | 7.61 | 7.52 | 7.62 |
| Dissolution temperature (Thr9) | 20.0 | °C | 2.0 | 8.0 | 32.0 | 20.0 | 20.0 | 20.0 | 20.0 | 20.0 | 20.0 | 32.0 | 20.0 | 8.0 | 20.0 | 20.0 | 20.0 | 20.0 | 20.0 | 32.0 | 20.0 |
| DMF activation total quantity (Thr9) | 4.74 | kg/kg | 2.00% | 4.17 | 5.31 | 4.75 | 4.75 | 4.75 | 4.14 | 5.27 | 4.80 | 4.75 | 4.75 | 4.76 | 4.74 | 4.72 | 4.74 | 4.77 | 4.75 | 4.81 | 4.78 |
| AA quantity (Thr9) | 2.00 | eq | 1.00% | 1.88 | 2.12 | 2.00 | 2.00 | 2.00 | 2.13 | 1.85 | 2.00 | 1.88 | 1.98 | 2.01 | 2.00 | 1.99 | 2.00 | 2.01 | 2.01 | 2.00 | 1.87 |
| PyBOP quantity (Thr9) | 2.00 | eq | 2.00% | 1.76 | 2.24 | 2.00 | 2.01 | 2.01 | 2.25 | 1.73 | 2.00 | 1.75 | 1.99 | 2.01 | 2.00 | 1.99 | 2.00 | 2.01 | 2.01 | 2.00 | 1.75 |
| Oxyma quantity (Thr9) | 2.00 | eq | 2.00% | 1.76 | 2.24 | 2.00 | 2.00 | 2.01 | 2.25 | 1.73 | 2.01 | 1.75 | 2.00 | 2.01 | 2.00 | 1.99 | 2.01 | 2.04 | 2.03 | 2.00 | 1.76 |
| DIPEA quantity (Thr9) | 3.00 | eq | 2.00% | 2.64 | 3.36 | 3.00 | 3.00 | 3.01 | 3.38 | 2.59 | 3.00 | 3.38 | 3.02 | 3.01 | 3.01 | 2.99 | 3.00 | 3.10 | 3.01 | 3.01 | 3.42 |
| Activation temperature (Thr9) | 5.0 | °C | 2.0 | -7.0 | 17.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 17.0 | 5.0 | -7.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 17.0 | 5.0 |
| Activation time (Thr9) | 20 | min | 10 | 0 | 80 | 20 | 20 | 20 | 80 | 0 | 20 | 80 | 20 | 20 | 20 | 20 | 20 | 20 | 20 | 20 | 20 |
| DMF coupling total quantity (Thr9) | 8.53 | kg/kg | 2.00% | 7.50 | 9.55 | 8.52 | 8.54 | 8.55 | 7.45 | 9.49 | 8.61 | 8.55 | 8.52 | 8.56 | 8.53 | 8.49 | 8.53 | 8.56 | 8.88 | 8.59 | 8.58 |
| Coupling temperature (Thr9) | 25.0 | °C | 2.0 | 13.0 | 37.0 | 22.3 | 25.0 | 25.0 | 25.0 | 25.0 | 25.0 | 33.0 | 25.0 | 17.0 | 25.0 | 25.0 | 25.0 | 25.0 | 20.0 | 33.0 | 25.0 |
| Coupling time (Thr9) | 240 | min | 10 | 180 | 300 | 240 | 240 | 240 | 300 | 180 | 240 | 300 | 240 | 240 | 600 | 240 | 240 | 240 | 240 | 240 | 240 |
| DMF post-coupling wash (Thr9, Ala8, Cys7, Thr6) | 7.58 | kg/kg | 2.00% | 6.67 | 8.49 | 7.59 | 7.58 | 7.61 | 7.70 | 7.50 | 7.59 | 7.57 | 7.58 | 7.61 | 6.66 | 7.56 | 7.63 | 7.60 | 7.58 | 7.58 | 7.58 |
| Dissolution temperature (Ala8, Thr6, Ala5, Thr4, Asn3) | 20.0 | °C | 2.0 | 8.0 | 32.0 | 20.0 | 20.0 | 20.0 | 20.0 | 20.0 | 19.8 | 32.0 | 20.0 | 8.0 | 20.0 | 20.0 | 20.0 | 20.0 | 20.0 | 32.0 | 20.0 |
| DMF activation total quantity (Ala8, Thr6, Ala5, Thr4, Asn3) | 6.63 | kg/kg | 2.00% | 5.84 | 7.43 | 6.63 | 6.63 | 6.63 | 5.83 | 7.39 | 6.63 | 6.67 | 6.64 | 6.66 | 6.66 | 6.61 | 6.65 | 6.65 | 6.64 | 6.63 | 6.69 |
| AA quantity (Ala8, Thr6, Ala5, Thr4, Asn3) | 2.00 | eq | 1.00% | 1.88 | 2.12 | 2.00 | 1.99 | 2.02 | 2.13 | 1.85 | 2.00 | 1.88 | 2.00 | 2.01 | 2.00 | 2.00 | 2.00 | 2.02 | 2.01 | 2.00 | 1.87 |
| Oxyma quantity (Ala8, Thr6, Ala5, Thr4, Asn3) | 2.00 | eq | 2.00% | 1.76 | 2.24 | 2.00 | 2.00 | 1.99 | 2.25 | 1.73 | 2.01 | 1.76 | 2.00 | 2.03 | 2.02 | 2.00 | 2.01 | 2.01 | 2.01 | 2.00 | 1.79 |
| DIC quantity (Ala8, Thr6, Ala5, Thr4, Asn3) | 2.20 | eq | 2.00% | 1.94 | 2.46 | 2.20 | 2.20 | 2.26 | 2.47 | 1.90 | 2.20 | 2.47 | 2.20 | 2.22 | 2.23 | 2.20 | 2.25 | 2.23 | 2.24 | 2.20 | 2.47 |
| Activation temperature (Ala8, Thr6, Ala5, Thr4, Asn3) | 20.0 | °C | 2.0 | 8.0 | 32.0 | 20.0 | 20.0 | 20.0 | 20.0 | 20.0 | 19.9 | 32.0 | 20.0 | 8.0 | 19.9 | 20.0 | 20.0 | 20.0 | 20.0 | 32.0 | 20.0 |
| Activation time (Ala8, Thr6, Ala5, Thr4, Asn3) | 20 | min | 10 | 0 | 80 | 20 | 20 | 20 | 80 | 0 | 20 | 81 | 20 | 20 | 20 | 20 | 20 | 20 | 20 | 20 | 20 |



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| Parmeter | Target | Units | Sigma | -6 sigma | +6 sigma | B1 | B2 | B3 | P3 | P4 | P5 | P6 | P8 | P9 | P10 | P11 | P12 | P13 | P14 | P15 | P16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| DMF coupling total quantity (Ala8, Thr6, Ala5, Thr4, Asn3) | 7.58 | kg/kg | 2.00% | 6.67 | 8.49 | 7.58 | 7.58 | 7.58 | 6.68 | 8.45 | 7.58 | 7.62 | 7.59 | 7.61 | 7.61 | 7.56 | 7.62 | 7.60 | 7.60 | 7.58 | 7.66 |
| Coupling temperature (Ala8, Thr6, Ala5, Thr4, Asn3) | 25.0 | °C | 2.0 | 13.0 | 37.0 | 25.0 | 25.0 | 25.0 | 25.0 | 25.0 | 25.0 | 33.0 | 25.0 | 17.0 | 25.0 | 25.0 | 25.0 | 25.0 | 25.0 | 33.0 | 25.0 |
| Coupling time (Ala8, Thr6, Ala5, Thr4, Asn3) | 240 | min | 10 | 180 | 300 | 240 | 240 | 240 | 300 | 180 | 240 | 300 | 240 | 240 | 600 | 240 | 240 | 240 | 240 | 240 | 240 |
| DMF dissolution total quantity (Cys7, Cys2) | 6.63 | kg/kg | 2.00% | 5.84 | 7.43 | 6.63 | 6.64 | 6.64 | 5.81 | 7.40 | 6.63 | 6.63 | 6.67 | 6.66 | 6.67 | 6.61 | 6.62 | 6.67 | 6.65 | 6.72 | 6.65 |
| AA quantity (Cys7, Cys2) | 2.00 | eq | 2.00% | 1.76 | 2.24 | 2.00 | 2.00 | 2.00 | 2.24 | 1.73 | 2.01 | 1.75 | 2.00 | 2.01 | 2.00 | 1.99 | 2.00 | 2.00 | 2.01 | 2.01 | 1.88 |
| Oxyma quantity (Cys7, Cys2) | 3.00 | eq | 1.00% | 2.82 | 3.18 | 3.00 | 3.00 | 3.01 | 3.22 | 2.76 | 3.01 | 2.81 | 3.00 | 3.01 | 3.00 | 2.99 | 2.99 | 3.02 | 3.04 | 3.02 | 2.61 |
| Dissolution temperature (Cys7, Cys2) [15] | 5.0 | °C | 2.0 | -7.0 | 17.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 17.0 | 5.0 | -7.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 17.0 | 5.0 |
| Dissolution time (Cys7, Cys2) | 5 | min | 10 | 0 | 65 | 5 | 5 | 5 | 65 | 0 | 5 | 65 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| DMF coupling total quantity (Cys7, Cys2) | 8.53 | kg/kg | 2.00% | 7.50 | 9.55 | 8.53 | 8.53 | 8.54 | 7.71 | 9.28 | 8.52 | 8.65 | 8.58 | 8.56 | 8.58 | 8.49 | 8.51 | 8.57 | 8.55 | 8.66 | 8.66 |
| DIC quantity (Cys7, Cys2) | 3.30 | eq | 2.00% | 2.90 | 3.70 | 3.30 | 3.30 | 3.31 | 3.72 | 2.85 | 3.30 | 3.72 | 3.30 | 3.31 | 3.30 | 3.28 | 3.29 | 3.38 | 3.40 | 3.35 | 3.71 |
| Coupling temperature (Cys7, Cys2) | 20.0 | °C | 2.0 | 8.0 | 32.0 | 20.0 | 20.0 | 20.0 | 19.9 | 20.0 | 20.0 | 28.0 | 20.0 | 12.0 | 20.0 | 20.0 | 20.0 | 20.0 | 20.0 | 28.0 | 20.0 |
| Coupling time (Cys7, Cys2) | 240 | min | 10 | 180 | 300 | 240 | 240 | 240 | 300 | 180 | 240 | 300 | 240 | 240 | 600 | 240 | 240 | 240 | 240 | 240 | 240 |
| DMF quantity post-coupling wash (Ala5, Thr4, Asn3, Cys2, gGlu1) | 9.48 | kg/kg | 2.00% | 8.34 | 10.61 | 9.48 | 9.49 | 9.49 | 9.48 | 9.37 | 9.49 | 9.49 | 9.49 | 9.49 | 8.29 | 9.47 | 9.46 | 9.48 | 9.48 | 9.48 | 9.48 |
| IPAC quantity - post-Fmoc deprotection wash (Thr4, Asn3, Cys2, gGlu1) | 8.73 | kg/kg | 2.00% | 7.68 | 9.77 | 8.77 | 8.74 | 8.79 | 8.73 | 7.57 | 8.73 | 8.73 | 8.71 | 8.77 | 8.75 | 8.72 | 8.71 | 8.73 | 8.77 | 8.72 | 8.74 |
| DMF - reswell (Thr4, Asn3, Cys2, gGlu1) | 9.48 | kg/kg | 2.00% | 8.34 | 10.61 | 9.40 | 9.49 | 9.51 | 9.52 | 9.30 | 9.48 | 9.49 | 9.48 | 8.32 | 9.48 | 9.45 | 9.46 | 9.49 | 9.49 | 9.49 | 9.48 |
| DMF activation total quantity (gGlu1) | 6.63 | kg/kg | 2.00% | 5.84 | 7.43 | 6.62 | 6.65 | 6.73 | 5.80 | 7.48 | 6.69 | 6.68 | 6.65 | 6.70 | 6.64 | 6.61 | 6.64 | 6.65 | 6.65 | 6.63 | 6.71 |
| Boc-L-Glu-OtBu quantity (gGlu1) | 2.00 | eq | 1.00% | 1.88 | 2.12 | 2.00 | 2.00 | 2.01 | 2.12 | 1.85 | 2.00 | 1.89 | 2.02 | 2.01 | 2.00 | 1.99 | 2.00 | 2.00 | 2.00 | 2.01 | 1.91 |
| Activation temperature (gGlu1) | 5.0 | °C | 2.0 | -7.0 | 17.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 17.0 | 5.0 | -7.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 17.0 | 5.0 |
| PyBOP quantity (gGlu1) | 2.00 | eq | 2.00% | 1.76 | 2.24 | 2.00 | 2.00 | 2.00 | 2.25 | 1.74 | 2.00 | 1.76 | 2.00 | 2.01 | 2.00 | 1.99 | 2.00 | 2.01 | 2.00 | 2.00 | 1.76 |
| Oxyma quantity (gGlu1) | 2.00 | eq | 2.00% | 1.76 | 2.24 | 2.00 | 2.00 | 2.00 | 2.25 | 1.74 | 2.00 | 1.75 | 1.99 | 2.01 | 2.00 | 2.00 | 2.01 | 2.03 | 2.00 | 2.01 | 1.80 |
| DIPEA quantity (gGlu1) | 3.00 | eq | 2.00% | 2.64 | 3.36 | 2.99 | 3.00 | 3.01 | 3.38 | 2.62 | 3.00 | 3.40 | 3.03 | 3.01 | 3.00 | 2.99 | 2.98 | 3.01 | 3.02 | 3.00 | 3.41 |
| Activation temperature (gGlu1) | 20.0 | °C | 2.0 | 8.0 | 32.0 | 20.0 | 20.0 | 20.0 | 20.0 | 20.0 | 20.0 | 32.0 | 20.0 | 8.0 | 20.0 | 20.0 | 20.0 | 20.0 | 20.0 | 32.0 | 20.0 |
| Activation time (gGlu1) | 20 | min | 10 | 0 | 80 | 20 | 20 | 20 | 80 | 0 | 20 | 80 | 20 | 20 | 20 | 20 | 21 | 20 | 20 | 20 | 20 |
| DMF coupling total quantity (gGlu1) | 7.58 | kg/kg | 2.00% | 6.67 | 8.49 | 7.55 | 7.60 | 7.68 | 6.63 | 8.59 | 7.65 | 7.63 | 7.60 | 7.67 | 7.59 | 7.55 | 7.61 | 7.60 | 7.61 | 7.60 | 7.71 |

[15] Referred to as “activation temperature” in the stats table



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| Parmeter | Target | Units | Sigma | -6 sigma | +6 sigma | B1 | B2 | B3 | P3 | P4 | P5 | P6 | P8 | P9 | P10 | P11 | P12 | P13 | P14 | P15 | P16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Coupling temperature (gGlu1) | 25.0 | °C | 2.0 | 13.0 | 37.0 | 25.0 | 25.0 | 25.0 | 25.0 | 25.0 | 25.0 | 33.0 | 25.0 | 17.0 | 25.0 | 25.0 | 25.0 | 25.0 | 25.0 | 33.0 | 25.0 |
| Coupling time (gGlu1) | 240 | min | 10 | 180 | 300 | 240 | 240 | 240 | 300 | 180 | 240 | 300 | 240 | 240 | 600 | 240 | 240 | 240 | 240 | 240 | 240 |
| DMF quantity for iodine solution prep | 3.79 | kg/kg | 2.00% | 3.34 | 4.25 | 3.79 | 3.79 | 3.80 | 3.78 | 3.75 | 3.82 | 3.79 | 4.04 | 3.82 | 3.81 | 3.30 | 4.25 | 3.31 | 4.27 | 3.87 | 3.81 |
| Iodine quantity | 5.00 | eq | 2.00% | 4.40 | 5.60 | 5.00 | 5.00 | 5.01 | 4.99 | 4.94 | 5.00 | 5.00 | 5.00 | 5.01 | 5.00 | 5.60 | 4.36 | 5.62 | 4.36 | 5.01 | 5.02 |
| DMF quantity for oxidation reaction | 11.37 | kg/kg | 2.00% | 10.01 | 12.74 | 11.37 | 11.38 | 11.40 | 11.36 | 11.24 | 11.37 | 11.37 | 11.40 | 11.42 | 11.39 | 9.98 | 12.68 | 9.93 | 12.79 | 11.40 | 11.39 |
| Oxidation temperature | -5.0 | °C | 2.0 | -17.0 | 7.0 | -5.0 | -5.0 | -5.0 | -5.0 | -5.0 | -5.0 | -5.0 | -5.0 | -5.0 | -5.0 | 5.0 | -9.0 | 5.0 | -15.0 | -5.0 | -5.0 |
| Iodine transfer time | 50 | min | 10 | 0 | 110 | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 40 | 60 | 100 | 20 | 40 | 40 |
| Post-oxidation wash temperature | -5.0 | °C | 2.0 | -17.0 | 7.0 | -5.0 | -5.0 | -5.0 | -5.0 | -5.0 | -5.0 | -5.0 | -5.0 | -5.0 | -5.0 | 5.0 | -7.0 | -5.0 | -5.0 | -5.0 | -5.0 |
| DMF quantity post-oxidation wash | 9.48 | kg/kg | 2.00% | 8.34 | 10.61 | 9.41 | 9.48 | 9.51 | 9.48 | 8.19 | 9.47 | 9.48 | 9.49 | 9.51 | 9.50 | 10.65 | 8.28 | 9.47 | 9.48 | 9.45 | 9.56 |
| IPAC quantity - deswell wash | 8.73 | kg/kg | 2.00% | 7.68 | 9.77 | 8.72 | 8.74 | 8.76 | 8.71 | 8.63 | 8.80 | 8.70 | 8.73 | 7.67 | 8.76 | 8.71 | 8.71 | 8.72 | 8.72 | 8.77 | 8.75 |
| MTBE quantity - deswell wash | 7.41 | kg/kg | 2.00% | 6.52 | 8.30 | 7.41 | 7.41 | 7.43 | 7.40 | 7.33 | 7.42 | 7.41 | 7.41 | 6.50 | 7.41 | 7.57 | 7.40 | 7.40 | 7.40 | 7.45 | 7.42 |
| Oxidation Exposure time | 50 | min | 10 | 0 | 110 | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 40 | 60 | 130 | 20 | 40 | 40 |
| Oxidation Reaction time | 0 | min | 10 | 0 | 60 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 30 | 0 | 0 | 0 |
| Drying jacket temperature[16] | 25 | °C | 2 | - | 37 | 25 | 25 and 42 | 25 | 25 | 25 | 25 | 25 | 25 | 25 | 25 | 25 | 25 | 25 | 25 | 25 | 25 |

[16] B2 wet cake was split and dried at two different temperatures. Data source for 42°C condition is in Technical Summary for LSN4182928 (ASYM-140021) Step 1 PAR Study, QDocs RPT-1194110



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The SPPS PAR study was designed around the parameters listed in **Table 26**. Sixteen experiments (B1 through P16) were executed: three baseline runs: B1, B2 (large scale) and B3, a set of edge-of-range bracketing runs (P3 through P12) and four deconvolution runs (P13 through P16). Each non-baseline experiment drives one or more parameters to the edge of its design window (Target ∓ 6σ, as defined in the **Table 26** notes; pink cells denote a parameter at its maximum and blue cells a parameter at its minimum relative to target) in order to establish the proven acceptable ranges and to elucidate the associated impurity mechanisms. Consistent with the **Table 26** notes, resin re-swell, deFmoc and post-deFmoc conditions are bundled across the Thr9, Ala8, Cys7, Thr6, Ala5, Thr4, Asn3, Cys2 and gGlu1 cycles; coupling and activation conditions are bundled for the standard amino acids (Ala8, Thr6, Ala5, Thr4 and Asn3) and, separately, for the cysteine residues (Cys7 and Cys2). The design intent and the specific parameters perturbed in each experiment are described below.

B1 and B3 are the baseline SPPS experiments. All parameters are held at their targets to establish the representative baseline impurity and yield profile against which every bracketed experiment is compared (baseline yields 91.9% and 93.2%, respectively). B2 repeats the baseline at large scale, charging 200 g of CTC resin versus the 45 g target, to demonstrate that the impurity profile is comparable across scale.

- P3 – High conversion: coupling and activation are driven toward maximum conversion. Amino-acid, Oxyma/PyBOP and DIC charges, DIPEA equivalents, activation times (to 65–80 min) and coupling times (to 240–300 min) are all set to +6σ, the piperidine preparation is raised to ~25% (v/v), and the capping charges (DMF, MeOH, DIPEA), capping temperature (~37°C) and capping time (120 min) are maximized; the corresponding DMF activation- and coupling-solvent quantities are simultaneously taken to –6σ (minimum volume) to concentrate the reactions.
- P4 – Low coupling conversion: the coupling step is stressed toward incomplete reaction to probe deletion-impurity formation. Amino-acid, Oxyma/PyBOP and DIC equivalents are set to –6σ, activation time is floored at 0 min and coupling times reduced to their 180-min minimum, while the DMF activation and coupling volumes are simultaneously raised to +6σ (maximum dilution). Post-deFmoc IPAC and post-oxidation DMF washes are also reduced.



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- P5 – Low deFmoc conversion: the Fmoc-removal step is stressed toward incomplete deprotection. Piperidine strength is lowered to ~15% (v/v), the piperidine-in-DMF charge and deFmoc temperature (~13 °C) are set to –6σ and total deFmoc time reduced to its 60-min minimum; the capping charges, temperature and time are likewise minimized (capping time floored at 0 min) so that residual Fmoc and uncapped sites are maximized.
- P6 – Racemization / Epimerization / DKP: the parameters that promote stereochemical degradation and diketopiperazine formation are maximized. Activation and coupling temperatures (activation jacket ~37 °C, activation up to 32 °C, coupling up to 33 °C), activation times (to 80 min), coupling times (to 300 min), DeFmoc temperature (37 °C) and time (180 min) and DIC/DIPEA equivalents are all set to +6σ, while amino-acid and Oxyma/PyBOP equivalents are set to –6σ to prolong exposure of activated species.
- P8 – Low resin loading: the effect of a reduced substitution / loading level on downstream purity and yield is assessed. The set-point charges in **Table 26** remain at target (CTC resin 45 g), so the low-loading condition is realized through the resin lot rather than an at-limit parameter; the run yielded 93.2%.
- P9 – Cold coupling: the coupling/activation temperatures are driven to their –6σ minima throughout the bundled cycles (activation jacket ~13 °C, activation temperatures from +8 °C down to –7 °C for the Thr9 and Cys cycles). The reswell and IPAC/MTBE de-swell wash quantities are also reduced, testing coupling completeness and impurity growth under cold conditions.
- P10 – Coupling long hold: the coupling hold time is extended to its +6σ maximum across the cycles (up to 480–600 min) while the post-coupling DMF wash quantities are reduced to –6σ, evaluating impurity formation under prolonged coupling and minimal inter-step washing.
- P11 – $I_2$ oxidation byproducts: the iodine-mediated disulfide-formation step is stressed toward over-oxidation. Iodine charge (to 5.60 eq) and the post-capping and post-oxidation DMF wash quantities are set to +6σ, while the iodine-solution-prep and oxidation-reaction DMF quantities are set to –6σ (more concentrated), maximizing oxidation-related byproducts.



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- P12 – $I_2$ oxidation incomplete: the mirror image of P11, stressing under-oxidation. Iodine charge (4.36 eq) and the post-capping/post-oxidation washes are set to –6σ while the iodine-solution-prep and oxidation-reaction DMF quantities are raised to +6σ (more dilute), so that the disulfide oxidation is driven toward incompletion.

The iodine-oxidation and cysteine-isomer brackets co-vary several parameters, so four deconvolution runs (P13 through P16) separate them to assign criticality to individual parameters:

- P13 – $I_2$ oxidation deconvolution 1: isolates the high-oxidation levers, setting iodine charge (5.62 eq) and oxidation exposure time (130 min) to +6σ while holding the iodine-prep and reaction DMF quantities at –6σ, to identify which drive the byproducts seen in P11.
- P14 – $I_2$ oxidation deconvolution 2: the complementary run, raising the iodine-prep and oxidation-reaction DMF quantities to +6σ while lowering iodine charge to –6σ (4.36 eq), to resolve the incomplete-oxidation contribution seen in P12.
- P15 – Cys isomer deconvolution 1: isolates the temperature contribution to cysteine-isomer formation, driving activation and coupling temperatures across the bundled cycles to their +6σ maxima (activation jacket ~37°C, activation up to 32°C, coupling up to 33°C, Cys activation to 17°C).
- P16 – Cys isomer deconvolution 2: isolates the reagent and deFmoc contribution, setting DeFmoc temperature (37°C) and time (180 min) and the DIC/DIPEA equivalents to +6σ while lowering amino-acid, PyBOP and Oxyma equivalents to –6σ, to resolve the non-temperature drivers of cysteine isomerization.

## 9. Risk Analysis of Experimental Conditions

The initial risk assessment narrowed the process parameters to those identified as posing a potential risk to impacting relevant quality attributes in crude or in materials forward processed to the purified drug substance. Potential risks to drug substance quality attributes include impurities that form in Boc-Fragment 3.

Prior to the analysis to determine parameter risk classifications, baseline average is calculated. Three baseline experiments — B1, B2 (large scale) and B3 — were representative of the baseline



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SPPS process and used to determine the representative process baseline average values for calculating corresponding risk thresholds for each relevant quality attribute. B1 and B3 were run at the 45 g CTC resin target scale, while B2 was executed at 200 g CTC resin to provide baseline material for the cleavage PARs; all three held every process parameter within ±1.5σ of its target set point.

Table 27 summarizes the baseline impurity data for both specified and unspecified impurities: the average across the three baseline experiments (column 2) and the standard deviation from three baseline results. Some of the baseline experiments had results that were below the method reporting threshold. To be conservative, these baseline values were assumed to be equal to the method reporting threshold. For example, the impurities method reporting threshold is 0.1%. In cases where results were reported as <0.1%, a value of 0.1% was used to calculate the baseline average, and associated risk thresholds.

Table 27 gives the criticality thresholds (columns 4 to 6) and each experiment's result against those thresholds, for both specified and unspecified impurities. For details around the criticality threshold calculations see Appendix VI: Baseline correction and risk threshold calculations.



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**Table 27**. Summary of baseline average values, specifications and Low/Medium Risk threshold values for impurities.[17,18]

| Critical Quality Attributes[19] | Baseline Average (n=3) | Baseline Standard Deviation (n=3) | Specification (High Risk Threshold) | Low/ Medium Risk Threshold | No/Low Risk Threshold |
|---|---|---|---|---|---|
| L3 @ RRT 1.00 | 96.87 | 0.32 | NLT 90.0% | 93.43 | 96.18 |
| L3 Dimer @ RRT 1.34 | 0.98 | 0.04 | NMT 1.5% | 1.24 | 1.03 |
| L3 Dimer (single D-Cys) @ RRT 1.37 | 0.10 | 0.00 | NMT 0.5% | 0.30 | 0.14 |
| L3 Dimer (single D-Cys) @ RRT 1.39 | 0.10 | 0.00 | NMT 0.5% | 0.30 | 0.14 |
| L3 Imp @ RRT 0.49 | 0.11 | 0.02 | NMT 0.5% | 0.31 | 0.15 |
| L3 γGlu1(des-Boc) @ RRT 0.56 | 0.11 | 0.01 | NMT 0.5% | 0.30 | 0.14 |
| L3 des-Asn3 @ RRT 0.64 | 0.10 | 0.00 | NMT 0.5% | 0.30 | 0.14 |
| L3 M+262 Da @ RRT 0.69 | 0.13 | 0.06 | NMT 0.5% | 0.32 | 0.17 |
| L3 M+294 Da @ RRT 0.69 | 0.10 | 0.00 | NMT 0.5% | 0.30 | 0.14 |
| L3 M-257 Da @ RRT 0.74 | 0.15 | 0.03 | NMT 0.5% | 0.33 | 0.19 |
| L3 Ac-2-10 @ RRT 0.75 | 0.24 | 0.06 | NMT 0.5% | 0.37 | 0.26 |
| L3 Fmoc-2-10 des-Asn @ RRT 0.77 | 0.10 | 0.00 | NMT 0.5% | 0.30 | 0.14 |
| L3 M-522 Da @ RRT 0.81 | 0.10 | 0.00 | NMT 0.5% | 0.30 | 0.14 |
| L3 des-tBu @ RRT 0.82 | 0.11 | 0.01 | NMT 0.5% | 0.30 | 0.15 |
| L3 des-Thr4 @ RRT 0.82 | 0.11 | 0.01 | NMT 0.5% | 0.31 | 0.15 |
| L3 M-213 Da @ RRT 0.83 | 0.10 | 0.00 | NMT 0.5% | 0.30 | 0.14 |
| L3 des-tBu @ RRT 0.88 | 0.10 | 0.00 | NMT 0.5% | 0.30 | 0.14 |
| L3 endo-Ala @ RRT 0.90 | 0.10 | 0.00 | NMT 0.5% | 0.30 | 0.14 |
| L3 des-Thr9 @ RRT 0.90 | 0.16 | 0.04 | NMT 0.5% | 0.33 | 0.20 |
| L3 linear dithiol @ RRT 0.92 | 0.10 | 0.00 | NMT 0.5% | 0.30 | 0.14 |
| L3 endo-Ala @ RRT 0.95 | 0.10 | 0.00 | NMT 0.5% | 0.30 | 0.14 |
| L3 endo-Gly10 @ RRT 0.96 | 0.12 | 0.03 | NMT 0.5% | 0.31 | 0.16 |
| L3 D-Cys7 @ RRT 0.98 | 0.11 | 0.01 | NMT 0.5% | 0.30 | 0.15 |
| L3 Boc-L3-SS @ RRT 1.01 | 0.10 | 0.00 | NMT 0.5% | 0.30 | 0.14 |
| L3 M+46 Da @ RRT 1.01 | 0.10 | 0.00 | NMT 0.5% | 0.30 | 0.14 |
| L3 D-Cys2&7 @ RRT 1.03 | 0.10 | 0.00 | NMT 0.5% | 0.30 | 0.14 |

[17] R. Candeago Signals notebook, D4U-D03050-122

[18] Statistical Data Integrity Review Documentation for PAR Report Relating to Fragment L3 (LSN4182928) Step 1 SPPS Process of eloralintide (LY3841136) Drug Substance, QDocs AMT-178599

[19] D4U-D03050-106 - Signals



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| Critical Quality Attributes[19] | Baseline Average (n=3) | Baseline Standard Deviation (n=3) | Specification (High Risk Threshold) | Low/ Medium Risk Threshold | No/Low Risk Threshold |
|---|---|---|---|---|---|
| L3 endo-γGlu1 @ RRT 1.04 | 0.13 | 0.01 | NMT 0.5% | 0.32 | 0.17 |
| L3 Fmoc-2-10 @ RRT 1.06 | 0.10 | 0.00 | NMT 0.5% | 0.30 | 0.14 |
| L3 endo-Thr @ RRT 1.11 | 0.10 | 0.00 | NMT 0.5% | 0.30 | 0.14 |
| L3 M+48 Da @ RRT 1.31 | 0.10 | 0.00 | NMT 0.5% | 0.30 | 0.14 |

Table 28 and Table 29 below, along with Table 41 in Appendix V summarize the Boc-Fragment 3 SPPS studies for parameter risk classification including the study conditions and resulting CQA risks across the collective univariate, multivariate and bracketing studies. Table 28 and Table 29 identify, for each parameter classified as medium or high risk, the CQAs whose response to the PAR perturbation drove that classification. Table 28 lists the CQAs based on PAR experiment analysis by UV result while Table 29 lists the CQAs based on PAR experiment analysis by MS results. In each table, the impact of a PAR experiment on a given CQA is reported as N, L, M or H — negligible, low, medium or high impact — relative to the risk thresholds listed in Table 27. A result is negligible below the No/Low risk threshold, low between the No/Low and Low/Medium thresholds, medium between the Low/Medium threshold and the specification limit, and high at or beyond the specification limit. The corresponding experimental conditions are given in **Table 26** in which parameters held at target were controlled within ±1.5σ of their set point and parameters deliberately bracketed in a given experiment were driven to approximately ±6σ of target.



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Table 28. CQA Impact Results for Fragment L3 (LSN4182928) Step 1 SPPS Process – UV results [18,20]

| Experiment | CQA | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | L3 Imp @ RRT 0.49 | L3 Imp @ RRT 0.56 | L3 Imp @ RRT 0.64 | L3 Imp @ RRT 0.69 | L3 Imp @ RRT 0.74 | L3 Imp @ RRT 0.75 | L3 Imp @ RRT 0.77 | L3 Imp @ RRT 0.81 | L3 Imp @ RRT 0.82 | L3 Imp @ RRT 0.83 | L3 Imp @ RRT 0.88 | L3 Imp A @ RRT 0.90 | L3 Imp B @ RRT 0.90 | L3 Imp @ RRT 0.92 | L3 Imp @ RRT 0.93 | L3 Imp @ RRT 0.95 | L3 Imp @ RRT 0.96 | L3 Imp @ RRT 0.98 | L3 @ RRT 1.00 | L3 Imp A @ RRT 1.01 | L3 Imp B @ RRT 1.01 | L3 Imp @ RRT 1.02 | L3 Imp @ RRT 1.03 | L3 Imp @ RRT 1.04 | L3 Imp @ RRT 1.06 | L3 Imp @ RRT 1.11 | L3 Imp @ RRT 1.21 | L3 Imp @ RRT 1.31 | L3 Imp @ RRT 1.34 | L3 Imp @ RRT 1.37 | L3 Imp @ RRT 1.39 |
| B1 | N | N | N | N | N | L | N | N | N | N | N | N | L | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N |
| B2 | N | N | N | N | N | L | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N |
| B3 | N | N | N | L | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N |
| P3 | N | L | N | L | L | L | N | N | M | N | N | N | N | N | N | N | N | N | N | L | N | N | N | N | N | N | N | L | N | N |
| P4 | N | N | N | N | N | L | N | N | L | N | N | N | N | N | N | N | N | N | N | N | N | L | N | L | N | N | N | N | N | N |
| P5 | N | N | H | N | N | L | L | L | N | N | N | N | N | N | N | N | N | N | N | L | N | N | N | N | N | N | L | N | N | N |
| P6 | N | N | N | N | N | L | N | N | N | N | N | N | L | N | N | N | N | L | N | N | N | N | N | N | N | L | N | N | N | N |
| P8 | N | N | N | N | N | L | N | N | N | N | N | N | L | N | N | N | N | N | N | N | N | N | N | N | N | N | N | L | N | N |
| P9 | N | N | N | L | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | L | N | L | N | N | N | N | L | N | N |
| P10 | N | N | N | N | L | L | N | N | N | N | N | N | N | N | N | N | N | L | N | N | N | L | N | L | N | N | N | N | L | N | N |
| P11 | N | N | N | N | L | L | N | N | L | L | N | N | N | N | N | N | N | N | N | N | N | N | N | L | N | N | N | N | N | N | N |
| P12 | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | L | N | N | N | N | N | N | N | L | N | N |
| P13 | N | N | N | L | N | L | N | N | L | N | N | N | N | N | N | N | N | L | N | L | N | M | N | L | N | N | N | N | N | N | N |
| P14 | N | N | N | N | N | L | N | N | L | N | N | N | N | N | N | N | N | L | N | N | N | M | N | L | N | N | N | N | N | N | N |
| P15 | N | N | N | N | N | N | N | N | L | N | N | N | N | N | N | N | N | L | N | N | N | N | N | L | N | N | N | N | L | N | N |
| P16 | N | N | N | N | N | N | N | N | L | N | N | N | N | N | N | N | N | L | N | N | N | N | N | L | N | N | N | N | L | L | N |

[20] The grouping of CQAs in the Table is due to limitation of statistical tool to be able to expand and include of all the CQAs in one chart, and the grouping is irrelevant to classifying CQA by specified/unspecified impurities.



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Table 29. CQA Impact Results for Fragment L3 (LSN4182928) Step 1 SPPS Process – MS results[18]

| | CQA | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Experiment | L3 Imp @ RRT 0.49 | L3 gGlu1(des-Boc) @ RRT 0.56 | L3 des-Asn3 @ RRT 0.64 | L3 M+262 Da @ RRT 0.69 | L3 M+294 Da @ RRT 0.69 | L3 M-257 Da @ RRT 0.74 | L3 Ac-2-10 @ RRT 0.75 | L3 Fmoc-2-10 des-Asn @ RRT 0.77 | L3 M-522 Da @ RRT 0.81 | L3 des-tBu @ RRT 0.82 | L3 des-Thr4 @ RRT 0.82 | L3 M-213 Da @ RRT 0.83 | L3 des-tBu @ RRT 0.88 | L3 endo-Ala @ RRT 0.90 | L3 des-Thr9 @ RRT 0.90 | L3 linear dithiol @ RRT 0.92 | L3 endo-Ala @ RRT 0.95 | L3 endo-Gly10 @ RRT 0.96 | L3 D-Cys7 @ RRT 0.98 | L3 @ RRT 1.00 | L3 Boc-L3-SS @ RRT 1.01 | L3 M+46 Da @ RRT 1.01 | L3 D-Cys2&7 @ RRT 1.03 | L3 endo-gGlu1 @ RRT 1.04 | L3 Fmoc-2-10 @ RRT 1.06 | L3 endo-Thr @ RRT 1.11 | L3 M+48 Da @ RRT 1.31 | L3 Dimer @ RRT 1.34 | L3 Dimer (single D-Cys) @ RRT 1.37 | L3 Dimer (single D-Cys) @ RRT 1.39 |
| B1 | N | N | N | N | N | N | L | N | N | N | N | N | N | N | L | N | N | N | N | N | N | N | N | N | N | N | N | N | N |
| B2 | N | N | N | N | N | N | L | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N |
| B3 | N | N | N | L | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N |
| P3 | N | L | N | N | L | L | L | N | N | L | L | N | N | N | N | N | N | N | N | N | L | N | N | N | N | N | L | N | N |
| P4 | N | N | N | N | N | N | L | N | N | N | L | N | N | N | N | N | N | N | N | N | N | N | L | L | N | N | N | N | N |
| P5 | N | N | H | L | N | N | L | L | L | N | N | N | N | N | N | N | N | N | N | N | L | N | N | N | N | N | N | N | N |
| P6 | N | N | N | N | N | N | L | N | N | N | N | N | N | N | L | N | N | N | L | N | N | N | N | N | N | L | N | N | N |
| P8 | N | N | N | N | N | N | L | N | N | N | L | N | N | N | L | N | N | N | N | N | N | N | N | N | N | N | N | L | N |
| P9 | N | N | N | N | L | N | N | N | N | N | L | N | N | N | N | N | N | N | N | N | N | N | L | L | N | N | N | L | N |
| P10 | N | N | N | N | N | L | L | N | N | N | N | N | N | N | N | N | N | L | N | N | N | N | L | L | N | N | N | L | N |
| P11 | N | N | N | N | N | L | L | N | N | N | L | L | N | N | N | N | N | N | N | N | N | N | N | L | N | N | N | N | N |
| P12 | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | L | N | N | N | N | N | N | L | N |
| P13 | N | N | N | N | L | N | L | N | N | N | L | N | N | N | N | N | N | L | N | L | N | M | L | N | N | N | N | N | N |
| P14 | N | N | N | N | N | N | L | N | N | N | L | N | N | N | N | N | N | L | N | N | N | M | L | N | N | N | N | N | N |
| P15 | N | N | N | N | N | N | N | N | N | L | N | N | N | N | N | N | N | L | N | N | N | N | L | N | N | N | N | L | N |
| P16 | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | N | L | N | N | N | N | L | N | N | N | N | L | L | N |



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## 10. Limit Plot Analysis of SPPS PAR data[18]

Criticality analysis was performed against every Boc-Fragment 3 response reported in Table 41 for the sixteen baseline and parametric experiments defined in **Table 26**. Twenty-nine responses are presented as control limit plots in Figure 9 through Figure 37: main peak purity (Section 10.1), one high impact response (Section 10.2), one medium impact response (Section 10.3), nineteen low impact responses (Section 10.4) and seven negligible impact responses (Section 10.5). Only three experiments produced a response above the Low/Medium impact boundary: P5 for des-Asn3 at RRT 0.64, and P13 and P14 for M+46 Da at RRT 1.01.

For each individual control limit plot, the gray region depicts the range of results observed in the baseline runs and the dashed line represents the baseline average. Hollow circles (○) are used for the baseline runs, solid circles (●) for the parametric experiments, and asterisks (*) for results where the analyte of interest was below the reporting threshold of the method used. Parametric experiments that exceed the Negligible/Low impact boundary are labelled in the limit plots and their associated results are given in the figure title.

### 10.1. Purity

The specification limit for Boc-Fragment 3 main peak purity is NLT 90.0 %Area. Across the sixteen experiments defined in **Table 26**, main peak purity ranged from 95.19 %Area (P5) to 97.17 %Area for B3 as reported in **Table 35** and **Table 36**. The three baseline runs gave 96.54 %Area for B1, 96.89 %Area for B2 (large scale) and 97.17 %Area for B3, giving a baseline average of 96.8667 %Area and a baseline range of 96.54 %Area - 97.17 %Area. B2 charged 200 g of CTC resin (large scale baseline) against the 45 g baseline and returned a result inside the baseline range, confirming that main peak purity is not scale-sensitive over the range studied.

No experiment approached the Low/Medium impact boundary of 93.4333 %Area. The lowest observed result retains 5.19 %Area of margin to the NLT 90.0 %Area specification and 1.76 %Area of margin to the Low/Medium impact boundary, and the remaining thirteen experiments were at or above 96.23 %Area. Main peak purity is therefore a low impact response across the



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full parameter space explored, the PARs established in Table 31 are sufficient to control it, and no controls beyond the established PARs are required.

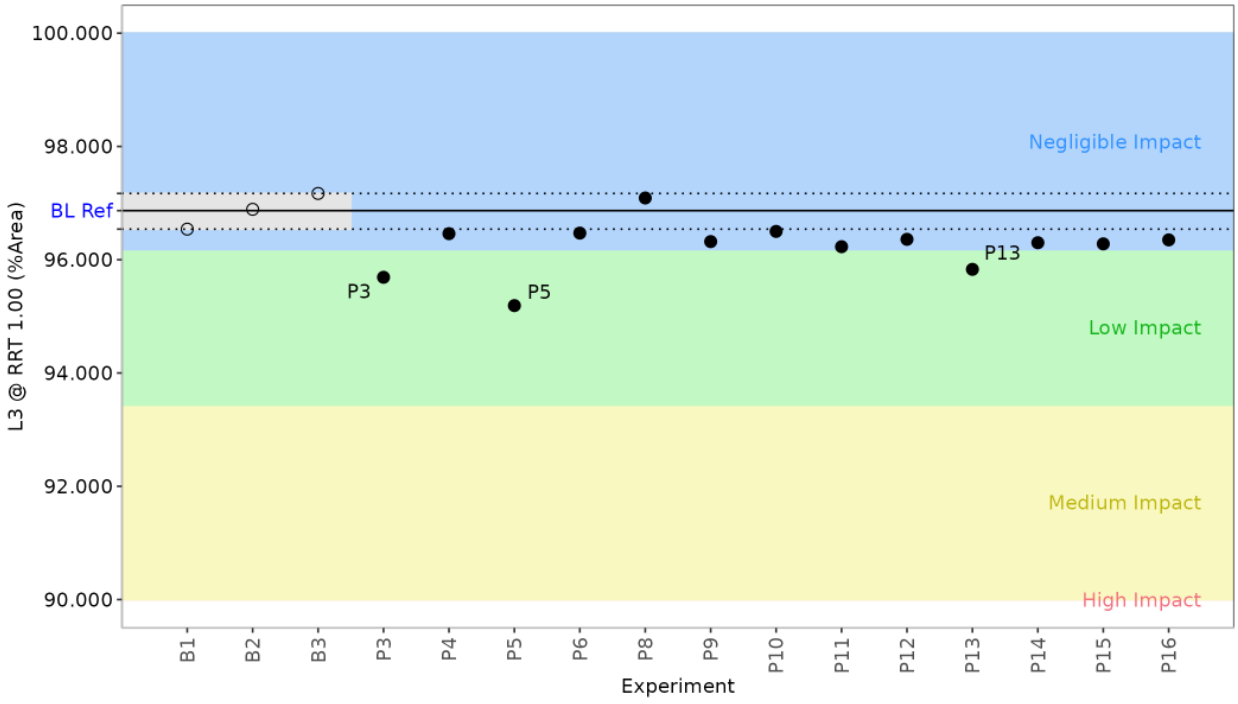



**Figure 9**. Limit plot for L3 @ RRT 1.00

### 10.2. High impact response: L3 des-Asn3 @ RRT 0.64

des-Asn(Trt)3 reached 0.6384 % in P5, exceeding the 0.5 % medium/high risk threshold. P5 represents a low-deFmoc-conversion challenge in which four Asn3 deFmoc parameters were simultaneously set at their lower PAR edges. No experimental point exists between the set point and the P5 corner, so the conforming boundary must be estimated by linear extrapolation between the two conditions.

The impurity is at RRT 0.64, confirmed by mass: des-Asn(Trt)3 requires loss of Asn plus Trt, matching the tabulated delta mass of −356.15. des-Asn3 arises from incomplete Fmoc removal at Thr4: residual Fmoc blocks the subsequent Asn3 coupling and the deletion propagates through the remaining sequence. Formation is a direct function of deFmoc conversion, controlled by the same four parameters varied in P5. Because the four parameters act through this common



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mechanism, they are treated as a single coupled excursion rather than as four independent effects. There is no evidence of baseline variability and no evidence of a scale effect for this impurity. Because the baseline results for this impurity were all below the reporting threshold, the 0.10 % bound is taken as the baseline anchor for the calculations below.

- Observed des-Asn3 (0.6384%) − 0.10%(baseline) equals to 0.5384 %,
- allowable rise to the 0.49[21] %Area as the conservative PAR range restriction from the baseline equals to 0.39 %.
- Linear proportional model: 0.39 % / 0.5384 % equals to 72.4 % of excursion distance.

The linear interpolation is carried forward as the conservative case. The interpolation targets an internal action limit of 0.49 %Area, which is below the NMT 0.50 %Area specification for an unspecified impurity, so that the narrowed PAR edge holds des-Asn3 just below the specification limit with a deliberate margin against an out-of-specification result, and returns des-Asn3 to a medium impact classification under the Appendix VI criticality definitions (Table **30**).

Table 30. Parameter restriction summary for L3 des-Asn3 Impurity[22]

| Parameter | Set point | Min value studied | Min value interpolated |
|---|---|---|---|
| deFmoc temperature (pre-Asn3) | 25.0 °C | 13.0 °C | 16.3 °C |
| Piperidine composition (pre-Asn3) | 20.0 % | 15.0 % | 16.4 %<br>(use -6σ since it is more restrictive: 16.6 %) |

[21] For conservative PAR restriction, the target limit is set to 98% of 0.50%, which is 0.49%. This is a technical decision.

[22] Riccardo Candeago Signals notebook: D4U-D03050-132



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| Parameter | Set point | Min value studied | Min value interpolated |
|---|---|---|---|
| Piperidine in DMF (pre-Asn3) | 7.44 kg/kg | 6.55 kg/kg | 6.78 kg/kg |
| Total deFmoc time (pre-Asn3) | 120 min | 60 min | 77 min |

Because the four parameters moved together, these define a coupled corner, not four independently validated ranges. des-Asn3 is well controlled at set point, undetected above 0.10 % across three baseline runs including large scale — a margin of at least 5× below the 0.5 % threshold. The impurity rises sharply only at the low-deFmoc corner represented by P5. At the narrowed PAR edges, the linear model predicts des-Asn3 at 0.5 %Area, which coincides with the NMT 0.5 %Area specification and leaves no margin to specification at the PAR edge; The response is classified as medium impact under the Appendix VI criticality definitions. The narrowed PARs are supported by the current data and are appropriate for process validation at target conditions; confirmatory data at the PAR edges will be generated as part of ongoing process verification to further strengthen the justification for operation at the range limits. Because these edges are derived by interpolation rather than demonstrated experimentally, any subsequent widening of the four pre-Asn3 deFmoc parameters should be supported by a confirmatory experiment run at the proposed PAR corner with the remaining parameters held at set point.


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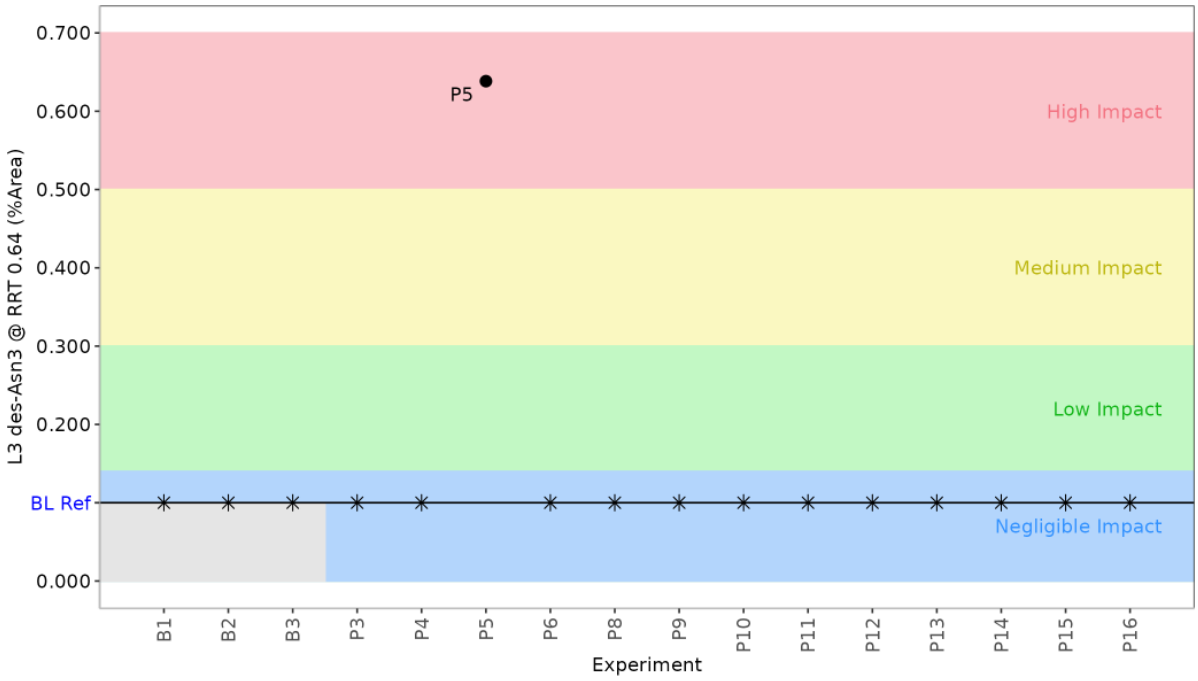



**Figure 10**. Limit plot for L3 des-Asn3 @ RRT 0.64

### 10.3. Medium impact response: L3 M+46 Da @ RRT 1.01

The M+46 Da response at RRT 1.01 exceeded the Low/Medium impact boundary of 0.3000 %Area in two experiments, P13 (0.3384 %Area) and P14 (0.3135 %Area), and remained in the low impact band in P4 (0.1516 %Area) and P9 (0.1672 %Area) (**Figure 11**). All baseline runs were at the reporting threshold. Both medium impact results are well below the specification of NMT 0.500 %Area, and the margin between the observed maximum and the specification is approximately 0.16 %Area. The univariate model did not identify any medium or high impact parameters for this response, so the classification is driven by the observed experimental values rather than by a modelled parameter effect. The response is retained in the deconvolution analysis in Section 11 to confirm which of the perturbed parameters in P13 and P14 are responsible before the associated PARs are finalized.

Comparison of the oxidation-step set points in **Table 26** isolates iodine contact time as the responsible parameter. P11 and P13 share the same high-oxidation direction (iodine 5.60 and 5.62 eq at +6σ, iodine-solution-prep DMF 3.30 and 3.31 kg/kg and oxidation-reaction DMF 9.98 and 9.93 kg/kg at −6σ), and P12 and P14 share the same low-oxidation direction (iodine 4.36 eq



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at −6σ, prep DMF 4.25 and 4.27 kg/kg and reaction DMF 12.68 and 12.79 kg/kg at +6σ). M+46 Da remained at the reporting threshold in P11 and P12 but rose to 0.3384 %Area and 0.3135 %Area in P13 and P14. Iodine stoichiometry and oxidation-solution concentration are therefore not the drivers of this response; the only parameters that differ between the two pairs are the iodine solution transfer time and the oxidation exposure time.

The two elevated results sit at opposite extremes of the iodine exposure profile. P13 combined a 100 min transfer with a 30 min oxidation rection time, giving approximately 130 min of total iodine contact against the 50 min set point (iodine transfer time w/ no oxidation reaction time), which favors cumulative over-oxidation of the peptide. P14 used a 20 min transfer, delivering the iodine charge at approximately 0.22 eq/min against the 0.10 eq/min set-point rate, which creates a transient local iodine excess during addition even though the total charge was at its −6σ minimum and the oxidation solution was at its most dilute. Because the response is elevated both by prolonged cumulative exposure and by rapid addition, a one-sided upper limit on transfer time would not control it.

Limiting the iodine solution transfer time to 40–100 min is directly supported by the experimental set. P11 (40 min transfer, approximately 40 min total iodine contact) and P12 (60 min transfer, approximately 60 min total contact) held M+46 Da at the baseline level while iodine charge and both DMF quantities were simultaneously at their ±6σ limits, so the proposed range is bracketed on both sides by experiments in which every other oxidation parameter was stressed to its extreme. Constraining the transfer time to 40–100 min, with total iodine contact time (transfer plus exposure) not exceeding 100 min, keeps the process within the demonstrated region, preserves the approximately 0.16 %Area margin to the NMT 0.500 %Area specification, and returns the response to low impact without narrowing the PARs established for iodine charge, oxidation-solution concentration, or oxidation temperature.


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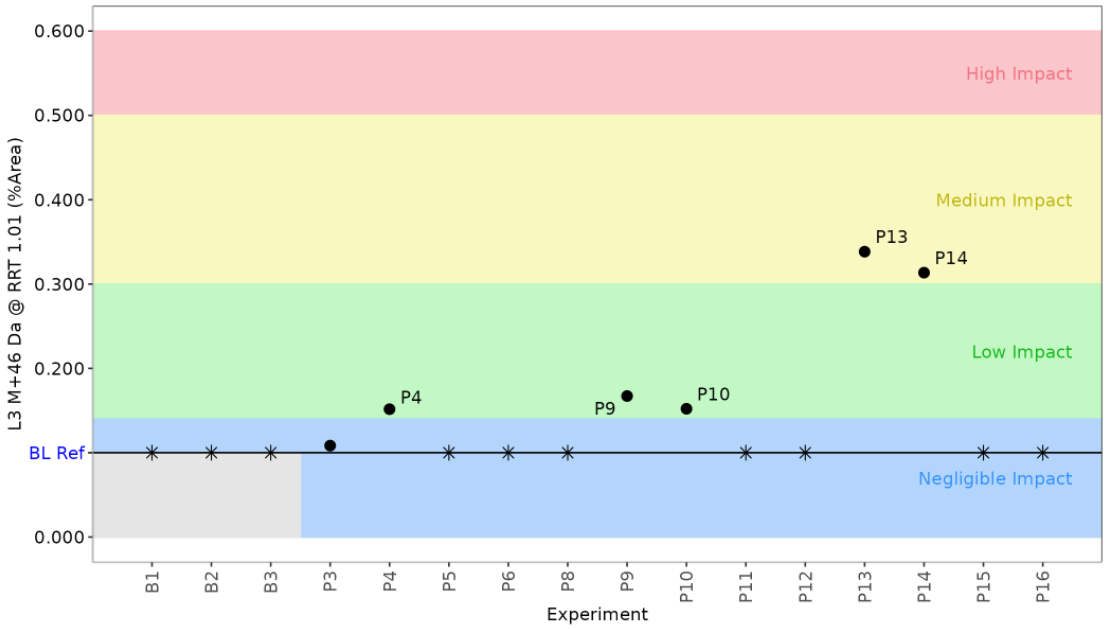



**Figure 11**. Limit plot for L3 M+46 Da @ RRT 1.01

### 10.4. Low impact responses

Nineteen responses had at least one experiment above the Negligible/Low impact boundary but no experiment above the Low/Medium impact boundary (Figure 12 through Figure 30). In every case the highest observed result retains substantial margin to the specification, and in most cases the experiments in the low impact band are those carrying the most aggressive parameter settings in **Table 26**. Against the NMT 0.500 %Area specifications the largest single excursion was L3 Ac-2-10 @ RRT 0.75 in P3 (0.3589 %Area), followed by L3 des-Thr9 @ RRT 0.90 in P6 (0.3136 %Area). L3 Dimer @ RRT 1.34 reached 1.1250 %Area in P8, but that response is assessed against a specification of NMT 1.500 %Area and remains below its Low/Medium impact boundary of 1.2399 %Area. The acetyl-capped 2-10 response at RRT 0.75 was above its Negligible/Low boundary of 0.2647 %Area in ten of the sixteen experiments, including two of the three baseline runs, which reflects a baseline level close to the criticality threshold (baseline average 0.2385 %Area) rather than a parametric effect; no experiment exceeded the Low/Medium impact boundary of 0.3693 %Area, so no medium impact parameter is assigned for



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this response. The established PARs are sufficient to control all of the responses in this group and no additional controls are proposed. The individual limit plots are given below.

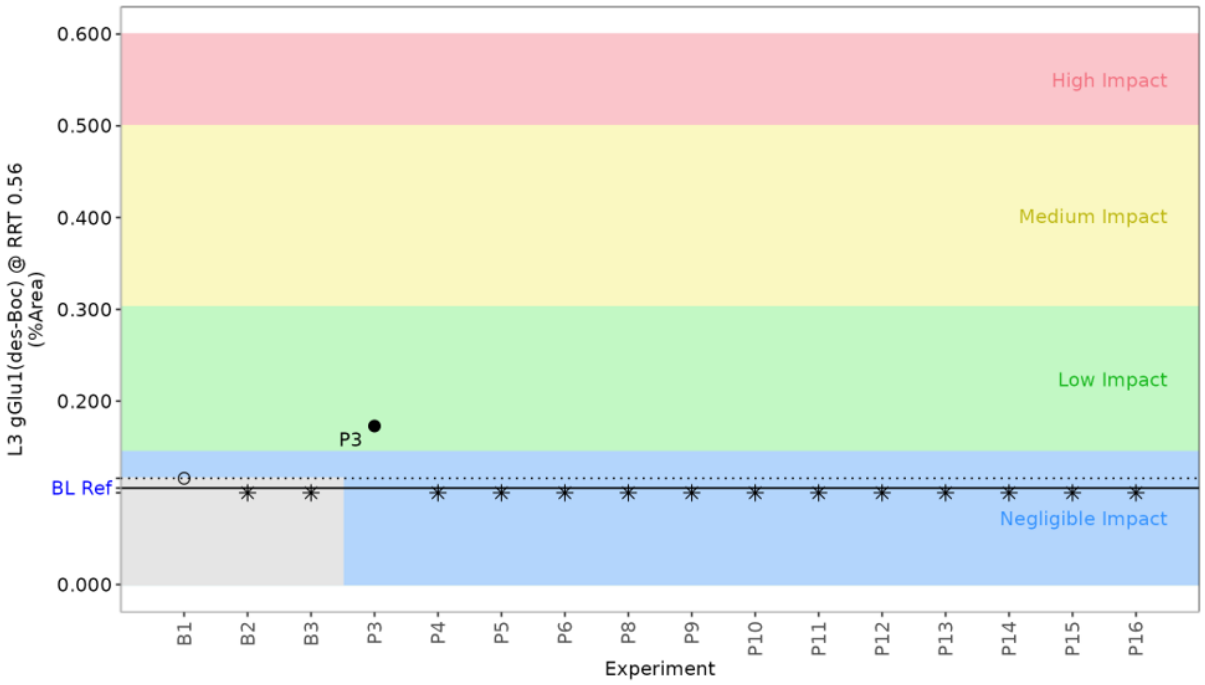



**Figure 12**. Limit plot for L3 gGlu1(des-Boc) @ RRT 0.56

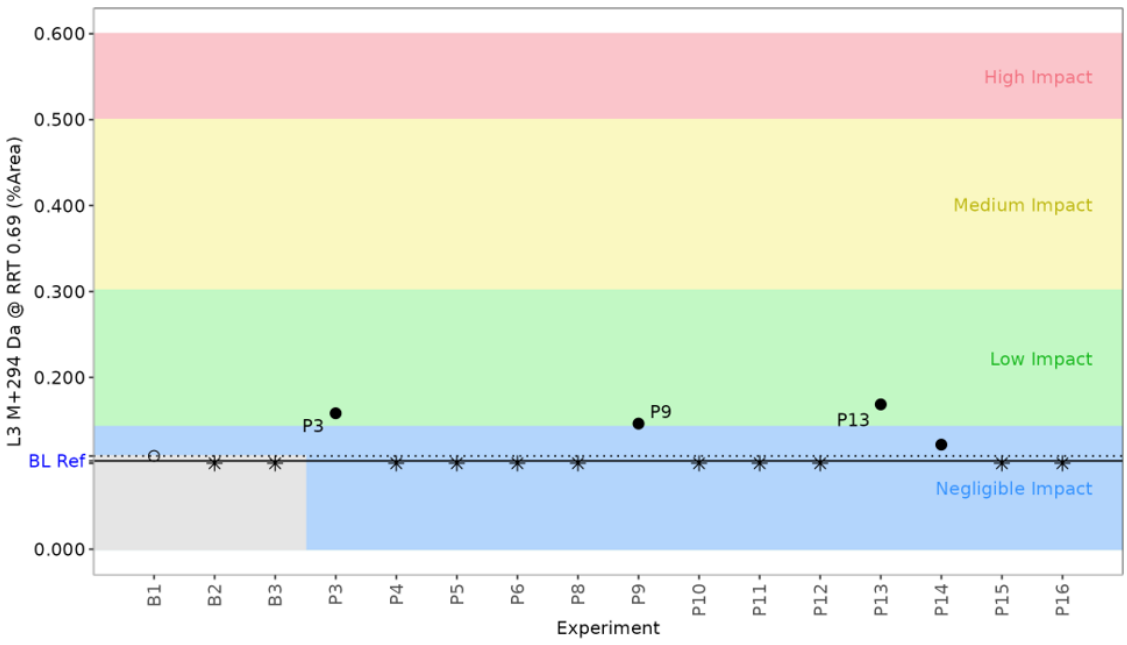



**Figure 13**. Limit plot for L3 M+294 Da @ RRT 0.69



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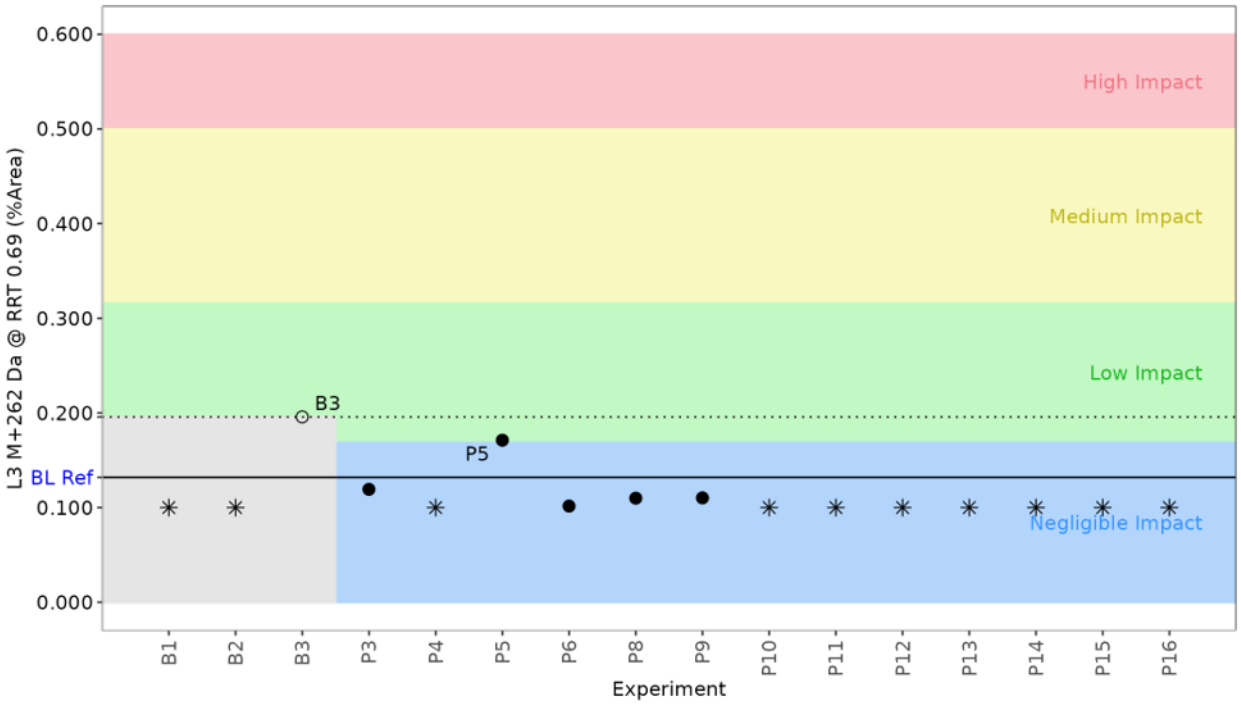



**Figure 14**. Limit plot for L3 M+262 @ RRT 0.69

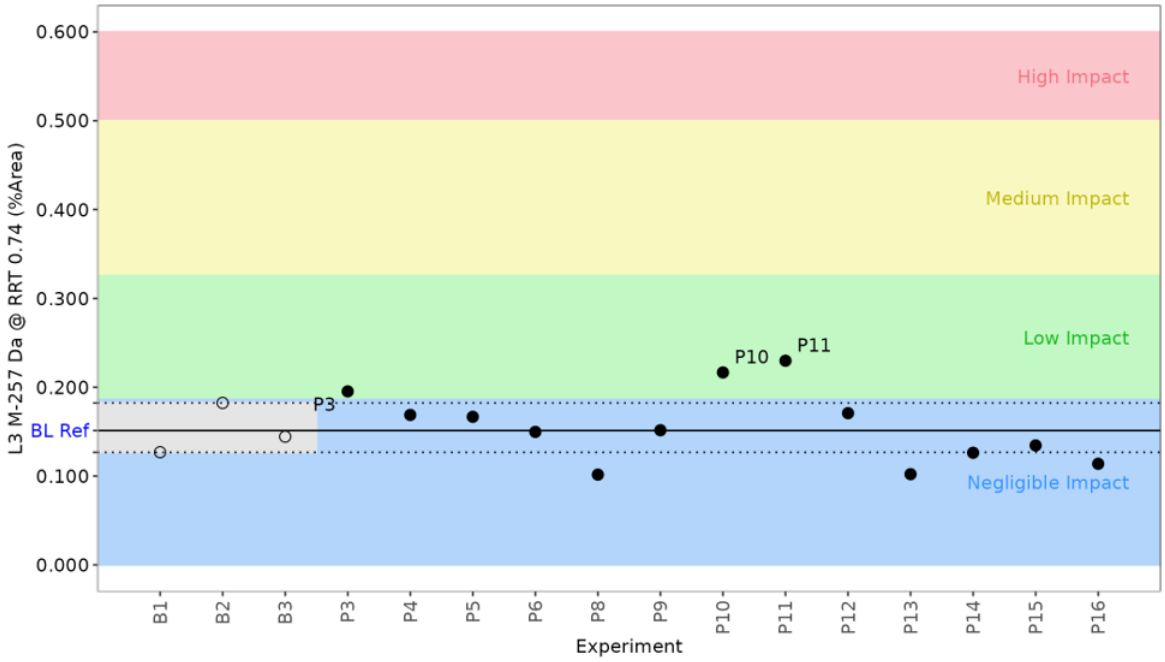



**Figure 15**. Limit plot for L3 M-257 @ RRT 0.74



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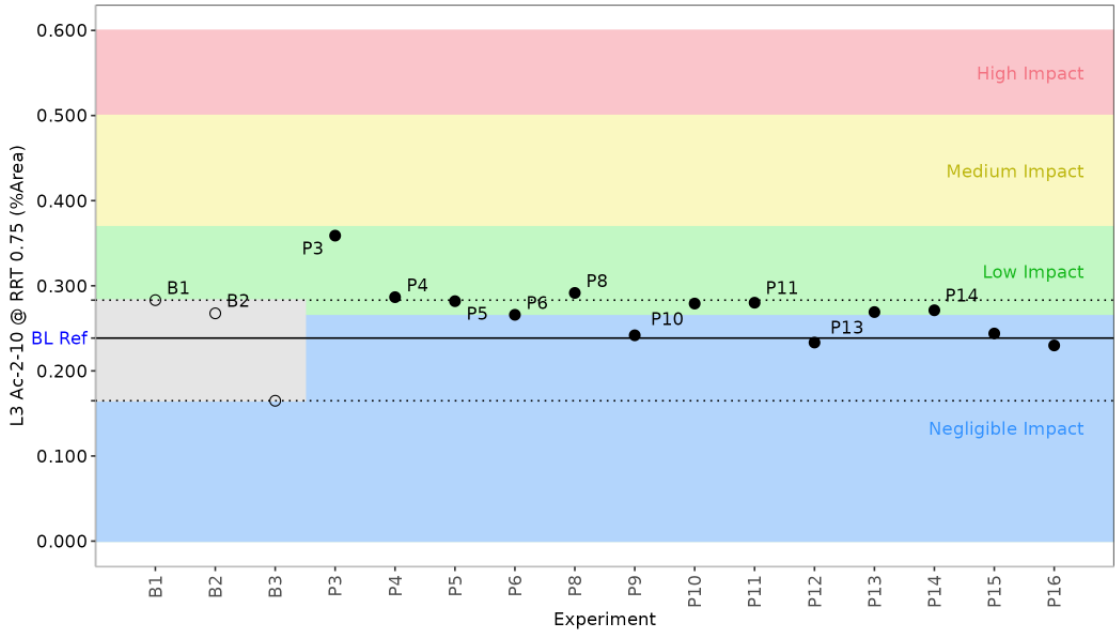



**Figure 16**. Limit plot for L3 Ac-2-10 @ RRT 0.75

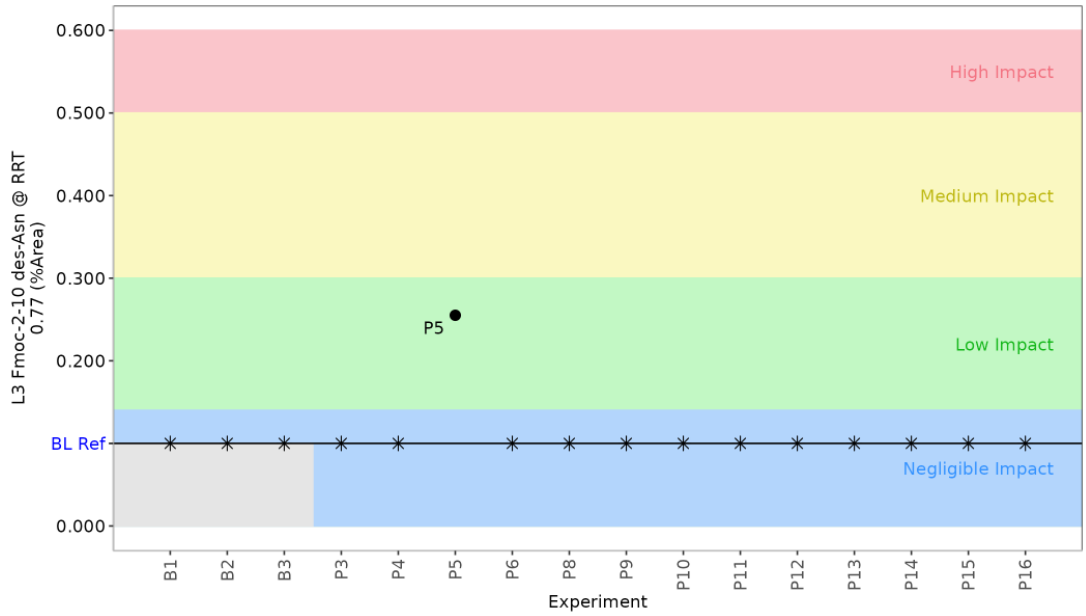



**Figure 17.** Limit plot for L3 Fmoc -2-10 des-Asn @ RRT 0.77



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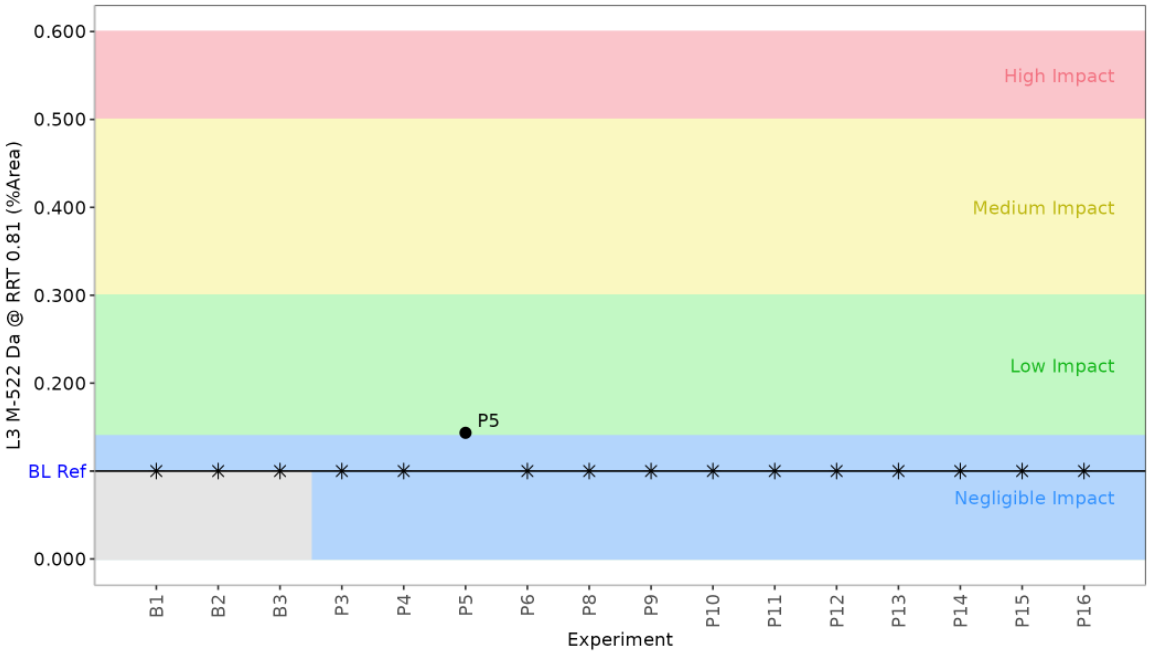



**Figure 18**. Limit plot for L3 M-522 Da @ RRT 0.81

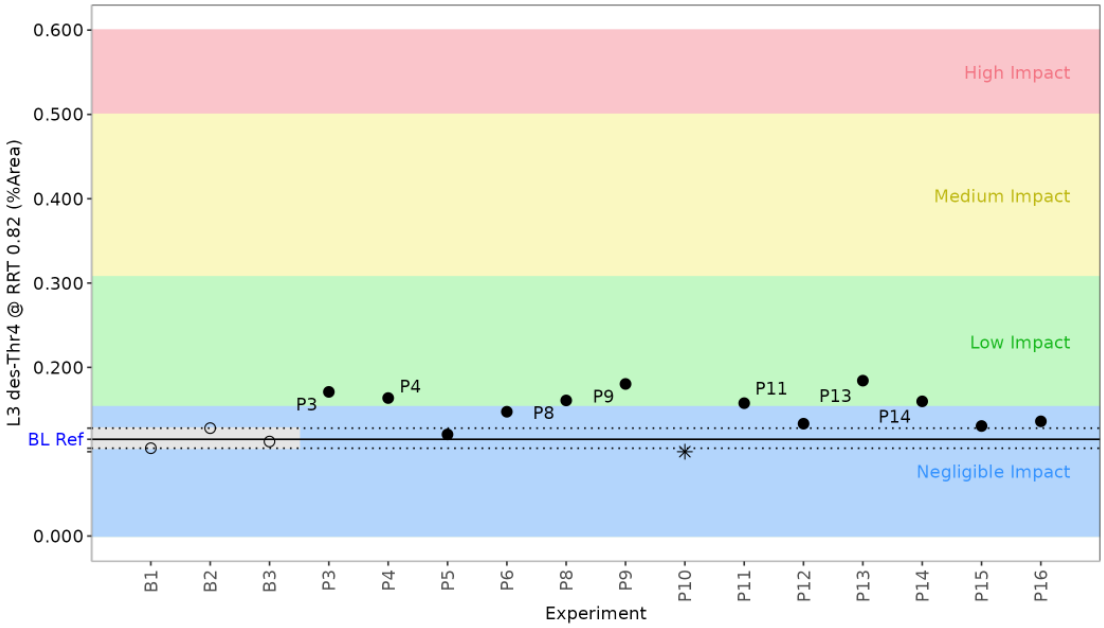



**Figure 19**. Limit plot for L3 des-Thr4 @ RRT 0.82



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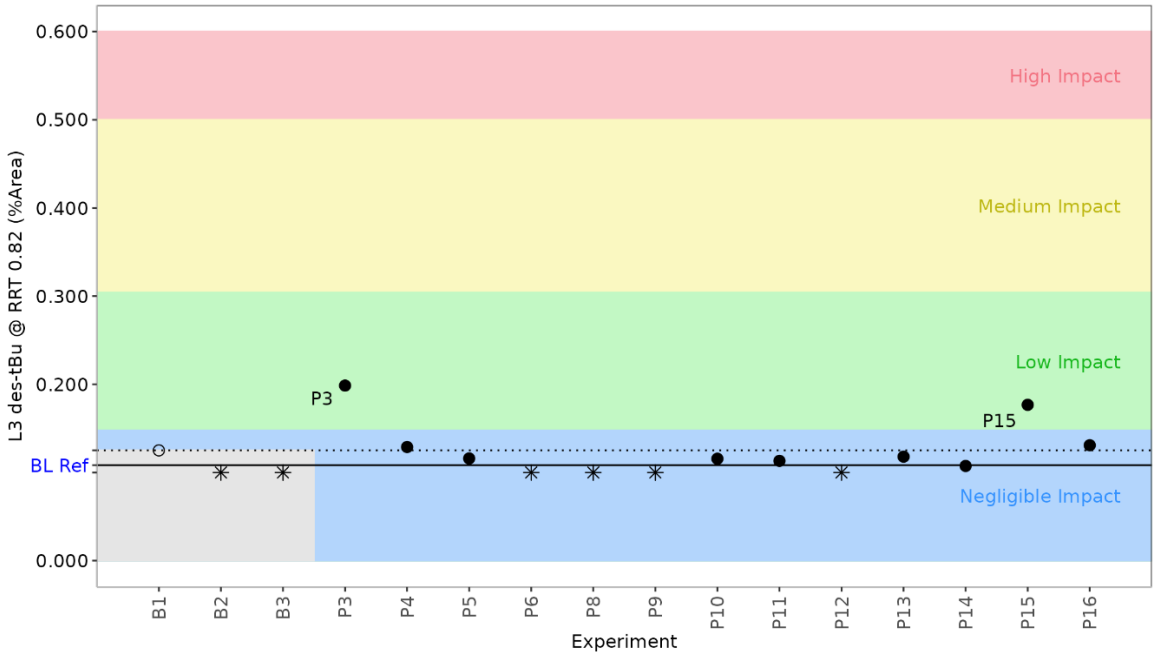



**Figure 20**. Limit plot for L3 des-tBu @ RRT 0.82

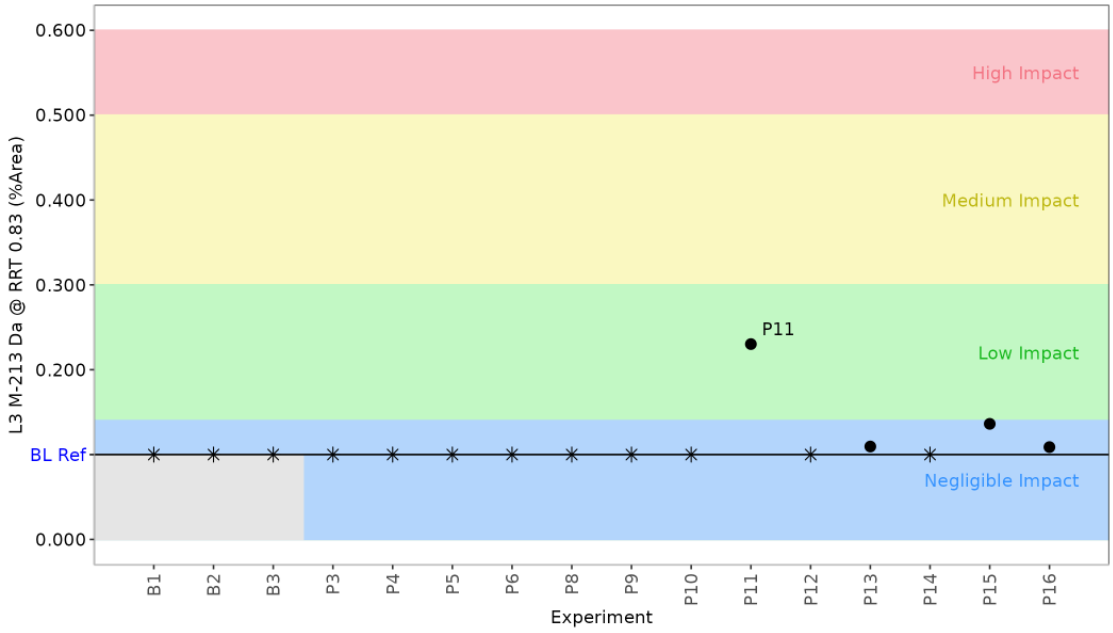



**Figure 21**. Limit plot for L3 M-213 @ RRT 0.83



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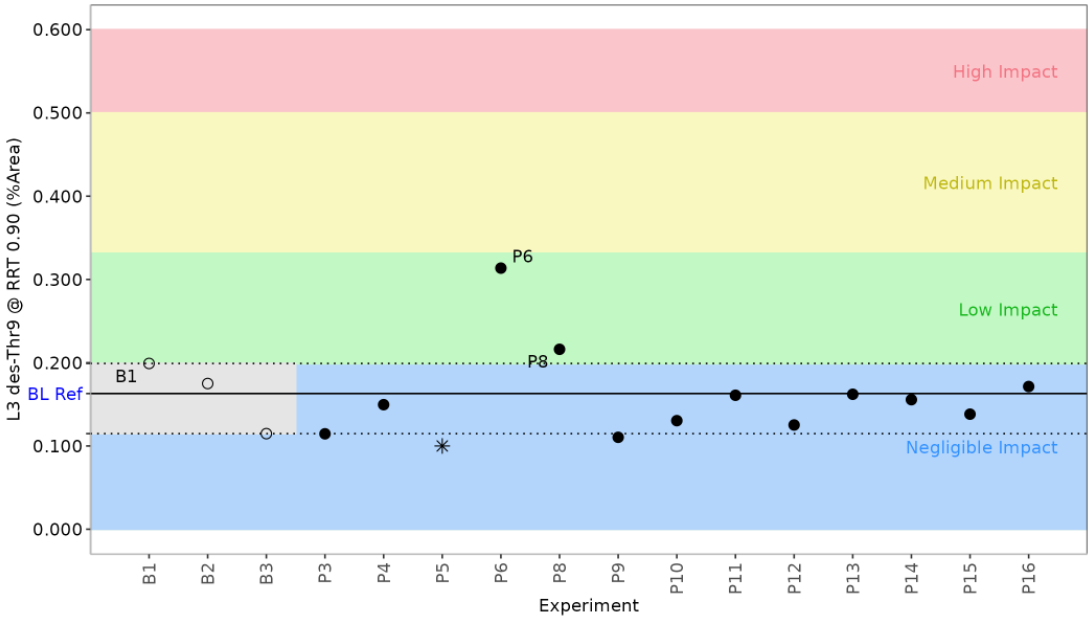



**Figure 22**. Limit plot for L3 des-Thr9 @ RRT 0.90

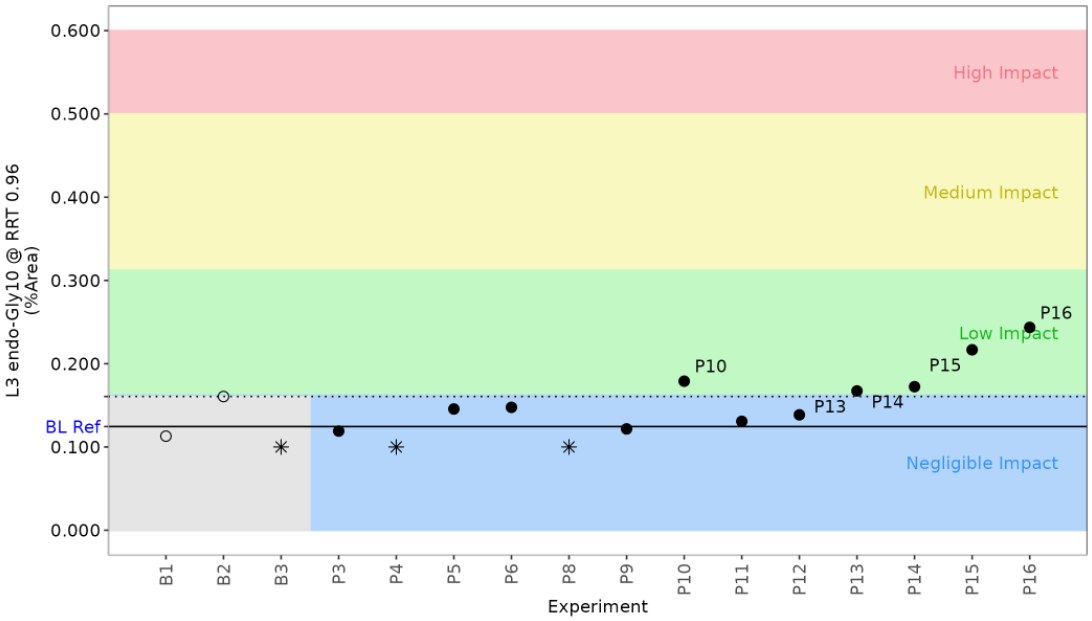



**Figure 23**. Limit plot for L3 endo-Gly10 @ RRT 0.96



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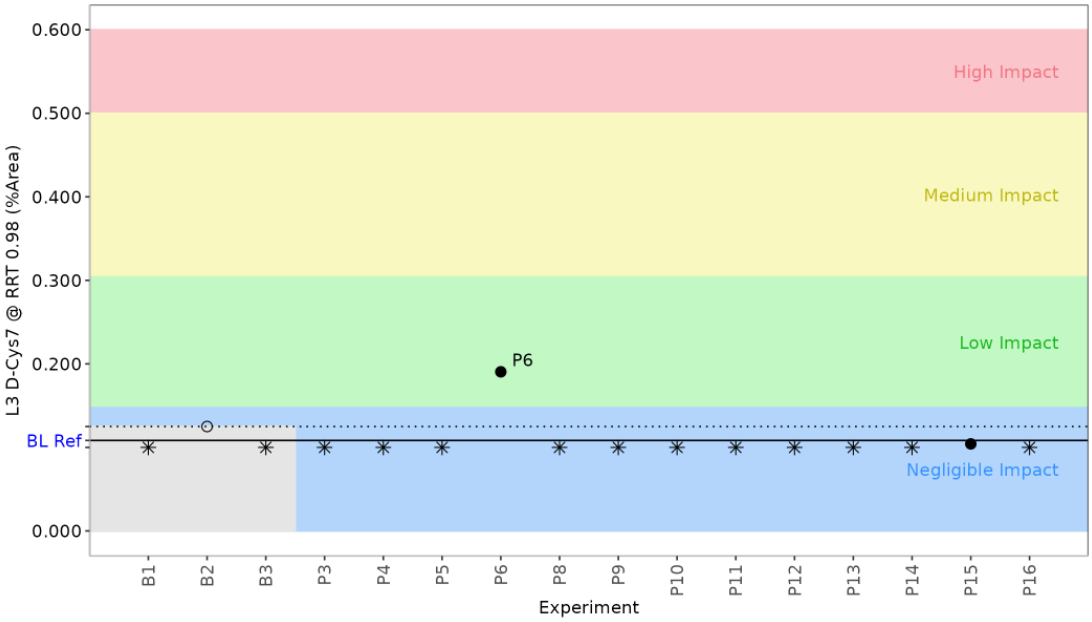



**Figure 24**. Limit plot for L3 D-Cys7 isomer @ RRT 0.98

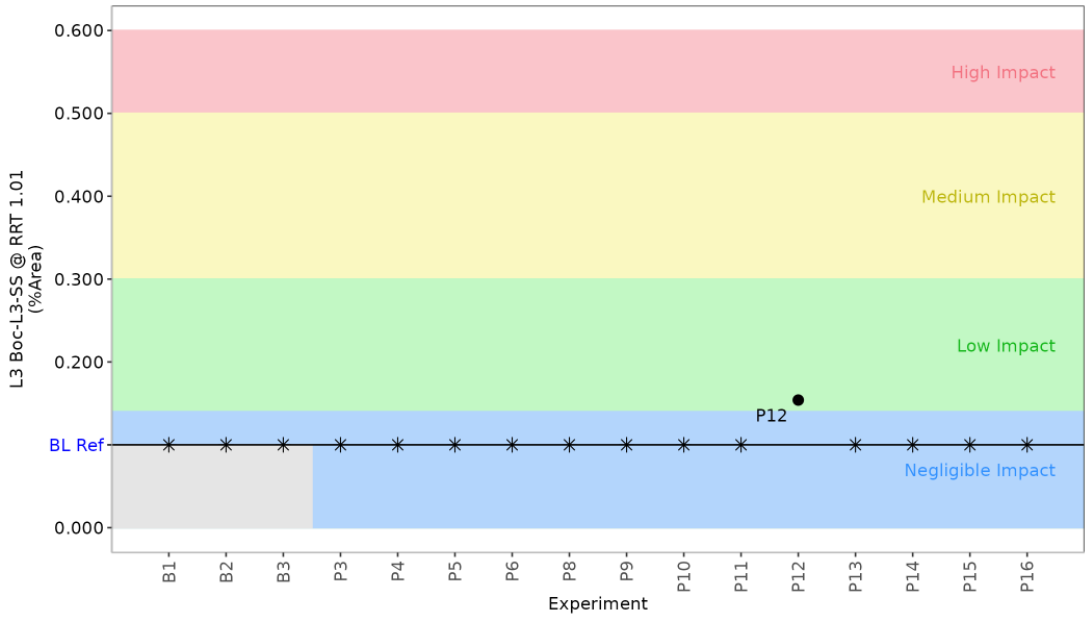



**Figure 25**. Limit plot for L3 Boc-L3-SS @ RRT 1.01



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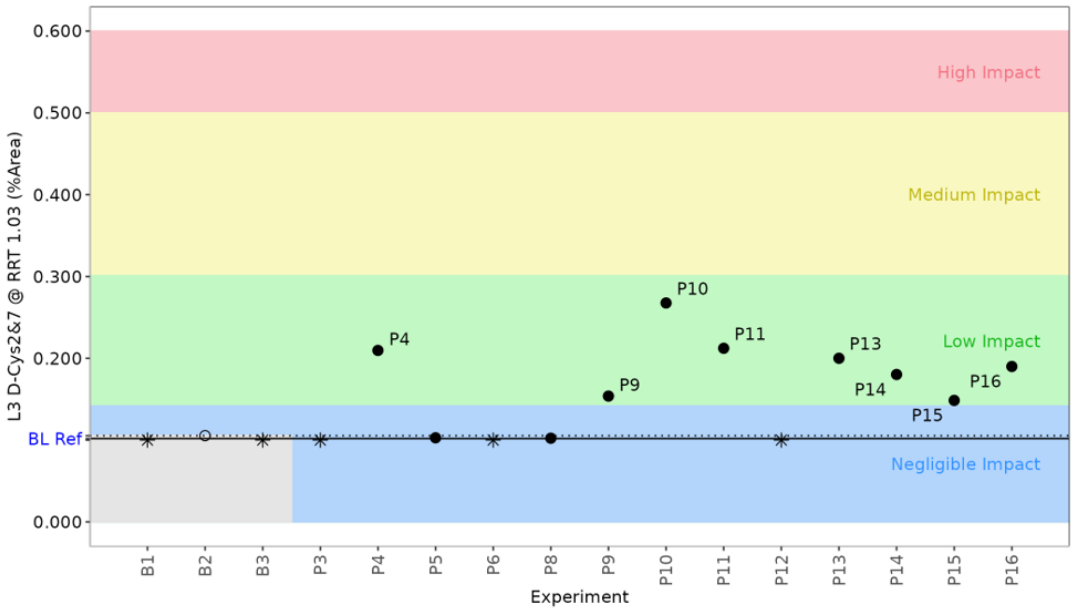



**Figure 26**. Limit plot for L3 isomer(D-Cys2&7) @ RRT 1.03

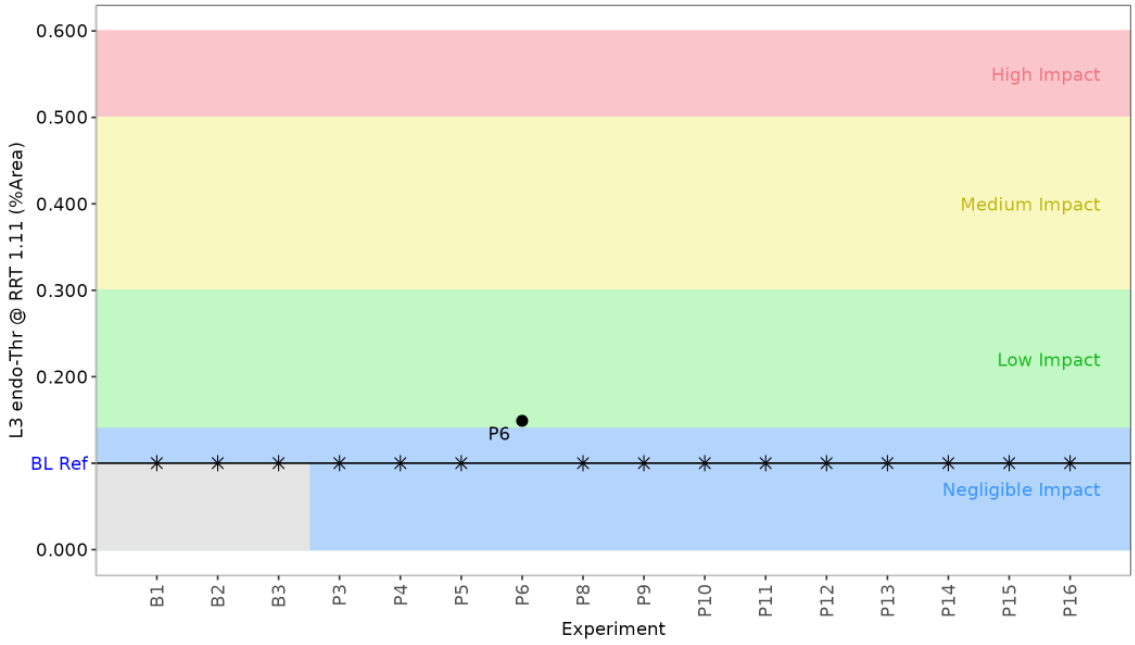



**Figure 27**. Limit plot for L3 endo-Thr @ RRT 1.11



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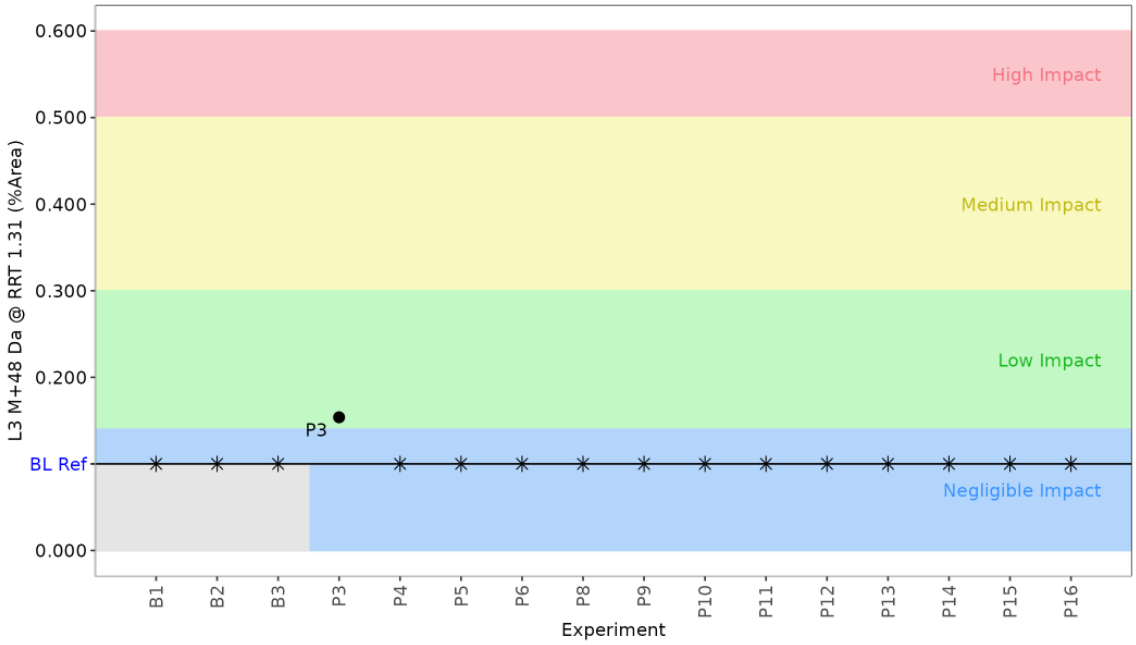



**Figure 28**. Limit plot for L3 M+48 Da @ RRT 1.31

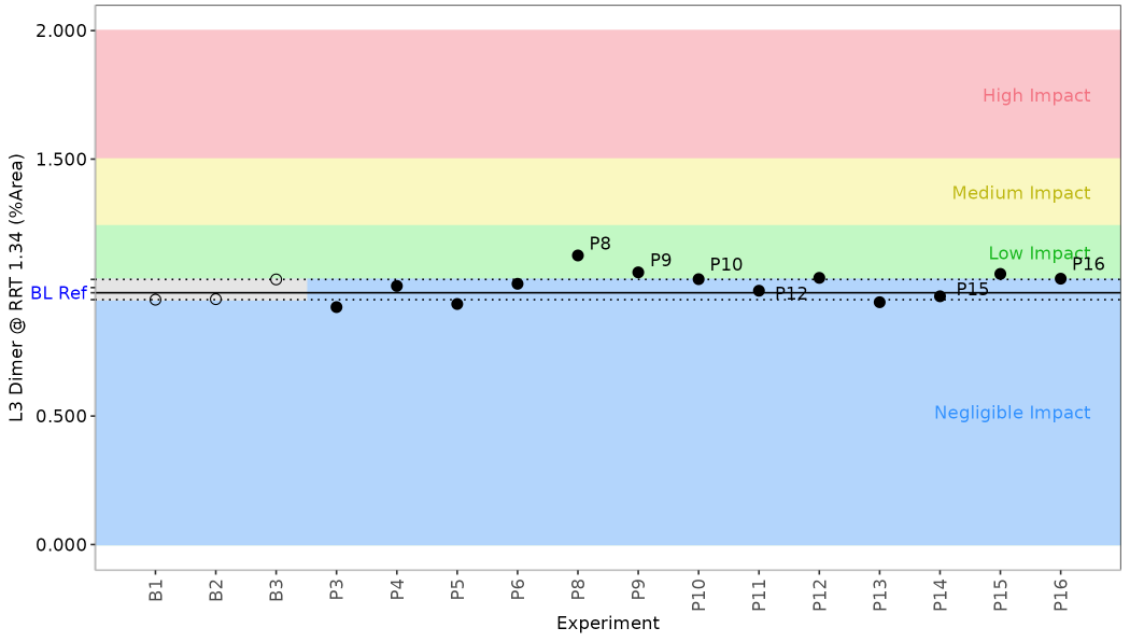



**Figure 29**. Limit plot for L3 Dimer @ RRT 1.34



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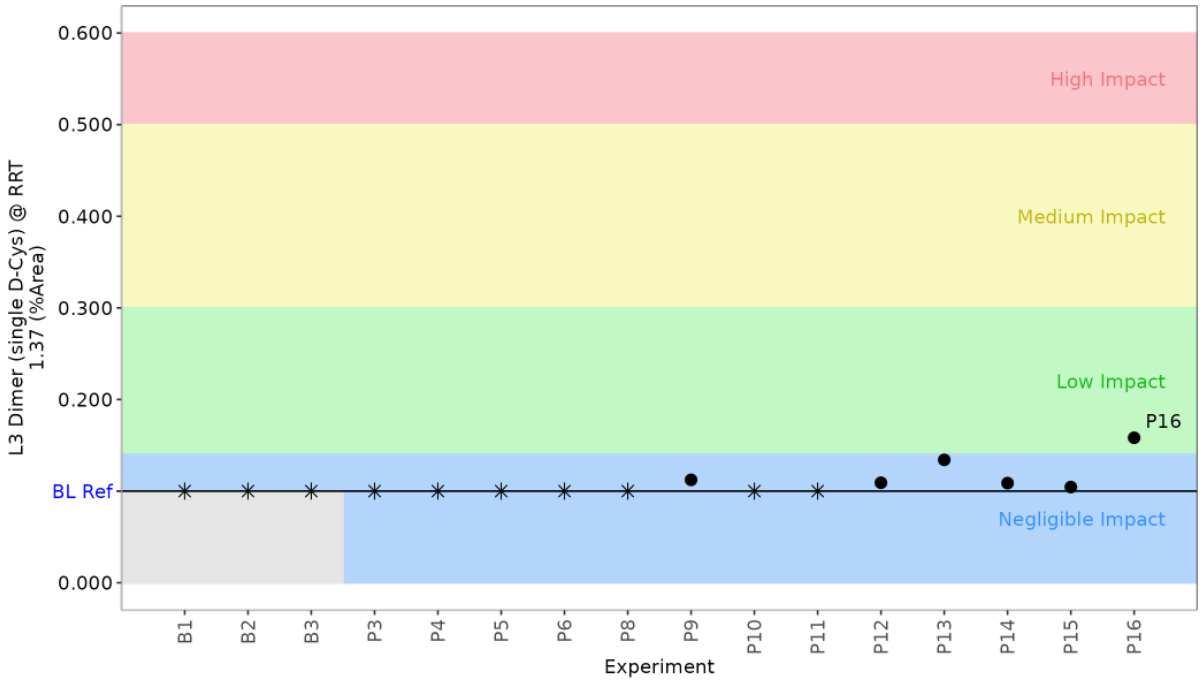



**Figure 30**. Limit plot for L3 Dimer (D-Cys2) @ RRT 1.37

### 10.5. Negligible impact responses

Seven responses were negligible impact in every baseline and parametric experiment, with all results at or below the Negligible/Low impact boundary; in most cases every result was at the reporting threshold of the method. The dimer (D-Cys7) response at RRT 1.39 was likewise negligible in all experiments and no limit plot is presented for it because no result was above the reporting threshold. These responses are not sensitive to any of the parameters perturbed in the study and require no control beyond the established PARs. The individual limit plots are given below for completeness in Figure 31 through Figure 37.



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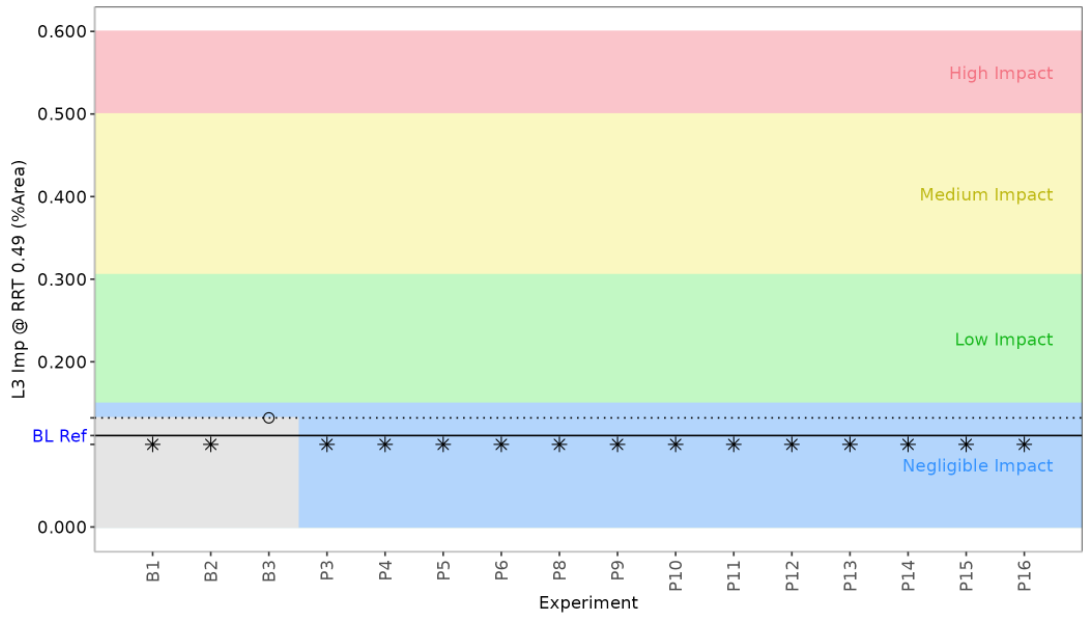



**Figure 31**. Limit plot for L3 Imp @ RRT 0.49

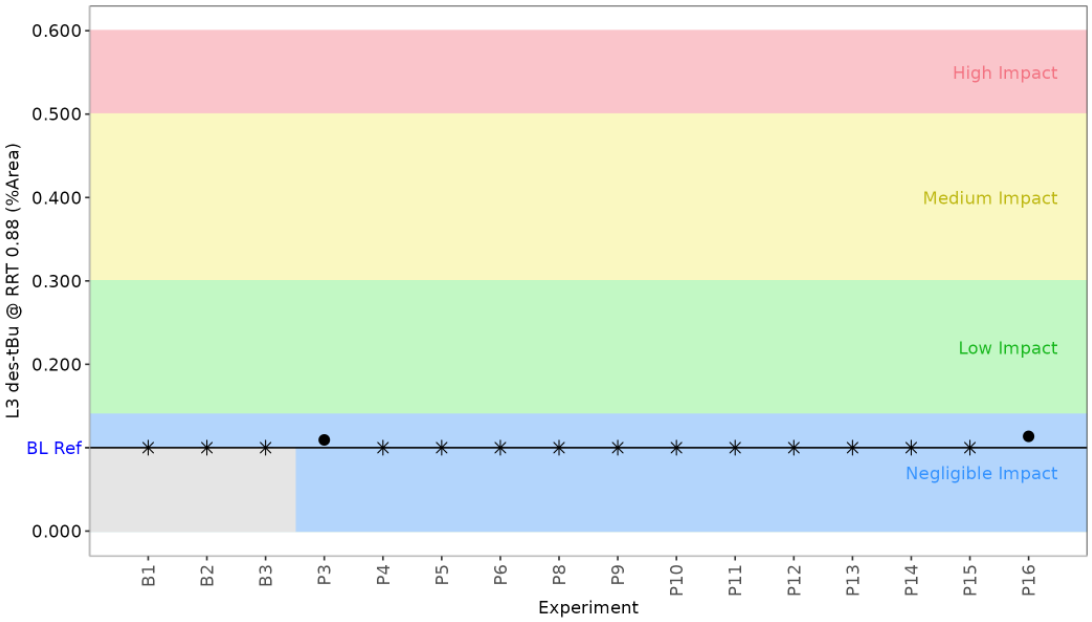



**Figure 32**. Limit plot for L3 Des-tBu @ RRT 0.88



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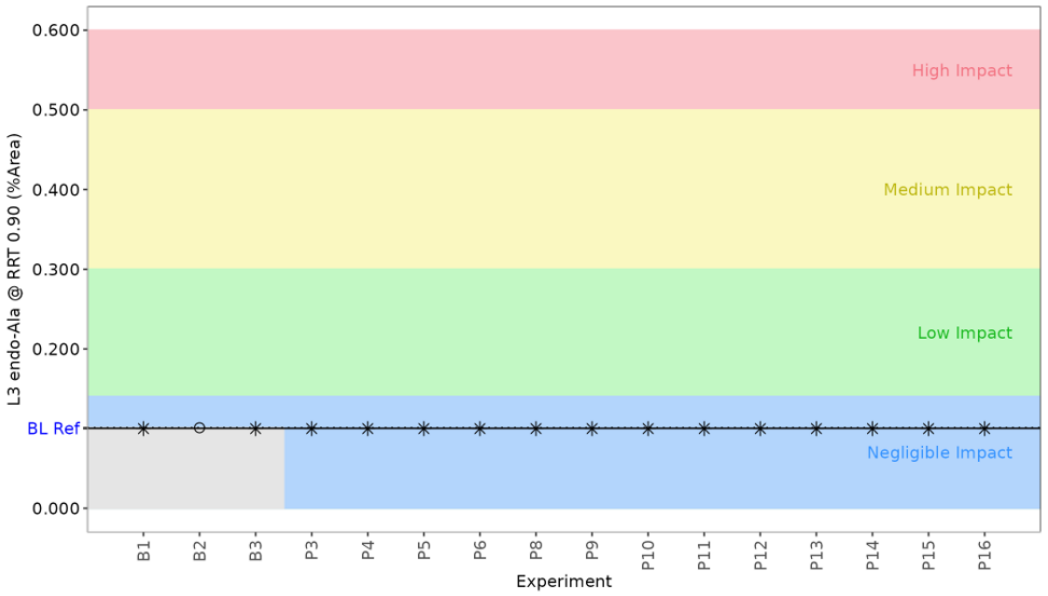



**Figure 33**. Limit plot for L3 endo Ala @ RRT 0.90

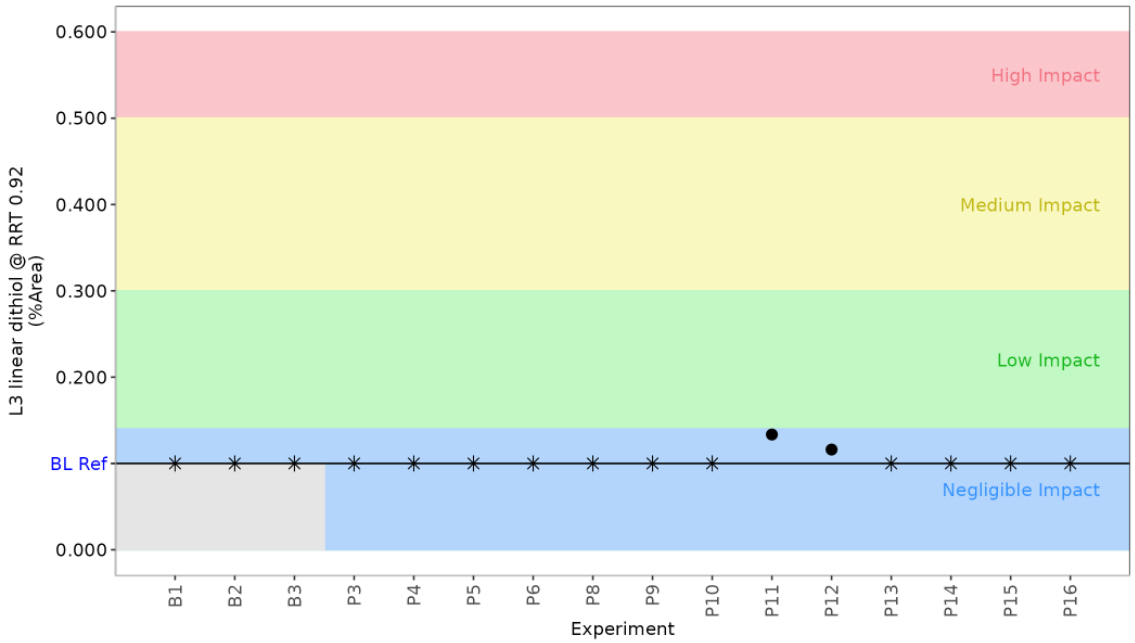



**Figure 34**. Limit plot for L3 linear dithiol @ RRT 0.92



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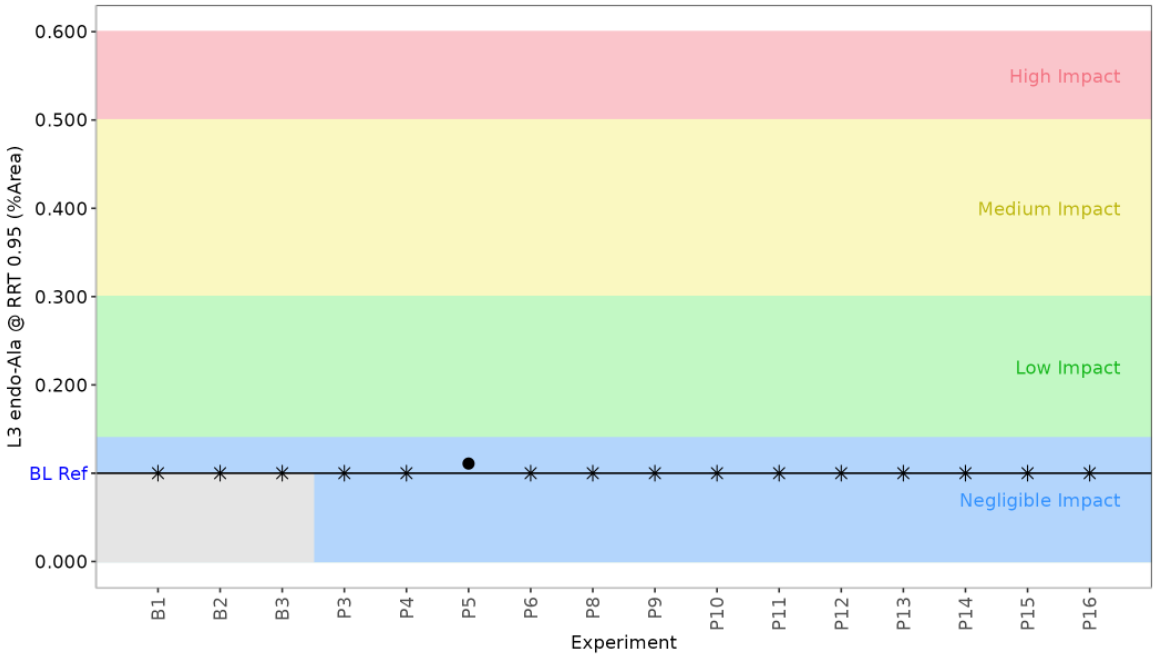



**Figure 35**. Limit plot for L3 endo Ala @ RRT 0.95

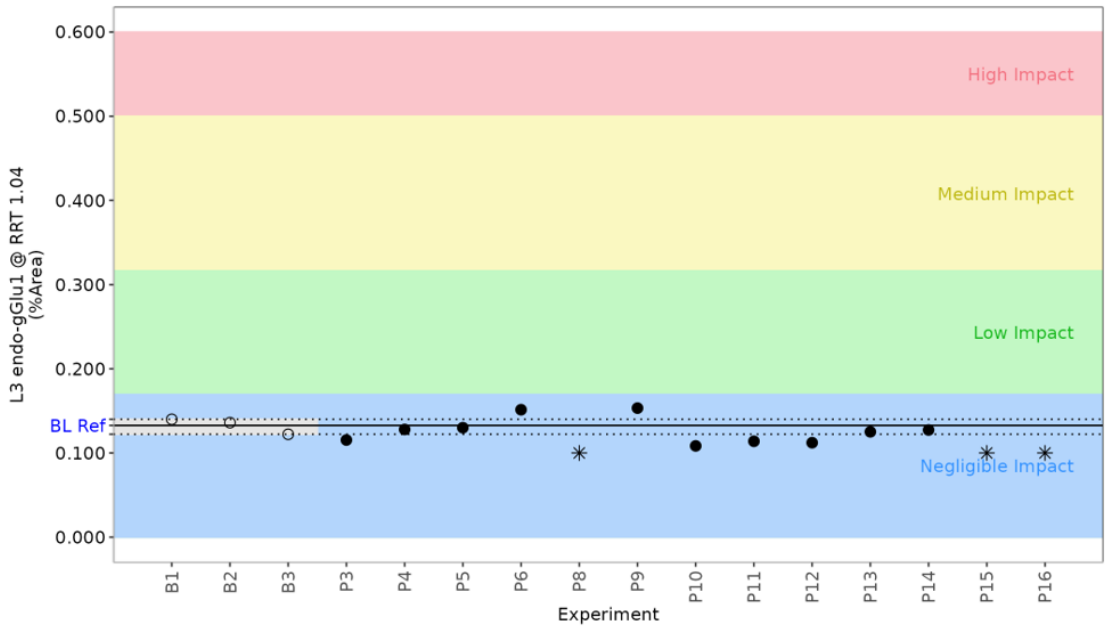



**Figure 36**. Limit plot for L3 endo-γGlu1 @ RRT 1.04



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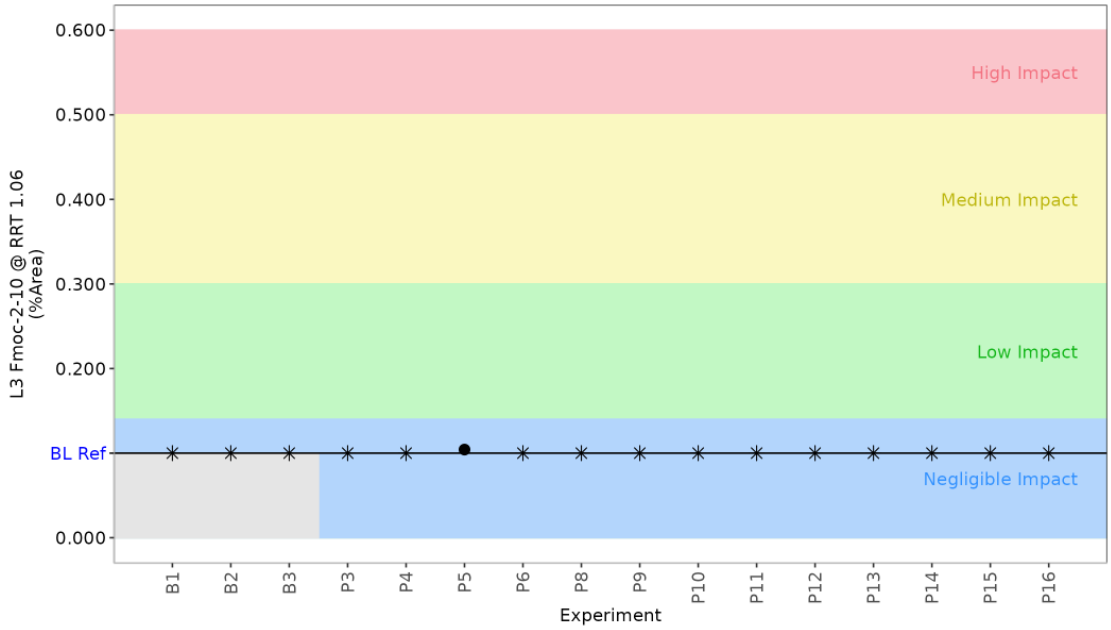



**Figure 37**. Limit plot for L3 Fmoc-2-10 @ RRT 1.06

## 11. Deconvolution analysis and intermediate parameter criticality analysis

As shown by the limit plots in Section 10, two responses exceeded a medium or high impact threshold across the sixteen PAR experiments: L3 des-Asn3 @ RRT 0.64, which reached 0.6384 %Area in P5 and was classified as high impact, and L3 M+46 Da @ RRT 1.01, which reached 0.3384 %Area in P13 and 0.3135 %Area in P14 and was classified as medium impact. All remaining responses were low or negligible and are controlled by the PARs established in Table 31. No dedicated follow-up deconvolution experiments were executed for this program; both elevated responses were deconvoluted analytically from within the existing sixteen-experiment set — P5 by interpolation between the set point and the P5 corner, and P13 and P14 by paired comparison against the oxidation experiments P11 and P12. This section summarizes each deconvolution and the resulting parameter criticality assignments.

### 11.1. Analysis of P5: high impact response L3 des-Asn3 @ RRT 0.64

P5 returned L3 des-Asn3 @ RRT 0.64 at 0.6384 %Area, above the 0.5000 %Area medium/high impact boundary and was therefore classified as high impact. P5 was a low-deFmoc-conversion challenge in which four Cys2 Fmoc-deprotection parameters — deFmoc temperature, piperidine



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composition, piperidine-in-DMF charge quantity and total deFmoc time — were simultaneously set at their lower PAR edges, with capping time floored at 0 min.

Because all four parameters act on des-Asn3 through the same mechanism, incomplete Fmoc removal at Thr4 blocking the subsequent Asn3 coupling, they were treated as a single coupled excursion rather than as four independent effects, and no experiment exists between the set point and the P5 corner. Deconvolution was therefore performed by interpolation, targeting the 0.49 %Area internal action limit as described in Section 10.2, which gives narrowed lower limits of 16.3 °C, 16.6 %, 6.77 kg/kg and 77 min (Table **30**).

All four deFmoc parameters for Fmoc on Thr4 are classified as CPPs at these narrowed lower limits. Because the limits are derived by interpolation rather than demonstrated experimentally, none of the four is independently validated at its individual edge, and a confirmatory experiment at the narrowed corner with the remaining parameters at set point would be required before any of the four ranges is widened.



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### 11.2. Analysis of P11 to P14: medium impact response L3 M+46 Da @ RRT 1.01

L3 M+46 Da @ RRT 1.01 exceeded the 0.3000 %Area low/medium impact boundary in two experiments, P13 (0.3384 %Area) and P14 (0.3135 %Area), and was classified as medium impact. Both experiments stressed the iodine oxidation step, and the univariate model did not identify any medium or high impact parameter for this response, so the elevated results were deconvoluted by direct comparison within the oxidation experiment set rather than by a follow-up run.

P11 and P13 share the same high-oxidation direction, and P12 and P14 the same low-oxidation direction, with respect to iodine charge, iodine-solution-preparation DMF quantity and oxidation-reaction DMF quantity. M+46 Da remained at the reporting threshold in P11 and P12 but rose in P13 and P14, which eliminates iodine stoichiometry and oxidation-solution concentration as drivers and isolates the iodine solution transfer time and the total iodine contact time, as set out in Section 10.3.

Accordingly, iodine solution transfer time, oxidation temperature, and total iodine contact time are classified as CPPs, with transfer time constrained to 20–100 min and total iodine contact time not lower than 20 min. Iodine charge quantity, iodine-solution-preparation DMF quantity, oxidation-reaction DMF quantity are deconvoluted as non-CPPs and retain the ranges established in **Table 26**, having been held at their ±6σ limits in P11 and P12 without elevating the response.



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## 12. Summary of PARs and Final Risk Assessment

The final risk assessment outcome is summarized in **Table 31**. All parameters **t**hat were not perturbed at 6 sigma range are classified as CPPs due to intentional restrictions within 6 sigma. A follow-up study could be performed later to further extend this range. Additionally, deFmoc parameters for DeFmoc of Thr4 were restricted to mitigate the high risk of having L3 des-Asn3 impurity. A few parameters for disulfide bond formation (oxidation) were restricted to mitigate the high risk of having L3 M+46 Da impurity.

**Table 31.** List of parameters, PARs and Final Risk Category.[12]

| Unit Operation | Parameter | Target | Units | -6σ | +6σ | MIN PAR (studied) | MAX PAR (studied) | Final Risk Category (Min) | Final Risk Category (Max) |
|---|---|---|---|---|---|---|---|---|---|
| Gly 10 dissolution | Dissolution jacket temperature (Gly10) | 25.0 | °C | 13.0 | 37.0 | 12.9 | 37.0 | Non CPP | Non CPP |
| | DMF dissolution total quantity (Gly10)[23] | 6.63 | kg/kg | 5.84 | 7.43 | 5.81 | 7.43 | Non CPP | Non CPP |
| | AA quantity (Gly10) | 2.25 | eq | 1.98 | 2.52 | 1.95 | 2.53 | Non CPP | Non CPP |
| | DIPEA quantity (Gly10) | 6.00 | eq | 5.28 | 6.72 | 5.18 | 6.76 | Non CPP | Non CPP |
| | Activation temperature (Gly10) | 15.0 | °C | 3.0 | 27.0 | 2.9 | 27.0 | Non CPP | Non CPP |
| | Activation time (Gly10) | 5 | min | 0 | 65 | 0 | 65 | — | Non CPP |
| Gly10 Loading | DMF coupling/loading total quantity (Gly10) | 7.58 | kg/kg | 6.67 | 8.49 | 6.64 | 8.49 | Non CPP | Non CPP |
| | Coupling/loading temperature (Gly10) | 25.0 | °C | 13.0 | 37.0 | **17.0** | **33.0** | **CPP** | **CPP** |

[23] Referred to as “DMF activation” in the stats table



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| Unit Operation | Parameter | Target | Units | -6σ | +6σ | MIN PAR (studied) | MAX PAR (studied) | Final Risk Category (Min) | Final Risk Category (Max) |
|---|---|---|---|---|---|---|---|---|---|
| | Coupling/loading time (Gly10) | 180 | min | 120 | 240 | 120 | 480 | Non CPP | Non CPP |
| | DMF post-coupling wash (Gly10) | NLT 7.58 | kg/kg | 6.67 | — | 6.63 | — | Non CPP | — |
| Resin Capping | DMF - Capping total quantity | 8.53 | kg/kg | 7.50 | 9.55 | 7.48 | 9.63 | Non CPP | Non CPP |
| | MeOH - Capping quantity | 10.00 | eq | 8.80 | 11.20 | 8.75 | 11.41 | Non CPP | Non CPP |
| | DIPEA quantity | 5.00 | eq | 4.40 | 5.60 | 4.37 | 5.61 | Non CPP | Non CPP |
| | DIPEA addition temperature | 20.0 | °C | 8.0 | 32.0 | **15.0** | **25.0** | **CPP** | **CPP** |
| | Capping temperature | 25.0 | °C | 13.0 | 37.0 | 13.0 | 37.1 | Non CPP | Non CPP |
| | Capping time | 60 | min | 0 | 120 | 0 | 120 | Non CPP | Non CPP |
| Post capping washing | DMF post-capping wash | NLT 7.58 | kg/kg | 6.67 | — | 6.63 | — | Non CPP | — |
| deFmoc | Piperidine composition in DMF – pre-Asn3 | 20.00 | % | 16.61 | 23.39 | 16.61 | 24.99 | Non CPP | Non CPP |
| | Piperidine composition in DMF - all other AA | 20.00 | % | 16.61 | 23.39 | 14.99 | 24.99 | Non CPP | Non CPP |
| | Piperidine in DMF quantity - Pre-Asn3 | 7.44 | kg/kg | 6.55 | 8.34 | **6.78** | 8.40 | **CPP** | Non CPP |
| | Piperidine in DMF quantity – Pre-Thr9 | 7.44 | kg/kg | 6.55 | 8.34 | **6.58** | 8.39 | **CPP** | Non CPP |



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| Unit Operation | Parameter | Target | Units | -6σ | +6σ | MIN PAR (studied) | MAX PAR (studied) | Final Risk Category (Min) | Final Risk Category (Max) |
|---|---|---|---|---|---|---|---|---|---|
| | Piperidine in DMF quantity – Pre-(Ala8, Cys7, Thr6, Ala5, Thr4, Cys2, gGlu1) | 7.44 | kg/kg | 6.55 | 8.34 | 6.54 | 8.38 | Non CPP | Non CPP |
| | DeFmoc temperature – pre-Asn3 | 25.0 | °C | 13.0 | 37.0 | **16.3** | 37.0 | **CPP** | Non-CPP |
| | DeFmoc temperature - All other AA | 25.0 | °C | 13.0 | 37.0 | 13.0 | 37.0 | Non CPP | Non CPP |
| | Total DeFmoc time – pre-Asn3 | 120 | min | 60 | 180 | **77** | 180 | **CPP** | Non-CPP |
| | Total DeFmoc time – pre-Thr6 | 120 | min | 60 | 180 | **62** | 180 | **CPP** | Non-CPP |
| | Total DeFmoc time - All others | 120 | min | 60 | 180 | 60 | 180 | Non CPP | Non-CPP |
| Post deFmoc washing | IPAC - post-deFmoc wash (Thr9, Ala8, Cys7, Thr6, Ala5) | NLT 6.98 | kg/kg | 6.14 | — | 6.04 | — | Non CPP | — |
| Re-swelling | DMF - reswell (Thr9, Ala8, Cys7, Thr6, Ala5) | NLT 7.58 | kg/kg | 6.67 | — | 6.72 | — | Non CPP | — |
| AA Activation | DMF activation total quantity (Thr9) | 4.74 | kg/kg | 4.17 | 5.31 | 4.14 | 5.27 | Non CPP | CPP |
| | AA quantity (Thr9) | 2.00 | eq | 1.88 | 2.12 | 1.85 | 2.13 | Non CPP | Non CPP |
| | PyBOP quantity (Thr9) | 2.00 | eq | 1.76 | 2.24 | 1.73 | 2.25 | Non CPP | Non CPP |



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| Unit Operation | Parameter | Target | Units | -6σ | +6σ | MIN PAR (studied) | MAX PAR (studied) | Final Risk Category (Min) | Final Risk Category (Max) |
|---|---|---|---|---|---|---|---|---|---|
| | Oxyma quantity (Thr9) | 2.00 | eq | 1.76 | 2.24 | 1.73 | 2.25 | Non CPP | Non CPP |
| | DIPEA quantity (Thr9) | 3.00 | eq | 2.64 | 3.36 | 2.59 | 3.42 | Non CPP | Non CPP |
| | Activation temperature (Thr9) | 5.0 | °C | -7.0 | 17.0 | -7.0 | 17.0 | Non CPP | Non CPP |
| | Activation time (Thr9) | 20 | min | 0 | 80 | 0 | 80 | Non CPP | Non CPP |
| Thr9 coupling | DMF coupling total quantity (Thr9) | 8.53 | kg/kg | 7.50 | 9.55 | 7.45 | **9.49** | Non CPP | **CPP** |
| | Coupling temperature | 25.0 | °C | 13.0 | 37.0 | **17.0** | **33.0** | **CPP** | **CPP** |
| | Coupling time | 240 | min | 180 | 300 | 180 | 600 | Non CPP | Non CPP |
| Post coupling washing | DMF post-coupling wash (Thr9, Ala8, Cys7, Thr6) | NLT 7.58 | kg/kg | 6.67 | — | 6.68 | — | Non CPP | — |
| Coupling solution activation | Dissolution temperature (Ala8, Thr6, Ala5, Thr4, Asn3) | 20.0 | °C | 8.0 | 32.0 | 8.0 | 32.0 | Non CPP | Non CPP |
| | DMF activation total quantity (Ala8, Thr6, Ala5, Thr4, Asn3) | 6.63 | kg/kg | 5.84 | 7.43 | 5.83 | 7.38 | Non CPP | **CPP** |
| | AA quantity (Ala8, Thr6, Ala5, Thr4, Asn3) | 2.00 | eq | 1.88 | 2.12 | 1.86 | 2.12 | Non CPP | Non CPP |
| | Oxyma quantity (Ala8, Thr6, Ala5, Thr4, Asn3) | 2.00 | eq | 1.76 | 2.24 | 1.74 | 2.25 | Non CPP | Non CPP |
| | DIC quantity (Ala8, Thr6, Ala5, Thr4, Asn3) | 2.20 | eq | 1.94 | 2.46 | 1.90 | 2.47 | Non CPP | Non CPP |
| | Activation time (Ala8, Thr6, Ala5, Thr4, Asn3) | 20 | min | 0 | 80 | 0 | 80 | — | Non CPP |



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| Unit Operation | Parameter | Target | Units | -6σ | +6σ | MIN PAR (studied) | MAX PAR (studied) | Final Risk Category (Min) | Final Risk Category (Max) |
|---|---|---|---|---|---|---|---|---|---|
| Coupling | DMF coupling total quantity (Ala8, Thr6, Ala5, Thr4, Asn3) | 7.58 | kg/kg | 6.67 | 8.49 | 6.68 | **8.43** | Non CPP | **CPP** |
| | Coupling temperature (Ala8, Thr6, Ala5, Thr4, Asn3) | 25.0 | °C | 13.0 | 37.0 | **17.1** | **33.0** | **CPP** | **CPP** |
| | Coupling time (Ala8, Thr6, Ala5, Thr4, Asn3) | 240 | min | 180 | 300 | 180 | 600 | Non CPP | Non CPP |
| Cys activation/dissolution | DMF dissolution total quantity (Cys7, Cys2) | 6.63 | kg/kg | 5.84 | 7.43 | 5.81 | **7.38** | Non CPP | **CPP** |
| | AA quantity (Cys7, Cys2) | 2.00 | eq | 1.76 | 2.24 | **1.85** | **2.13** | **CPP** | **CPP** |
| | Oxyma quantity (Cys7, Cys2) | 3.00 | eq | 2.82 | 3.18 | 2.61 | 3.22 | Non CPP | Non CPP |
| | Activation temperature (Cys7, Cys2) | 5.0 | °C | -7.0 | 17.0 | -7.0 | 17.0 | Non CPP | Non CPP |
| | Activation time (Cys7, Cys2) | 5 | min | 0 | 65 | 0 | 65 | — | Non CPP |
| Cys coupling | DMF coupling total quantity (Cys7, Cys2) | 8.53 | kg/kg | 7.50 | 9.55 | **7.71** | **9.26** | **CPP** | **CPP** |
| | DIC quantity (Cys7, Cys2) | 3.30 | eq | 2.90 | 3.70 | 2.85 | 3.72 | Non CPP | Non CPP |
| | Coupling temperature (Cys7, Cys2) | 20.0 | °C | 8.0 | 32.0 | **12.0** | **28.0** | **CPP** | **CPP** |
| | Coupling time (Cys7, Cys2) | 240 | min | 180 | 300 | 180 | 600 | Non CPP | Non CPP |
| Post coupling washing | DMF post-coupling wash (Ala5, Thr4, Asn3, Cys2, gGlu1) | NLT 9.48 | kg/kg | 8.34 | — | 8.32 | — | Non CPP | — |
| Post deFmoc washing | IPAC - post-deFmoc wash (Thr4, Asn3, Cys2, gGlu1) | NLT 8.73 | kg/kg | 7.68 | — | 7.59 | — | Non CPP | — |



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| Unit Operation | Parameter | Target | Units | -6σ | +6σ | MIN PAR (studied) | MAX PAR (studied) | Final Risk Category (Min) | Final Risk Category (Max) |
|---|---|---|---|---|---|---|---|---|---|
| Re-swelling | DMF - reswell (Thr4, Asn3, Cys2, gGlu1) | NLT 9.48 | kg/kg | 8.34 | — | 8.34 | — | Non CPP | — |
| gGlu1 activation | DMF activation total quantity (gGlu1) | 6.63 | kg/kg | 5.84 | 7.43 | 5.80 | 7.48 | Non CPP | Non CPP |
| | gGlu1 quantity | 2.00 | eq | 1.88 | 2.12 | 1.85 | 2.12 | Non CPP | Non CPP |
| | Dissolution temperature (gGlu1)[24] | 5.0 | °C | -7.0 | 17.0 | -7.0 | 17.0 | Non CPP | Non CPP |
| | PyBOP quantity (gGlu1) | 2.00 | eq | 1.76 | 2.24 | 1.74 | 2.25 | Non CPP | Non CPP |
| | Oxyma quantity (gGlu1) | 2.00 | eq | 1.76 | 2.24 | 1.74 | 2.25 | Non CPP | Non CPP |
| | DIPEA quantity (gGlu1) | 3.00 | eq | 2.64 | 3.36 | 2.62 | 3.41 | Non CPP | Non CPP |
| | Activation temperature (gGlu1)[25] | 20.0 | °C | 8.0 | 32.0 | 8.0 | 32.0 | Non CPP | Non CPP |
| | Activation time (gGlu1) | 20 | min | 0 | 80 | 0 | 80 | — | Non CPP |
| gGlu1 coupling | DMF coupling total quantity (gGlu1) | 7.58 | kg/kg | 6.67 | 8.49 | 6.63 | 8.59 | Non CPP | Non CPP |
| | Coupling temperature (gGlu1) | 25.0 | °C | 13.0 | 37.0 | **17.0** | **33.0** | **CPP** | **CPP** |
| | Coupling time (gGlu1) | 240 | min | 180 | 300 | 180 | 600 | Non CPP | Non CPP |

[24] Parameter name is “activation TEMP 1_gGlu1 activation_11.3.4” in stats table

[25] Parameter name is “Activation TEMP 2_gGlu1 activation_11.3.9” in stats table



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| Unit Operation | Parameter | Target | Units | -6σ | +6σ | MIN PAR (studied) | MAX PAR (studied) | Final Risk Category (Min) | Final Risk Category (Max) |
|---|---|---|---|---|---|---|---|---|---|
| Oxidative Trt Removal | DMF quantity (for iodine solution) [26] | 3.79 | kg/kg | 3.34 | 4.25 | 3.30 | 4.27 | Non CPP | Non CPP |
| | Iodine quantity | 5.00 | eq | 4.40 | 5.60 | 4.36 | 5.62 | Non CPP | Non CPP |
| | DMF quantity (for oxidation reaction) | 11.37 | kg/kg | 10.01 | 12.74 | 9.93 | 12.79 | Non CPP | Non CPP |
| | Oxidation temperature | -5.0 | °C | -17.0 | 7.0 | **-15.0** | **5.0** | **CPP** | **CPP** |
| | Iodine transfer time[27] | 50 | min | 0 | 110 | **20** | **100** | **CPP** | **CPP** |
| Post oxidation washing | Post-oxidation wash temperature | -3.0[28] | °C | -17.0 | 7.0 | **-7.0** | **5.0** | **CPP** | **CPP** |
| | DMF post-oxidation wash | NLT 9.48 | kg/kg | 8.34 | — | 8.19 | — | Non CPP | — |
| Deswell - IPAC | IPAC - deswell wash | NLT 8.73 | kg/kg | 7.68 | — | 7.67 | — | Non CPP | — |
| Deswell - MTBE | MTBE - deswell wash | NLT 7.41 | kg/kg | 6.52 | — | 6.51 | — | Non CPP | — |
| Oxidative Trt Removal | Oxidation Exposure time[29] | 50 | min | 0 | 110 | **20** | 130 | **CPP** | Non CPP |
| | Oxidation Reaction time[29] | 0 | min | 0 | 60 | 0 | **30** | Non CPP | **CPP** |

[26] Referred to as “DMF - iodine soln prep_Oxidative Trt Removal_12.1.1” in stats table

[27] Per Section 11.2, iodine solution transfer time is restricted to 20-100 min; the range shown is the range studied.

[28] Stats table reports target as -5.0°C (see parameter ““Post-oxidation wash TEMP 1_Post oxidation washing_12.2.1””), however the target was set at -3.0°C to have a broader range on the lower end.

[29] Per Section 11.2, total iodine contact time (transfer plus oxidation reaction time) is restricted to not less than 20 min.



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| Unit Operation | Parameter | Target | Units | -6σ | +6σ | MIN PAR (studied) | MAX PAR (studied) | Final Risk Category (Min) | Final Risk Category (Max) |
|---|---|---|---|---|---|---|---|---|---|
| Drying[30] | Drying jacket temperature | NMT 25 | °C | — | 37 | — | 42 | — | Non CPP |

[30] B2 wet cake was split and dried at two different temperatures. Data source for target and MAX PAR is in Technical Summary for LSN4182928 (ASYM-140021) Step 1 PAR Study, QDocs RPT-1194110



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## 12.1. Justification for One-sided PARs with a PAR Less than 6σ

Several parameters are one-sided and are listed in **Table 32** followed by the rationale.

Parameters are set as one-sided when the other end is a limit of practicality or no risk.

**Table 32.** Summary of one-sided PAR parameters[12]

| Unit Operation | Parameter | Target | Units | PAR limit | Final parameter risk criticality |
|---|---|---|---|---|---|
| Gly 10 dissolution | Activation time (Gly10) | 5 | min | NMT 65 | Non CPP |
| Gly10 Loading | DMF post-coupling wash (Gly10) | NLT 7.58 | kg/kg | NLT 6.63 | Non CPP |
| Post capping washing | DMF post-capping wash | NLT 7.58 | kg/kg | NLT 6.63 | Non CPP |
| Post deFmoc washing | IPAC - post-deFmoc wash (Thr9, Ala8, Cys7, Thr6, Ala5) | NLT 6.98 | kg/kg | NLT 6.04 | Non CPP |
| Re-swelling | DMF - reswell (Thr9, Ala8, Cys7, Thr6, Ala5) | NLT 7.58 | kg/kg | NLT 6.72 | Non CPP |
| AA Activation | Activation time (Thr9) | 20 | min | NMT 80 | Non CPP |
| Post coupling washing | DMF post-coupling wash (Thr9, Ala8, Cys7, Thr6) | NLT 7.58 | kg/kg | NLT 6.68 | Non CPP |
| Coupling solution activation | Activation time (Ala8, Thr6, Ala5, Thr4, Asn3) | 20 | min | NMT 80 | Non CPP |
| Cys activation/dissolution | Activation time (Cys7, Cys2) | 5 | min | NMT 65 | Non CPP |
| Post coupling washing | DMF post-coupling wash (Ala5, Thr4, Asn3, Cys2, gGlu1) | NLT 9.48 | kg/kg | NLT 8.32 | Non CPP |
| Post deFmoc washing | IPAC - post-deFmoc wash (Thr4, Asn3, Cys2, gGlu1) | NLT 8.73 | kg/kg | NLT 7.59 | Non CPP |
| Re-swelling | DMF - reswell (Thr4, Asn3, Cys2, gGlu1) | NLT 9.48 | kg/kg | NLT 8.34 | Non CPP |
| gGlu1 activation | Activation time (gGlu1) | 20 | min | NMT 80 | Non CPP |
| Post oxidation washing | DMF post-oxidation wash | NLT 9.48 | kg/kg | NLT 8.19 | Non CPP |
| Deswell - IPAC | IPAC - deswell wash | NLT 8.73 | kg/kg | NLT 7.67 | Non CPP |



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| Unit Operation | Parameter | Target | Units | PAR limit | Final parameter risk criticality |
|---|---|---|---|---|---|
| Deswell - MTBE | MTBE - deswell wash | NLT 7.41 | Kg/kg | NLT 6.51 | Non CPP |
| Drying | Drying jacket temperature | NMT 25 | °C | NMT 42 | Non CPP |

Activation time: These parameters are one-sided with an upper limit only. No lower limit is required because activation begins as soon as the coupling reagents are combined, and the reagent addition and activated-solution transfer operations at scale necessarily take longer than the 0 min low end of the range studied, so a zero-activation time cannot be realized at scale. The upper limits — NMT 65 min for the Gly10 dissolution and the Cys7/Cys2 couplings, NMT 80 min for gGlu1, and NMT 80 min for the standard amino acids (Ala8, Thr6, Ala5, Thr4 and Asn3) — are retained to bound activated-ester decomposition and epimerization. All five parameters were classified low risk.

DMF post-coupling washes : These washes are one-sided with a lower limit only, set at the minimum quantity studied (NLT 6.63 for Gly10, NLT 6.68 kg/kg for Thr9, Ala8, Cys7, Thr6, and 8.32 kg/kg for Ala5, Thr4, Asn3, Cys2, gGlu1). No upper limit is required because additional DMF only improves removal of residual activated amino acid, coupling reagent and urea by-products, and the resin is filtered between charges, so an excess charge carries no quality risk. The risk lies entirely on the low side, where insufficient solvent leaves reactive species on the resin. All three parameters were classified low risk.

IPAc post-deFmoc washes : These washes are one-sided with a lower limit only. Sufficient IPAc is required to displace piperidine and dibenzofulvene from the resin before the next coupling, whereas an additional charge only improves that displacement and presents no quality risk, so no upper limit is set. Both parameters were classified low risk.

DMF re-swell charges : The re-swell steps are one-sided with a lower limit only (NLT 6.72 kg/kg for Thr9, Ala8, Cys7, Thr6, Ala5 and 8.34 kg/kg for Thr4, Asn3, Cys2, gGlu1). A minimum DMF charge is needed to fully re-swell the resin and restore access to the growing peptide chain; solvent in excess of that charge only increases swelling and carries no quality risk. Both parameters were classified low risk.



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De-swell washes: Both are one-sided with a lower limit only (NLT 7.67 kg/kg for IPAC and 6.51 kg/kg for MTBE). These washes displace DMF and shrink the resin prior to drying; additional solvent improves solvent exchange and presents no quality risk, so no upper limit is required. Both parameters were classified low risk.

## 13. In-Process Controls

To ensure a successful process, certain in-process controls (IPCs) will be leveraged. Additional samples are planned to understand process throughout and categorized as FIOs. Details about each IPC are listed in Table 33.

**Table 33.** In-process controls for the eloralintide Boc-Fragment 3 (LSN4182928) SPPS process preparation.[31, 32]

| Test | Processing point | Acceptance Criteria | IPC or FIO |
|---|---|---|---|
| Ninhydrin Colorimetric Test | Amino Acid Coupling Reaction completion | Negative | IPC |
| L3(Trt)2 Residue post oxidation washing | Oxidation wash | ≤ 0.5% | IPC |
| Loss on Drying | Post SPPS Drying | ≤ 8%[33] | IPC |

[31] Technical Summary for LSN4182928 (ASYM-140021) Step 1 PAR Study, QDocs RPT-1194110

[32] Technical summary (RPT-1194110) reports appearance as an IPC, however per QDocs RPT-1197131, "Change Control Form Change-CPo140021-030106.01", appearance is no longer considered an IPC.

[33] Technical Summary (QDocs RPT-1194110) reports LOD ≤ 10.0%, however we suggest a more conservative value of LOD ≤ 8.0% based on pivotal batch lot used for long term refrigerated stability, which had LOD 8.4%. Source: QDoc MPR-242380, "Asymchem Life Sciences L3 Step 1 BPR-CPo140021-01-01-02", p. 157, sampling number CPo140021-01-01-02-3.12.14-01.



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| | | | |
|---|---|---|---|
| Residual Piperidine (headspace GC, AM-140021-020) | Post Fmoc-Deprotection Wash, Prior to Coupling | Report | FIO |

Separated from the IPCs above, a for-information-only (FIO) test is performed for residual piperidine. Following Fmoc deprotection, the resin is washed four times with isopropyl acetate at 20–25 °C, and the final wash is sampled and analyzed for residual piperidine by headspace GC prior to charging the next coupling reaction. This FIO test is intended to be run on the first few production batches only, to confirm that the wash procedure adequately removes residual piperidine to below the acceptance limit; once that confirmation is obtained, the FIO test will be discontinued.

## 14. Hold Points

Hold points associated with the process were determined outside of the parametric studies, and are discussed below. A successful hold point was defined as producing Boc-Fragment 3, LSN4182928, that can be processed downstream into acceptable drug substance with expected purity and not necessarily zero change in the crude API impurity profile. The limits discussed are not known points of failure but limits of knowledge. The coupling reaction hold was demonstrated at a 25.0 °C target temperature for 480 min on the Gly10 cycle and for 600 min on all other residues, with no medium or high impact CQA response observed at either hold time. Boc-Fragment 3 on resin was demonstrated stable after a wet cake was dried for 2 days at 42 °C, which bounds in-process holds at ambient and elevated temperature, and for 1 year at 2 to 8 °C, which supports long-term storage of the peptidyl resin. The peptidyl wet resin that was dried at 42 °C for 48 h was not carried forward to Step 3. However, all the dimers and trimers observed in Step 2 are known to be fully reduced by TCEP in Step 3 to the linear dithiol monomer. Stable hold point results are summarized in Table 34.

**Table 34.** Stable hold points[31]

| Parameter | Temperature (°C) | Hold time |
|---|---|---|
| Coupling reaction hold (for Gly10) | 25.0 | 480 min |



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| | | |
|---|---|---|
| Coupling reaction hold (all other residues) | 25.0 | 600 min |
| Drying wet cake | 42 | 48 h |
| Dry resin Stability | 2 to 8 | 12 months |



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## Appendix I: Analytical Data for Baseline and PAR Experiments

**Table 35.** UPLC-UV impurity data for the L3 Step 1 baseline and PAR experiments.[34, 35]

| Analyte / Impurity | B1 | B2 | P3 | P4 | P5 | P6 | B3 | P8 | P9 | P10 | P11 | P12 | P13 | P14 | P15 | P16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| L3 Imp @ RRT 0.49 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.13 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 |
| L3 Imp @ RRT 0.56 | 0.12 | 0.10 | 0.17 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 |
| L3 Imp @ RRT 0.64 | 0.10 | 0.10 | 0.10 | 0.10 | 0.64 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 |
| L3 Imp @ RRT 0.69 | 0.18 | 0.10 | 0.28 | 0.10 | 0.17 | 0.10 | 0.20 | 0.11 | 0.26 | 0.10 | 0.10 | 0.11 | 0.26 | 0.18 | 0.10 | 0.10 |
| L3 Imp @ RRT 0.74 | 0.13 | 0.18 | 0.20 | 0.17 | 0.17 | 0.15 | 0.14 | 0.10 | 0.15 | 0.22 | 0.23 | 0.17 | 0.10 | 0.13 | 0.13 | 0.11 |
| L3 Imp @ RRT 0.75 | 0.28 | 0.27 | 0.36 | 0.29 | 0.28 | 0.27 | 0.16 | 0.29 | 0.24 | 0.28 | 0.28 | 0.23 | 0.27 | 0.27 | 0.24 | 0.23 |
| L3 Imp @ RRT 0.77 | 0.10 | 0.10 | 0.10 | 0.10 | 0.26 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 |
| L3 Imp @ RRT 0.81 | 0.10 | 0.10 | 0.10 | 0.10 | 0.14 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 |
| L3 Imp @ RRT 0.82 | 0.23 | 0.20 | 0.37 | 0.29 | 0.24 | 0.19 | 0.20 | 0.21 | 0.24 | 0.21 | 0.27 | 0.21 | 0.30 | 0.27 | 0.31 | 0.27 |
| L3 Imp @ RRT 0.83 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.23 | 0.10 | 0.11 | 0.10 | 0.14 | 0.11 |
| L3 Imp @ RRT 0.88 | 0.10 | 0.10 | 0.11 | 0.10 | 0.13 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.11 |

[34] Candeago, R. Signal Notebook: D4U-D03050-106, “LAARA L3 SPPS Stats Table (equiv and kg-kg) - UV-V9-23SEP2026”, last modified Sep 23, 2026, 05:20 PM

[35] Reported as area %. For all the non-detect values (ND), 0.10% were given as LOQ



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| Analyte / Impurity | B1 | B2 | P3 | P4 | P5 | P6 | B3 | P8 | P9 | P10 | P11 | P12 | P13 | P14 | P15 | P16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| L3 Imp A @ RRT 0.90 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 |
| L3 Imp B @ RRT 0.90 | 0.20 | 0.18 | 0.11 | 0.15 | 0.10 | 0.31 | 0.11 | 0.22 | 0.11 | 0.13 | 0.16 | 0.13 | 0.16 | 0.16 | 0.14 | 0.17 |
| L3 Imp @ RRT 0.92 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.13 | 0.12 | 0.10 | 0.10 | 0.10 | 0.10 |
| L3 Imp @ RRT 0.93 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.11 | 0.10 | 0.10 | 0.10 | 0.10 |
| L3 Imp @ RRT 0.95 | 0.10 | 0.10 | 0.10 | 0.10 | 0.11 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 |
| L3 Imp @ RRT 0.96 | 0.11 | 0.16 | 0.12 | 0.10 | 0.15 | 0.15 | 0.10 | 0.10 | 0.12 | 0.18 | 0.13 | 0.14 | 0.17 | 0.17 | 0.22 | 0.24 |
| L3 Imp @ RRT 0.98 | 0.10 | 0.13 | 0.10 | 0.10 | 0.10 | 0.19 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 |
| L3 @ RRT 1.00 | 96.54 | 96.89 | 95.69 | 96.46 | 95.19 | 96.47 | 97.17 | 97.09 | 96.32 | 96.50 | 96.23 | 96.36 | 95.83 | 96.30 | 96.28 | 96.35 |
| L3 Imp A @ RRT 1.01 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.15 | 0.10 | 0.10 | 0.10 | 0.10 |
| L3 Imp B @ RRT 1.01 | 0.10 | 0.10 | 0.11 | 0.15 | 0.10 | 0.10 | 0.10 | 0.10 | 0.17 | 0.15 | 0.10 | 0.10 | 0.34 | 0.31 | 0.10 | 0.10 |
| L3 Imp @ RRT 1.02 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.11 | 0.10 | 0.12 | 0.10 | 0.10 | 0.10 | 0.12 | 0.10 |
| L3 Imp @ RRT 1.03 | 0.10 | 0.11 | 0.10 | 0.21 | 0.10 | 0.10 | 0.10 | 0.10 | 0.15 | 0.27 | 0.21 | 0.10 | 0.20 | 0.18 | 0.15 | 0.19 |
| L3 Imp @ RRT 1.04 | 0.14 | 0.14 | 0.12 | 0.13 | 0.13 | 0.15 | 0.12 | 0.10 | 0.15 | 0.11 | 0.11 | 0.11 | 0.13 | 0.13 | 0.10 | 0.10 |



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| Analyte / Impurity | B1 | B2 | P3 | P4 | P5 | P6 | B3 | P8 | P9 | P10 | P11 | P12 | P13 | P14 | P15 | P16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| L3 Imp @ RRT 1.06 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 |
| L3 Imp @ RRT 1.11 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.15 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 |
| L3 Imp @ RRT 1.21 | 0.10 | 0.10 | 0.10 | 0.10 | 0.14 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 |
| L3 Imp @ RRT 1.31 | 0.10 | 0.10 | 0.15 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 |
| L3 Imp @ RRT 1.34 | 0.95 | 0.95 | 0.92 | 1.01 | 0.94 | 1.01 | 1.03 | 1.13 | 1.06 | 1.03 | 0.99 | 1.04 | 0.94 | 0.97 | 1.05 | 1.03 |
| L3 Imp @ RRT 1.37 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.11 | 0.10 | 0.10 | 0.11 | 0.13 | 0.11 | 0.10 | 0.16 |
| L3 Imp @ RRT 1.39 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.12 |

**Table 36.** UPLC-MS impurity data for the L3 Step 1 baseline and PAR experiments.[36, 35]

| Analyte / Impurity | B1 | B2 | P3 | P4 | P5 | P6 | B3 | P8 | P9 | P10 | P11 | P12 | P13 | P14 | P15 | P16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| L3 Imp @ RRT 0.49 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.13 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 |
| L3 gGlu1(des-Boc) @ RRT 0.56 | 0.12 | 0.10 | 0.17 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 |
| L3 des-Asn3 @ RRT 0.64 | 0.10 | 0.10 | 0.10 | 0.10 | 0.64 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 |
| L3 M+262 Da @ RRT 0.69 | 0.10 | 0.10 | 0.12 | 0.10 | 0.17 | 0.10 | 0.20 | 0.11 | 0.11 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 |

[36] Candeago, R. Signal Notebook: D4U-D03050-106, "LAARA L3 SPPS Stats Table (equiv and kg-kg) - MS-V9-23SEP2026 ", last modified Sep 23, 2026, 05:20 PM



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| Analyte / Impurity | B1 | B2 | P3 | P4 | P5 | P6 | B3 | P8 | P9 | P10 | P11 | P12 | P13 | P14 | P15 | P16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| L3 M+294 Da @ RRT 0.69 | 0.11 | 0.10 | 0.16 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.15 | 0.10 | 0.10 | 0.10 | 0.17 | 0.12 | 0.10 | 0.10 |
| L3 M-257 Da @ RRT 0.74 | 0.13 | 0.18 | 0.20 | 0.17 | 0.17 | 0.15 | 0.14 | 0.10 | 0.15 | 0.22 | 0.23 | 0.17 | 0.10 | 0.13 | 0.13 | 0.11 |
| L3 Ac-2-10 @ RRT 0.75 | 0.28 | 0.27 | 0.36 | 0.29 | 0.28 | 0.27 | 0.16 | 0.29 | 0.24 | 0.28 | 0.28 | 0.23 | 0.27 | 0.27 | 0.24 | 0.23 |
| L3 Fmoc-2-10 des-Asn @ RRT 0.77 | 0.10 | 0.10 | 0.10 | 0.10 | 0.26 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 |
| L3 M-522 Da @ RRT 0.81 | 0.10 | 0.10 | 0.10 | 0.10 | 0.14 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 |
| L3 des-tBu @ RRT 0.82 | 0.13 | 0.10 | 0.20 | 0.13 | 0.12 | 0.10 | 0.10 | 0.10 | 0.10 | 0.12 | 0.11 | 0.10 | 0.12 | 0.11 | 0.18 | 0.13 |
| L3 des-Thr4 @ RRT 0.82 | 0.10 | 0.13 | 0.17 | 0.16 | 0.12 | 0.15 | 0.11 | 0.16 | 0.18 | 0.10 | 0.16 | 0.13 | 0.18 | 0.16 | 0.13 | 0.14 |
| L3 M-213 Da @ RRT 0.83 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.23 | 0.10 | 0.11 | 0.10 | 0.14 | 0.11 |
| L3 des-tBu @ RRT 0.88 | 0.10 | 0.10 | 0.11 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.11 |
| L3 endo-Ala @ RRT 0.90 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 |
| L3 des-Thr9 @ RRT 0.90 | 0.20 | 0.18 | 0.11 | 0.15 | 0.10 | 0.31 | 0.11 | 0.22 | 0.11 | 0.13 | 0.16 | 0.13 | 0.16 | 0.16 | 0.14 | 0.17 |
| L3 linear dithiol @ RRT 0.92 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.13 | 0.12 | 0.10 | 0.10 | 0.10 | 0.10 |
| L3 endo-Ala @ RRT 0.95 | 0.10 | 0.10 | 0.10 | 0.10 | 0.11 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 |
| L3 endo-Gly10 @ RRT 0.96 | 0.11 | 0.16 | 0.12 | 0.10 | 0.15 | 0.15 | 0.10 | 0.10 | 0.12 | 0.18 | 0.13 | 0.14 | 0.17 | 0.17 | 0.22 | 0.24 |
| L3 D-Cys7 @ RRT 0.98 | 0.10 | 0.13 | 0.10 | 0.10 | 0.10 | 0.19 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 |
| L3 @ RRT 1.00 | 96.54 | 96.89 | 95.69 | 96.46 | 95.19 | 96.47 | 97.17 | 97.09 | 96.32 | 96.50 | 96.23 | 96.36 | 95.83 | 96.30 | 96.28 | 96.35 |



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| Analyte / Impurity | B1 | B2 | P3 | P4 | P5 | P6 | B3 | P8 | P9 | P10 | P11 | P12 | P13 | P14 | P15 | P16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| L3 Boc-L3-SS @ RRT 1.01 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.15 | 0.10 | 0.10 | 0.10 | 0.10 |
| L3 M+46 Da @ RRT 1.01 | 0.10 | 0.10 | 0.11 | 0.15 | 0.10 | 0.10 | 0.10 | 0.10 | 0.17 | 0.15 | 0.10 | 0.10 | 0.34 | 0.31 | 0.10 | 0.10 |
| L3 D-Cys2&7 @ RRT 1.03 | 0.10 | 0.11 | 0.10 | 0.21 | 0.10 | 0.10 | 0.10 | 0.10 | 0.15 | 0.27 | 0.21 | 0.10 | 0.20 | 0.18 | 0.15 | 0.19 |
| L3 endo-gGlu1 @ RRT 1.04 | 0.14 | 0.14 | 0.12 | 0.13 | 0.13 | 0.15 | 0.12 | 0.10 | 0.15 | 0.11 | 0.11 | 0.11 | 0.13 | 0.13 | 0.10 | 0.10 |
| L3 Fmoc-2-10 @ RRT 1.06 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 |
| L3 endo-Thr @ RRT 1.11 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.15 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 |
| L3 M+48 Da @ RRT 1.31 | 0.10 | 0.10 | 0.15 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 |
| L3 Dimer @ RRT 1.34 | 0.95 | 0.95 | 0.92 | 1.01 | 0.94 | 1.01 | 1.03 | 1.13 | 1.06 | 1.03 | 0.99 | 1.04 | 0.94 | 0.97 | 1.05 | 1.03 |
| L3 Dimer (single D-Cys) @ RRT 1.37 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.11 | 0.10 | 0.10 | 0.11 | 0.13 | 0.11 | 0.10 | 0.16 |
| L3 Dimer (single D-Cys) @ RRT 1.39 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.12 |

**Table 37.** D-Enantiomer purity results for baseline and PAR experiments. [31]

| D-Enantiomer (% area) | B1 | B2 (25 °C) | B2 (42 °C) | P3 | P4 | P5 | P6 | P7 | P8 | P9 | P10 | P11 | P12 | P13 | P14 | P15 | P16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Ala | 0.02 | 0.03 | 0.03 | 0.02 | 0.03 | 0.02 | 0.08 | 0.02 | 0.02 | 0.05 | 0.03 | 0.05 | 0.04 | 0.04 | 0.03 | 0.04 | 0.06 |
| Asp | 0.09 | 0.02 | 0.02 | 0.09 | 0.09 | 0.09 | 0.07 | 0.01 | 0.11 | 0.13 | 0.03 | 0.11 | 0.11 | 0.07 | 0.07 | 0.08 | 0.10 |
| Cys | 0.07 | 0.13 | 0.12 | 0.08 | 0.07 | 0.07 | 0.03 | 0.01 | 0.07 | 0.09 | 0.07 | 0.08 | 0.07 | 0.12 | 0.11 | 0.09 | 0.12 |
| Glu | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 | 0.03 | ND | 0.03 | 0.06 | 0.01 | 0.08 | 0.04 | 0.02 | 0.02 | 0.02 | 0.02 |



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| D-Enantiomer (% area) | B1 | B2 (25 °C) | B2 (42 °C) | P3 | P4 | P5 | P6 | P7 | P8 | P9 | P10 | P11 | P12 | P13 | P14 | P15 | P16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Thr | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 | 0.02 | 0.06 | 0.01 | 0.02 | 0.01 | 0.04 | 0.02 | 0.01 | 0.01 | 0.01 | 0.01 |



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## Appendix II: Lot Reference Guide to PAR experiment

**Table 38.** Lot reference table for SPPS PAR lots [31]

| Experiment # | Description | SPPS Lot # | SPPS PAR→Step2 Cleavage Lot # | SPPS PAR→Step3 dithiolacetal cyclization lot # |
|---|---|---|---|---|
| B1 | Baseline 1 | NBK012005-1 | NBK012670-1 | NBK013001-21 |
| B2 | Baseline, large scale for cleavage PARs | NBK012005-2 | NBK012670-13 (25 °C); NBK012670-14 (42 °C) | NBK013001-22 |
| P3 | High conversion build | NBK012005-3 | NBK012670-2 | NBK013001-23 |
| P4 | Low coupling conversion build | NBK012005-4 | NBK012670-4 | NBK013001-27 |
| P5 | Low deFmoc conversion build | NBK012005-5 | NBK012670-5 | NBK013001-24 |
| P6 | Racemization / epimerization / DKP build | NBK012005-6 | NBK012670-6 | NBK013001-25 |
| B3 | Baseline 2 | NBK012005-7 | NBK012670-7 | NBK013001-28 |
| P8 | Alternate supplier resin | NBK012005-8 | NBK012670-8 | NBK013001-26 |
| P9 | Cold coupling | NBK012005-9 | NBK012670-9 | NBK013001-29 |
| P10 | Coupling long hold | NBK012005-10 | NBK012670-10 | NBK013001-30 |
| P11 | I2 oxidation, higher risk of incomplete oxidation | NBK012005-11 | NBK012670-11 | NBK013001-31 |
| P12 | I2 oxidation, risk of incomplete oxidation | NBK012005-12 | NBK012670-12 | NBK013001-32 |
| P13 | I2 oxidation deconvolution 1 | NBK012005-13 | NBK012670-15 | NBK013001-33 |



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| Experiment # | Description | **SPPS Lot #** | **SPPS PAR→Step2 Cleavage Lot #** | **SPPS PAR→Step3 dithiolacetal cyclization lot #** |
|---|---|---|---|---|
| P14 | I2 oxidation deconvolution 2 | NBK012005-14 | NBK012670-16 | NBK013001-34 |
| P15 | Cys isomer deconvolution 1 | NBK012005-15 | NBK012670-17 | NBK013001-35 |
| P16 | Cys isomer deconvolution 2 | NBK012005-16 | NBK012670-18 | NBK013001-36 |



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## Appendix III. Coupling and Deprotection Reaction Time Targets and PARs

**Table 39.** Coupling and deprotection reaction time targets and PARS range studied

| Parameter | Residue(s) | Target (min) | PAR Low (Min Studied) | PAR High (Max Studied) | σ-Coverage (Low / High) |
|---|---|---|---|---|---|
| Coupling time | Gly10 (resin loading) | 180 | 120 | 480 | −6σ / +30σ |
| Coupling time | Thr9 | 240 | 180 | 600 | −6σ / +36σ |
| Coupling time | Ala8, Thr6, Ala5, Thr4, Asn3 | 240 | 180 | 600 | −6σ / +36σ |
| Coupling time | Cys7, Cys2 | 240 | 180 | 600 | −6σ / +36σ |
| Coupling time | gGlu1 | 240 | 180 | 600 | −6σ / +36σ |
| Total deFmoc time | Thr9, Ala8, Cys7, Thr6, Ala5, Thr4, Asn3, Cys2, gGlu1 | 120 | 60 | 180 | −6σ / +6σ |
| Iodine transfer time (oxidative Trt removal) | Cys7/Cys2 disulfide | 50 | 20 | 100 | −3σ / +5σ |
| Oxidation exposure time | Cys7/Cys2 disulfide | 50 | 20 | 130 | −3σ / +8σ |
| Oxidation reaction time | Cys7/Cys2 disulfide | 0 | 0 | 30 | 0σ / +3σ |



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## Appendix IV. Stability of peptide on resin intermediate upon storage at 5°C[37]

**Table 40.** Storage stability of Boc-Fragment 3 dry peptide resin[38]

| Component / impurity | RRT | 1 month | 6 months | 12 months |
|---|---|---|---|---|
| Lot / campaign reference | – | Step 2 Campaign 01 batch 02 | 7846-03 RS Process | 3001-02 RS Process |
| Combined purity (product + dimer + trimer, area%) | – | 98.82 | 98.99 | 98.66 |
| Ratio of product / dimer | – | 4.0:1 | 4.0:1 | 4.2:1 |
| Boc-L3-SS (main peak, area%) | 1.00 | 77.59 | 78.02 | 78.12 |
| L3 M-419 Da | 0.75 | 0.10 | 0.06 | 0.06 |
| L3 endo-Gly10 | 0.95 | 0.24 | 0.10 | 0.12 |
| L3 Imp @ RRT 0.96 | 0.96 | – | 0.17 | – |
| L3 D-Cys2 | 1.04 | – | 0.06 | 0.03 |
| L3 Imp @ RRT 1.23 | 1.23 | 0.16 | – | 0.03 |
| L3 Imp @ RRT1.30 | 1.30 | 0.10 | 0.0 | – |
| L3 Dimer 1 | 1.34 | 8.45 | 8.71 | 8.09 |
| L3 Dimer 2 | 1.36 | 10.96 | 10.68 | 10.55 |
| L3 Trimer 1 | 1.57 | 0.53 | 0.56 | 0.75 |
| L3 Trimer 2 | 1.59 | 1.29 | 1.02 | 1.15 |

[37] L3-Technical Summary for LSN4182928 (ASYM-140021) Step 1 PAR Study, QDocs RPT-1194110, addendum 1.0_Supplementary Stability Data

[38] The analysis was done based on step 2 disulfide product instead of step 3 dithioacetal product. Therefore, RRT and name of impurity may not be same as other analysis.



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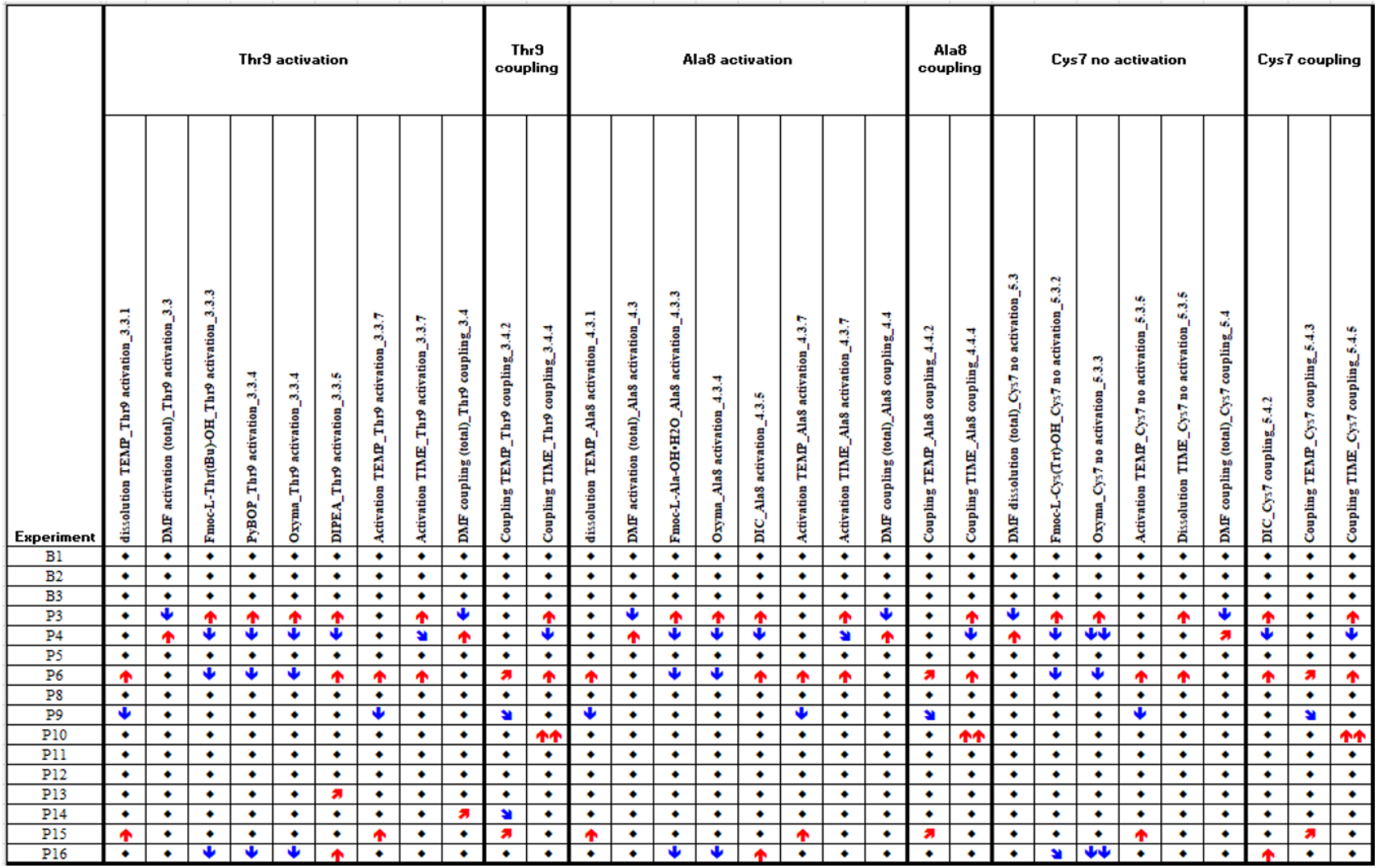
## Appendix V: Boc-Fragment 3 SPPS Parameter Risk Classification

**Table 41.** PAR Experiments for Fragment L3 (LSN4182928) Step 1 – Process parameter Assessment Study Summary[18]

| | Prepare 100V X% Piperidine/DMF Bulk Solution | Gly 10 dissolution | | | | | | | Gly10 Loading | | Post coupling washing | | | Resin Capping | | | | | | Post capping washing | deFmoc (pre Thr9) | | | Post deFmoc washing | | Re-swelling | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Experiment | % Piperidine (v/v)_Prepare 100V X% Piperidine/DMF Bulk Solution_1.0. | dissolution jacket TEMP_Gly 10 dissolution_2.1.1 | DMF activation (total)_Gly 10 dissolution_2.1 | Fmoc-Gly-OH_Gly 10 dissolution_2.1.3 | DIPEA_Gly 10 dissolution_2.1.5 | Activation TEMP_Gly 10 dissolution_2.1.6 | Activation TIME_Gly 10 dissolution_2.1.6 | DMF coupling (total)_Gly10 Loading_2.2 | Coupling TEMP_Gly10 Loading_2.2.2 | Coupling TIME_Gly10 Loading_2.2.4 | DMF post-coupling wash_Post coupling washing_2.3.1 | DMF post-coupling wash 1_Post coupling washing_3.5.1 | DMF post-coupling wash 1_Post coupling washing_7.5.1 | DMF - Capping (total)_Resin Capping_2.4 | MeOH - Capping_Resin Capping_2.4.2 | DIPEA_Resin Capping_2.4.3 | DIPEA addition TEMP_Resin Capping_2.4.4 | Capping TEMP_Resin Capping_2.5.1 | Capping TIME_Resin Capping_2.5.2 | DMF post-capping wash 1_Post capping washing_2.6.1 | piperidine in DMF_deFmoc (pre Thr9)_3.1.1 | DeFmoc TEMP_deFmoc (pre Thr9)_3.1.2 | Total DeFmoc TIME_deFmoc (pre Thr9)_3.1.2 | IPAC - post-deFmoc wash 1_Post deFmoc washing_3.2.1 | IPAC - post-deFmoc wash 1_Post deFmoc washing_8.2.1 | DMF - reswell 1_Reswelling_3.2.5 | DMF - reswell 1_Reswelling_8.2.5 |
| B1 | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ |
| B2 | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ |
| B3 | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ |
| P3 | ↑↑ | ◆ | ↓ | ↑ | ↑ | ◆ | ↑ | ↓ | ◆ | ↑ | ◆ | ◆ | ◆ | ↑ | ↑ | ↑ | ↗ | ↑ | ↑ | ◆ | ↑ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ |
| P4 | ◆ | ◆ | ↑ | ↓ | ↓ | ◆ | ◆ | ↑ | ◆ | ↓ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ↓ | ↓ | ◆ | ◆ |
| P5 | ↓↓ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ↓ | ↓ | ↓ | ↘ | ↓ | ↓ | ◆ | ↓ | ◆ | ↓ | ◆ | ◆ | ◆ | ◆ |
| P6 | ◆ | ↑ | ◆ | ↓ | ↓ | ↑ | ↑ | ◆ | ↗ | ↑ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ↑ | ↑ | ◆ | ◆ | ◆ | ◆ |
| P8 | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ |
| P9 | ◆ | ↓ | ◆ | ◆ | ◆ | ↓ | ◆ | ◆ | ↘ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ↓ | ↓ |
| P10 | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ↑↑ | ↓ | ↓ | ↓ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ |
| P11 | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ↑ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ |
| P12 | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ↓ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ |
| P13 | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ |
| P14 | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ |
| P15 | ◆ | ↑ | ◆ | ◆ | ◆ | ↑ | ◆ | ◆ | ↗ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ |
| P16 | ◆ | ◆ | ◆ | ↓ | ↓ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ↑ | ↑ | ◆ | ◆ | ◆ |



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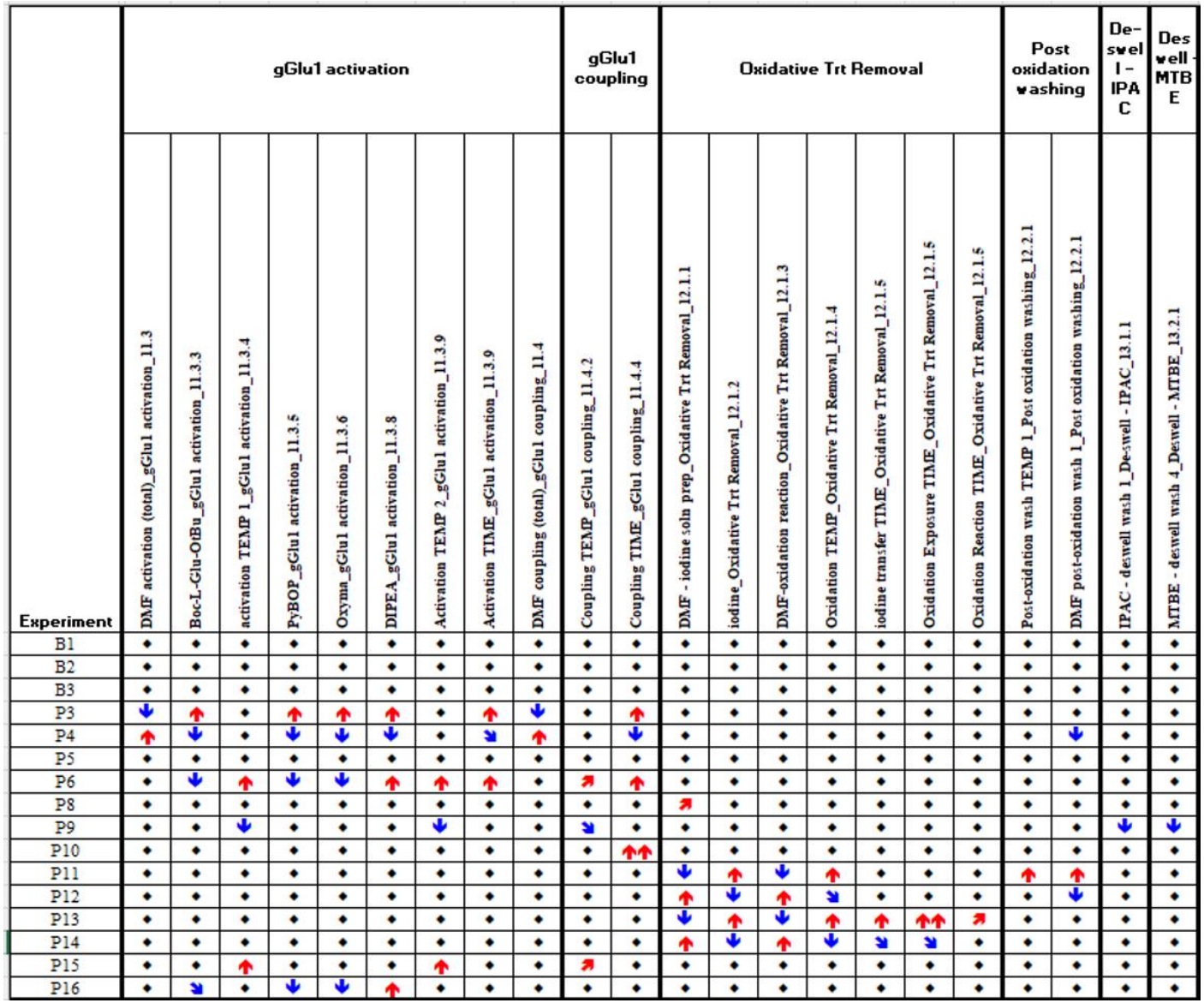


| Experiment | Thr9 activation | | | | | | | | | Thr9 coupling | | Ala8 activation | | | | | | | | Ala8 coupling | | Cys7 no activation | | | | | | Cys7 coupling | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | dissolution TEMP_Thr9 activation_3.3.1 | DMF activation (total)_Thr9 activation_3.3 | Fmoc-L-Thr(tBu)-OH_Thr9 activation_3.3.3 | PyBOP_Thr9 activation_3.3.4 | Oxyma_Thr9 activation_3.3.4 | DIPEA_Thr9 activation_3.3.5 | Activation TEMP_Thr9 activation_3.3.7 | Activation TIME_Thr9 activation_3.3.7 | DMF coupling (total)_Thr9 coupling_3.4 | Coupling TEMP_Thr9 coupling_3.4.2 | Coupling TIME_Thr9 coupling_3.4.4 | dissolution TEMP_Ala8 activation_4.3.1 | DMF activation (total)_Ala8 activation_4.3 | Fmoc-L-Ala-OH•H2O_Ala8 activation_4.3.3 | Oxyma_Ala8 activation_4.3.4 | DIC_Ala8 activation_4.3.5 | Activation TEMP_Ala8 activation_4.3.7 | Activation TIME_Ala8 activation_4.3.7 | DMF coupling (total)_Ala8 coupling_4.4 | Coupling TEMP_Ala8 coupling_4.4.2 | Coupling TIME_Ala8 coupling_4.4.4 | DMF dissolution (total)_Cys7 no activation_5.3 | Fmoc-L-Cys(Trt)-OH_Cys7 no activation_5.3.2 | Oxyma_Cys7 no activation_5.3.3 | Activation TEMP_Cys7 no activation_5.3.5 | Dissolution TIME_Cys7 no activation_5.3.5 | DMF coupling (total)_Cys7 coupling_5.4 | DIC_Cys7 coupling_5.4.2 | Coupling TEMP_Cys7 coupling_5.4.3 | Coupling TIME_Cys7 coupling_5.4.5 |
| B1 | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ |
| B2 | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ |
| B3 | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ |
| P3 | ◆ | ↓ | ↑ | ↑ | ↑ | ↑ | ◆ | ↑ | ↓ | ◆ | ↑ | ◆ | ↓ | ↑ | ↑ | ↑ | ◆ | ↑ | ↓ | ◆ | ↑ | ↓ | ↑ | ↑ | ◆ | ↑ | ↓ | ↑ | ◆ | ↑ |
| P4 | ◆ | ↑ | ↓ | ↓ | ↓ | ↓ | ◆ | ↘ | ↑ | ◆ | ↓ | ◆ | ↑ | ↓ | ↓ | ↓ | ◆ | ↘ | ↑ | ◆ | ↓ | ◆ | ↑ | ↓ | ↓↓ | ◆ | ◆ | ↗ | ↓ | ↓ |
| P5 | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ |
| P6 | ↑ | ◆ | ↓ | ↓ | ↓ | ↑ | ↑ | ↑ | ◆ | ↗ | ↑ | ↑ | ◆ | ↓ | ↓ | ↑ | ↑ | ↑ | ◆ | ↗ | ↑ | ◆ | ↓ | ↓ | ↑ | ↑ | ◆ | ↑ | ↗ | ↑ |
| P8 | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ |
| P9 | ↓ | ◆ | ◆ | ◆ | ◆ | ◆ | ↓ | ◆ | ◆ | ↘ | ◆ | ↓ | ◆ | ◆ | ◆ | ◆ | ↓ | ◆ | ◆ | ↘ | ◆ | ◆ | ◆ | ◆ | ◆ | ↓ | ◆ | ◆ | ↘ | ◆ |
| P10 | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ↑↑ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ↑↑ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ↑↑ |
| P11 | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ |
| P12 | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ |
| P13 | ◆ | ◆ | ◆ | ◆ | ◆ | ↗ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ |
| P14 | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ↗ | ↘ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ |
| P15 | ↑ | ◆ | ◆ | ◆ | ◆ | ◆ | ↑ | ◆ | ◆ | ↗ | ◆ | ↑ | ◆ | ◆ | ◆ | ◆ | ↑ | ◆ | ◆ | ↗ | ◆ | ◆ | ◆ | ◆ | ◆ | ↑ | ◆ | ◆ | ↗ | ◆ |
| P16 | ◆ | ◆ | ↓ | ↓ | ↓ | ↑ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ↓ | ↓ | ↑ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ↘ | ↓↓ | ◆ | ◆ | ◆ | ↑ | ◆ |



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| Experiment | gGlu1 activation | | | | | | | | | gGlu1 coupling | | Oxidative Trt Removal | | | | | | | Post oxidation washing | | De-swell - IPAC | Deswell - MTBE |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | DMF activation (total)_gGlu1 activation_11.3 | Boc-L-Glu-OtBu_gGlu1 activation_11.3.3 | activation TEMP 1_gGlu1 activation_11.3.4 | PyBOP_gGlu1 activation_11.3.5 | Oxyma_gGlu1 activation_11.3.6 | DIPEA_gGlu1 activation_11.3.8 | Activation TEMP 2_gGlu1 activation_11.3.9 | Activation TIME_gGlu1 activation_11.3.9 | DMF coupling (total)_gGlu1 coupling_11.4 | Coupling TEMP_gGlu1 coupling_11.4.2 | Coupling TIME_gGlu1 coupling_11.4.4 | DMF - iodine soln prep_Oxidative Trt Removal_12.1.1 | iodine_Oxidative Trt Removal_12.1.2 | DMF-oxidation reaction_Oxidative Trt Removal_12.1.3 | Oxidation TEMP_Oxidative Trt Removal_12.1.4 | iodine transfer TIME_Oxidative Trt Removal_12.1.5 | Oxidation Exposure TIME_Oxidative Trt Removal_12.1.5 | Oxidation Reaction TIME_Oxidative Trt Removal_12.1.5 | Post-oxidation wash TEMP 1_Post oxidation washing_12.2.1 | DMF post-oxidation wash 1_Post oxidation washing_12.2.1 | IPAC - deswell wash 1_Deswell - IPAC_13.1.1 | MTBE - deswell wash 4_Deswell - MTBE_13.2.1 |
| B1 | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ |
| B2 | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ |
| B3 | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ |
| P3 | ↓ | ↑ | ◆ | ↑ | ↑ | ↑ | ◆ | ↑ | ↓ | ◆ | ↑ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ |
| P4 | ↑ | ↓ | ◆ | ↓ | ↓ | ↓ | ◆ | ↘ | ↑ | ◆ | ↓ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ↓ | ◆ | ◆ |
| P5 | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ |
| P6 | ◆ | ↓ | ↑ | ↓ | ↓ | ↑ | ↑ | ↑ | ◆ | ↗ | ↑ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ |
| P8 | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ↗ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ |
| P9 | ◆ | ◆ | ↓ | ◆ | ◆ | ◆ | ↓ | ◆ | ◆ | ↘ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ↓ | ↓ |
| P10 | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ↑↑ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ |
| P11 | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ↓ | ↑ | ↓ | ↑ | ◆ | ◆ | ◆ | ↑ | ↑ | ◆ | ◆ |
| P12 | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ↑ | ↓ | ↑ | ↘ | ◆ | ◆ | ◆ | ◆ | ↓ | ◆ | ◆ |
| P13 | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ↓ | ↑ | ↓ | ↑ | ↑ | ↑↑ | ↗ | ◆ | ◆ | ◆ | ◆ |
| P14 | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ↑ | ↓ | ↑ | ↓ | ↘ | ↘ | ◆ | ◆ | ◆ | ◆ | ◆ |
| P15 | ◆ | ◆ | ↑ | ◆ | ◆ | ◆ | ↑ | ◆ | ◆ | ↗ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ |
| P16 | ◆ | ↘ | ◆ | ↓ | ↓ | ↑ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ | ◆ |



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**Appendix VI: Baseline correction and risk threshold calculations**

Criticality levels were determined in the following manner for properties with a NLT specification value (e.g., assay): $High\ Risk\ Threshold = Specification\ Limit$

$$Low/Medium\ Risk\ Threshold = Baseline\ Average - 0.5 \times (Baseline\ Average - Specification\ Limit)$$

$$No/Low\ Risk\ Threshold = BaselineAverage - 0.1 \times (Baseline\ Average - Specification\ Limit)$$

Criticality levels were determined in the following manner for properties with a NMT specification value (e.g., impurities):

$$High\ Risk\ Threshold = Specification\ Limit$$

$$Low/Medium\ Risk\ Threshold = Baseline\ Average + 0.5 \times (Specification\ Limit - Baseline\ Average)$$

$$No/Low\ Risk\ Threshold = Baseline\ Average + \max\left[0.02, 0.1 \times (Specification\ Limit - Baseline\ Average)\right]$$

For the no/low risk threshold, if 10% of the difference between the baseline average and the specification limit is less than 0.015%, an absolute difference of 0.02% (when rounded to two decimal places) or more from the baseline process will be used as the threshold for differentiating between a no risk parameter and low risk parameter. This value corresponds to 10% of the difference from a baseline average of 0.00% and a specification limit of not more than 0.15% (ICH qualification level for an impurity with a maximum daily dose of ≤ 2 g/day). Furthermore, the magnitude of this threshold is equal to approximately two standard deviations of the measurement variability for a typical impurities method, which would provide approximately 95% confidence that the actual magnitude of the observed parameter effect is greater than the measurement variability. However, in cases where the baseline average is near the specification limit, adding 0.02% (absolute) to the baseline average may result in the parameter crossing the low/medium and/or medium/high risk classification thresholds. For example, if the baseline average for an impurity was 0.12% and the specification for that impurity was NMT 0.15%, a



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parameter effect of 0.02% would exceed 50% of the difference between the baseline average and the specification, resulting in that parameter being assigned a medium risk level, rather than a low risk level. In these instances, a parameter with this magnitude of an effect would be assigned as having either medium or high risk according to the above rules for those risk levels.



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## Appendix VII: Definitions used across partnering organizations.

**Table 42. Definitions used across partnering organizations.[18]**

| SMDD Notation | Statistical Team Notation | Observations |
|---|---|---|
| Critical Quality Attribute (CQA) | Quality Attribute | |
| Proven Acceptable Range Experiment (PAR Experiment) | Parametric Experiment | |
| No/Low-Risk Threshold | Low-Risk Threshold | In Statistical Team's notation, parameters are associated with risk, not impact. So, parameter risk is assessed against risk threshold |
| Low/Medium-Risk Threshold | Medium-Risk Threshold | |
| Medium/High-Risk Threshold | High-Risk Threshold | |
| No-Risk Parameter | Negligible-Impact Range | In Statistical Team's notation, experiments are associated with impact, not risk. So, experiments fall within impact ranges |
| Low-Risk Parameter | Low-Impact Range | |
| Medium-Risk Parameter | Medium-Impact Range | |



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## 15. Revisions

| Version | Date | Section | Change made | Reason for Revision |
|---|---|---|---|---|
| 1.0 | 9/23/2026 | N/A | N/A | New version |
| 2.0 | 10/15/2026 | 12 | Item# 74 in Table, Target temp. from -5°C to -3°C | The lower end of the PAR limit was too close to the target value, so the target temperature was increased by 2C. |
| 3.0 | 10/19/2026 | All | Change from LSN382099 to LSN4182928. | Header LSN was wrong. |
| 3.0 | 10/19/2026 | 12 | Item #78 in Table 31, change +6σ from 100 to 110. | +6σ was incorrect (1 σ = 10 min, target is 50 min, so target +6σ = 110 min). |
| 3.0 | 10/19/2026 | Signature page | Add additional DIR resources to sign this document (Rajesh A Kulkarni, Adam Finne, Shivaprasad Sankaraneni, David Zhe Yuan). | These DIR resources were not included in the previous versions (they were only listed the L3 DIR Links attachment). |
| 3.0 | 10/19/2026 | Appendix V | Table 39 changed to Table 40 | Table number was hardcoded. Change to cross-reference. |
| 3.0 | 10/19/2026 | 5 | "Figure 6. Decision tree for handling known process impurities based on occurrence and detectability" change figure number from hardcoded to cross-reference. | Figure 6's number was hardcoded, resulting in duplicate Figure 6. Changed to cross-reference. |
| 3.0 | 10/19/2026 | 8 | Change from Table 28 (hardcoded) to cross-reference Table 26 in text in two places: "Target ∓ 6σ, as defined in the Table 26 notes" and "Consistent with the Table 26 notes". | Cross-reference to wrong table. |
| 3.0 | 10/19/2026 | 11 | Text cites Table 28 for the narrowed des-Asn3 limits. Should be Table 30. "…which gives narrowed lower limits of 16.3 °C, 16.6 %, 6.77 kg/kg and 77 min (Table **30**)." | Hardcoded number instead of cross-reference. |
| 3.0 | 10/19/2026 | 7.3 | Text refers to Figure 7 instead of Figure 8. Corrected: "The SPPS equipment set-up is shown in Figure 8." | Figure number was hardcoded. |
| 3.0 | 10/19/2026 | 10 | was: "as control limit plots in Figure 7 through Figure 37" changed to: "control limit plots in Figure 9 through Figure 37" | Figure numbers were hardcoded, change to cross-reference. |
| 3.0 | 10/19/2026 | 10.4 | was: "(Figure 11 through Figure 29)" now: "(Figure 12 through Figure 30)" | Hardcoded figure number, changed to cross-reference. |
| 3.0 | 10/19/2026 | 10.5 | was "for completeness in Figure 30 through Figure 36." now: "for completeness in Figure 31 through Figure 37" | Hardcoded figure number, changed to cross-reference. |



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| 3.0 | 10/19/2026 | 10 | Section 10.1 was skipped. Renumber sections to include 10.1 | Section numbering skipped 10.1 (it started with 10.2). |
| 3.0 | 10/19/2026 | 8 | was "The set-point charges in Table 28 remain at target (CTC resin 45 g), "<br>now: "The set-point charges in Table 26 remain at target (CTC resin 45 g)" | Hardcoded figure number, changed to cross-reference. |
| 3.0 | 10/19/2026 | 10.1 | was "Across the sixteen experiments defined in Table 28, main peak purity ranged from 95.19 %Area (P5) to 97.17 %Area for B3 as reported in Table 34. "<br>now: "Across the sixteen experiments defined in Table 26, main peak purity ranged from 95.19 %Area (P5) to 97.17 %Area for B3 as reported in Table 35. " | Hardcoded figure numbers, changed to cross-reference. |
| 3.0 | 10/19/2026 | 6.5 | was "and the responses were analyzed for impact to the Boc-Fragment 3 specifications (Table 5). "<br>now "and the responses were analyzed for impact to the Boc-Fragment 3 specifications (Table 4). " | Hardcoded figure numbers, changed to cross-reference. |
| 3.0 | 10/19/2026 | 9 | was "along with Table 39 in Appendix V"<br>now "along with Table 40 in Appendix V" | Hardcoded figure numbers, changed to cross-reference. |
| 3.0 | 10/19/2026 | 4 | was "The entire SPPS process can be broken down into 8 sub-unit operations as depicted in Figure 5 and corresponding equipment flow diagram is shown in Figure 5."<br>now: "The entire SPPS process can be broken down into 8 sub-unit operations as depicted in Figure 7 and corresponding equipment flow diagram is shown in Figure 5." | Reference to Figure 5 twice with different meaning. |
| 3.0 | 10/19/2026 | 10 | was "parametric experiments defined in Table 28"<br>now: "parametric experiments defined in Table 26" | Hardcoded figure number, changed to cross-reference. |
| 3.0 | 10/19/2026 | 10.1 | was "the PARs established in Table 28 are sufficient to control it"<br>now: ""the PARs established in Table 31 are sufficient to control it"" | Hardcoded figure number, changed to cross-reference. |
| 3.0 | 10/19/2026 | 11 | was: "by the PARs established in Table 26"<br>now: "by the PARs established in Table 31" | Hardcoded figure number, changed to cross-reference. |
| 3.0 | 10/19/2026 | 5 | Assign section 5.1 to "Discussion of impurities observed during SPPS, cleavage, and dithioacetalization to form Boc-Fragment 3" and 5.2 to "L3 SPPS Control Strategy" | Sections had no numbering |
| 3.0 | 10/19/2026 | 12 | Footnote 21 & 22 was "Per Section 11.2, iodine solution transfer time is restricted to 20-100 min" and " ... total iodine contact time " | Hardcoded section number, changed to cross-reference. |



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| | | | now is "Per Section 11.3, iodine solution transfer time is restricted to 20-100 min" also ".... total iodine contact time " | |
| 3.0 | 10/19/2026 | 11.3 | was "total iodine contact time not lowering than 20 min" now "total iodine contact time not lower than 20 min" | Typo |
| 3.0 | 10/19/2026 | 11.3 | was "and oxidation temperature are deconvoluted as non-CPPs" delete "and oxidation temperature" | Oxidation temperature is a CPP on both sides |
| 3.0 | 10/19/2026 | 11.3 | was "iodine solution transfer time and total iodine contact time are classified as CPPs", now is "iodine solution transfer time, oxidation temperature, and total iodine contact time are classified as CPPs" | Oxidation temp is a CPP |
| 3.0 | 10/19/2026 | 10.2 | was "is classified as medium impact under the Appendix V criticality definitions" changed to "is classified as medium impact under the Appendix VI criticality definitions" | wrong section |
| 3.0 | 10/19/2026 | 6.10 | Table 18, 20, 24, caption: change from "Summary of the initial risk assessment" to "Parametric studies for " | wrong caption |
| 3.0 | 10/19/2026 | 3, 6, 7, 11, 12 | Fix section numbering from 12.2 -> 12.1 , also for section 3, 6, 7, 11 | wrong section number |
| 3.0 | 10/19/2026 | 10.2 | was "under the Appendix V criticality definitions" now "under the Appendix VI criticality definitions" | wrong section number |
| 3.0 | 10/19/2026 | 5.2 | was "risk level categorized as defined in Section 6" change to "risk level categorized as defined in Section 2" | wrong section number |
| 3.0 | 10/19/2026 | 10 | was "and nine negligible impact responses " now "and seven negligible impact responses " | wrong number of negligible responses |
| 3.0 | 10/19/2026 | 10 | was "Thirty-one responses are presented as " now "Twenty-nine responses are presented as" | wrong number of responses |
| 3.0 | 10/19/2026 | 10.5 | was "Nine responses were negligible impact in every " now "Seven responses were negligible impact in every " | wrong number of responses |
| 3.0 | 10/19/2026 | 8 | was "consisting of eighteen experiments " now "consisting of sixteen experiments " | wrong number of experiments |



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| 3.0 | 10/19/2026 | Appendix II | Table 37 footnote reference change from 2, 9 to 23 | Wrong reference. Refer to AsymChem tech report |
| 3.0 | 10/19/2026 | 6.13 | delete "In addition, several process targets had not undergone rigorous optimization due to the speed of development." | Repetition of the same sentence |
| 3.0 | 10/20/2026 | 8 | Table 26 (overview of executed PAR experiment conditions for SPPS process), change the Short Name to Parameter name, and by help of D4U-D03050-106 > AARA L3 SPPS Stats Table (equiv and kg-kg) - MS Summary Table - V9.xlsx, comparing columns A and B in “summary table”, rewrite the parameters names to be unique. | Parameter names are not unique. |
| 3.0 | 10/20/2026 | 8 | Table 26 (overview of executed PAR experiment conditions for SPPS process), delete “Batch scale (mmol resin)” | it is the bases (1 eq) and does not add information. |
| 3.0 | 10/20/2026 | 8 | Table 26 (overview of executed PAR experiment conditions for SPPS process), fix the number of decimals (0 for min, 1 for °C, 2 for others). | Wrong number of decimals displayed, does not match with source document. |
| 3.0 | 10/20/2026 | 8 | Change the following values:<br><br>Piperidine in DMF quantity (deFmoc) (Thr9, Ala8, Cys7, Thr6, Ala5, Thr4, Asn3, Cys2, gGlu1), P3 7.49 -> 8.39; P4 8.05 -> 7.40.<br><br>Oxyma quantity (gGlu1), P4 0.07 -> 1.74. | The values were wrong because table was using v. 8 of the stats table (which contained a typo in the raw data provided by AsymChem), while the correct table should have been v. 9 (the most recent one) |
| 3.0 | 10/20/2026 | 12 | Table 31. Change parameter names for “Activation jacket temperature (Gly10)” to “Dissolution jacket temperature (Gly10)”, “DMF activation total quantity (Gly10)” to “DMF dissolution total quantity (Gly10)”, “DMF coupling total quantity (Gly10)” change to “DMF coupling/loading total quantity (Gly10)”., “Coupling temperature (Gly10)” change to “Coupling/loading temperature (Gly10)”. “Coupling time (Gly10)” change to “Coupling/loading time (Gly10)” | Correct name is dissolution for Gly10 and loading, not activation and coupling. |
| 3.0 | 10/20/2026 | 12 | Table 31. Include drying jacket temperature. | Drying jacket temperature was not part of “Table 31. List of parameters, PARs and Final Risk Category” |
| 3.0 | 10/20/2026 | 6.12 | Table 24. Parametric studies for the resin de-swell and drying step. Update Drying temp max studied NMT 42.0°C | Max incorrectly reported max studied NMT 42.0°C. |



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| 3.0 | 10/20/2026 | 14 | Change footnote reference from "28" to cross-reference to Asymchem Tech report (**Table 34.** Stable hold points [23]). Also change phrasing from "Wet cake resin stability at elevated temperature (drying study)" to "Dry cake resin stability at elevated temperature (drying study)". | Wrong reference.<br><br>Dry cake is not specific enough, it was not clear if the resin was initially wet. Asymchem tech report Table 5.1 reports correct value. |
| 3.0 | 10/20/2026 | 8 | Table 26 (overview of executed PAR experiment conditions for SPPS process), remove "post-deFmoc jacket TEMP 10.2.1", "transfer temperature", and "yield" rows | Not a PAR parameter (always nominal) and useIndicator=0 |
| 3.0 | 10/20/2026 | 8 | Table 26, oxidation exposure time, set target to 50 min (instead of 20 min), change +6 sigma and -6 sigma accordingly | Align with Table 31 target time |
| 3.0 | 10/20/2026 | 8 | Table 26, change sigmas for quantity charged to % | Align with approach used in other PAR reports |
| 3.0 | 10/20/2026 | 8 | Table 26, move column B3 such that all baselines are next to each one (B1, B2, B3) | Align with approach used in other PAR reports |
| 3.0 | 10/21/2026 | 12 | Table 31. Insert "DMF post-coupling wash (Gly10)". Delete first column (#) | Was not present. It is a non-bundled parameter with UseIndicator=1, get min, targets, sigmas from "DMF post-coupling wash_Post coupling washing_2.3.1" in "LAARA L3 SPPS Stats Table (equiv and kg-kg) - MS-V9-23SEP2026".<br><br># column deletion: not necessary. |
| 3.0 | 10/21/2026 | All | Change hardcoded references for Table, Figure, etc to Word's cross-references. | To avoid misnumbering references. |
| 3.0 | 10/21/2026 | App. III | **Table 38.** Coupling and deprotection reaction time targets and PARS range studied. Change oxidation exposure time from 40 to 50, max from 100 min to 130 min. Sigma coverage from -2σ /+9σ to $-3\sigma$ / $+8\sigma$. | Wrong values reported. |
| 3.0 | 10/21/2026 | App. I | Add Table 35. UPLC-UV impurity data for the L3 Step 1 baseline and PAR experiments. | It was not included |



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| 3.0 | 10/21/2026 | App. I | In Table 36. UPLC-MS impurity data for the L3 Step 1 baseline and PAR experiments. Change corresponding footnote from D4U-D03050-122 to D4U-D03050-106. Values reported as ND in the stats table are changed to 0.10 area% | Wrong reference, no blanks should be present, |
| 3.0 | 10/21/2026 | App. I | Add footnote “25 Reported as area %. For all the non-detect values (ND) , 0.10% were given as LOQ” for Table 35. UPLC-UV impurity data for the L3 Step 1 baseline and PAR experiments, and delete the wording at the end of UPLC-MS table: “Note: values are %Area. Peaks with a UV proportion < 0.1% are not counted. Where a UV peak comprises co-eluting species, the MS area% apportionment is reported for each species; co-eluting MS proportions that are relatively low are not counted, and MS proportions < 0.1% have no available information from MS/MS. Entries marked "(MS only)" are co-eluting MS-detected species with no separately reported UV peak. ND = not detected; a blank cell indicates the species was not detected or not counted in that experiment.” | Deletion information is not necessary |
| 3.0 | 10/21/2026 | 10.2 | “action limit of 0.49 %Area, set 2 % below the NMT 0.50 %Area” delete “set 2% below” | This causes confusion and does not add information |
| 3.0 | 10/21/2026 | 6.9 | Delete footnote 10 in upper value of target range studied for Oxyma charge in Table 18. Parametric studies for the Coupling reaction step. | There was no footnote to refer to. |
| 3.0 | 10/21/2026 | App. III | Table 39. Iodine transfer time (oxidative Trt removal). Min value was 40, correct is 20 min. | Wrong min value. Change based on “iodine transfer TIME_Oxidative Trt Removal_12.1.5” |
| 3.0 | 10/22/2026 | 8 | Table 26. Oxyma quantity (Cys7, Cys2) change sigma to 1% (was 2%). The +- 6 sigma were correct. AA quantity (Gly10) change sigma to 1% (was 2%), -6 and +6 sigma. | One sigma values were wrong for AA charges. |
| 3.0 | 10/22/2026 | | Table 32. Summary of one-sided PAR parameters. Delete column that has parameter range | Range was not studied since these are one-sided PARs |
| 3.0 | 10/22/2026 | 12 | Table 31. List of parameters, PARs and Final Risk Category. Change target from 32°C to 25°C. Drying upper end NMT 42°C. +6 sigma is 37.<br><br>Wording added to explain why upper limit 42°C although measured after soft cleavage, will pass spec after step 3. | Decision to use a single decimal instead of two is by Nil T. Gray.<br><br>Jenn Stockdill advised on dimers and trimers being converted to main product after step 3. |



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| 3.0 | 10/22/2026 | 6.10 | Table 20. Parametric studies for the resin wash after coupling. Include three categories of wash: post loading/coupling DMF wash quantity (Gly10); DMF post-coupling wash (Thr9, Ala8, Cys7, Thr6); DMF post-coupling wash (Ala5, Thr4, Asn3, Cys2, gGlu1), and corresponding target and range studied. | Only a single wash was included, however they do not correspond on the actual PAR runs performed. The washes were bundled in 3 different categories: Gly10 separate since it is a single wash, Thr9-Thr6 lower target, Ala5-gGlu1 higher target. |
| 3.0 | 10/22/2026 | 10 | Delete the description below each limit plot. | Not necessary to specify the area % of each point in the limit plots since this info is already in various tables in the document. |
| 3.0 | 10/22/2026 | | Table 30. Parameter restriction summary for L3 des-Asn3 Impurity. Piperidine in DMF change from to 6.77 kg/kg to 6.78 kg/kg.<br><br>Change footnote of that table to “Riccardo Candeago Signals notebook: D4U-D03050-132”, was “Byoung Min - 3084708-D01841-008”. | PAR restriction calculations were carried out again in D4U-D03050-132. |
| 3.0 | 10/22/2026 | | Table 41. PAR Experiments for Fragment L3 (LSN4182928) Step 1 – Process parameter Assessment Study Summary.<br>Replace with Table 4 from the latest stats table (QDocs v3.0).<br>Parameters that changed:<br>piperidine in DMF_deFmoc (pre Thr9)_3.1.1, P3, P4 changed.<br>Removed the N/A from Oxidation Exposure TIME_…12.1.5" and "Oxidation Reaction TIME_…12.1.5".<br>P4, "Oxyma_gGlu1 activation_11.3.6" changed. | Stats table was updated. |
| 3.0 | 10/22/2026 | | Insert “Appendix VII: Definitions used across partnering organizations.<br>Table 42. Definitions used across partnering organizations.” | Needed to explain nomenclature differences between stats team and SMDD |
| 3.0 | 10/23/2026 | | Table 31. DMF coupling total quantity (Cys7, Cys2), -6 sigma is changed to 7.50 instead of 7.51. | Pointed out by David Yuan, error in rounding. |



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| 3.0 | 10/23/2026 | | Table 33. In-process controls for the eloralintide Boc-Fragment 3 (LSN4182928) SPPS process preparation.<br><br>Add appearance IPC.<br><br>Change LOD limit from NMT 9% to NMT 10% | Align with tech report. |
| | | | Table 31. **List of parameters, PARs and Final Risk Category.**<br><br>Update DMF quantity pre-Asn3 to match corrected restrictions in D4U-D03050-132 (min 6.77 to 6.78 kg/kg) | Align with D4U-D03050-132 |
| | | | | |



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